ŚĀḶUVAṆṆKUPPAM EXCAVATIONS

2005-07



SATHYABHAMA BADHREENATH

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प्रत्नकीर्तिमपावृणु

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Cover : Front : Excavated Brick Temple Complex, Śāḷuvaṇkuppam

Back : View of the Excavated site from the south

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FOREWORD

Temples form an important component in the tangible heritage of India and have been studied for their architecture, sculpture and also for their religious, social and economic impact. The material of construction and the skills of engineering are other fields of study.

Temples in perishable materials like brick, timber and mortar existed is clear from the inscription of Mahēndravarman̤ I of the Pallava dynasty in the Mandagapattu cave temple in Villupuram district, Tamil Nadu and also from the Sangam classics.

To a large extent, the Pallavas of Kāñchīpuram were the forerunners in the experimentation and development of temple architecture in Tamil Nadu and most of their pioneering efforts are to be seen in present Kāñchīpuram district. While it was known through inscriptions that some of the brick temples were converted into stone, no temple constructed of brick had been reported.

The tsunami of 2004, which struck the coast of south India uncovered an inscription and indirectly, a temple constructed of brick. This temple of brick at Śāḷuvaṉkuppam near Māmallapuram has been diligently exposed despite being buried in sandy soil. The type of construction, the spatial growth of the temple, the inscriptions, the terracotta and pottery found at this site have been systematically studied.

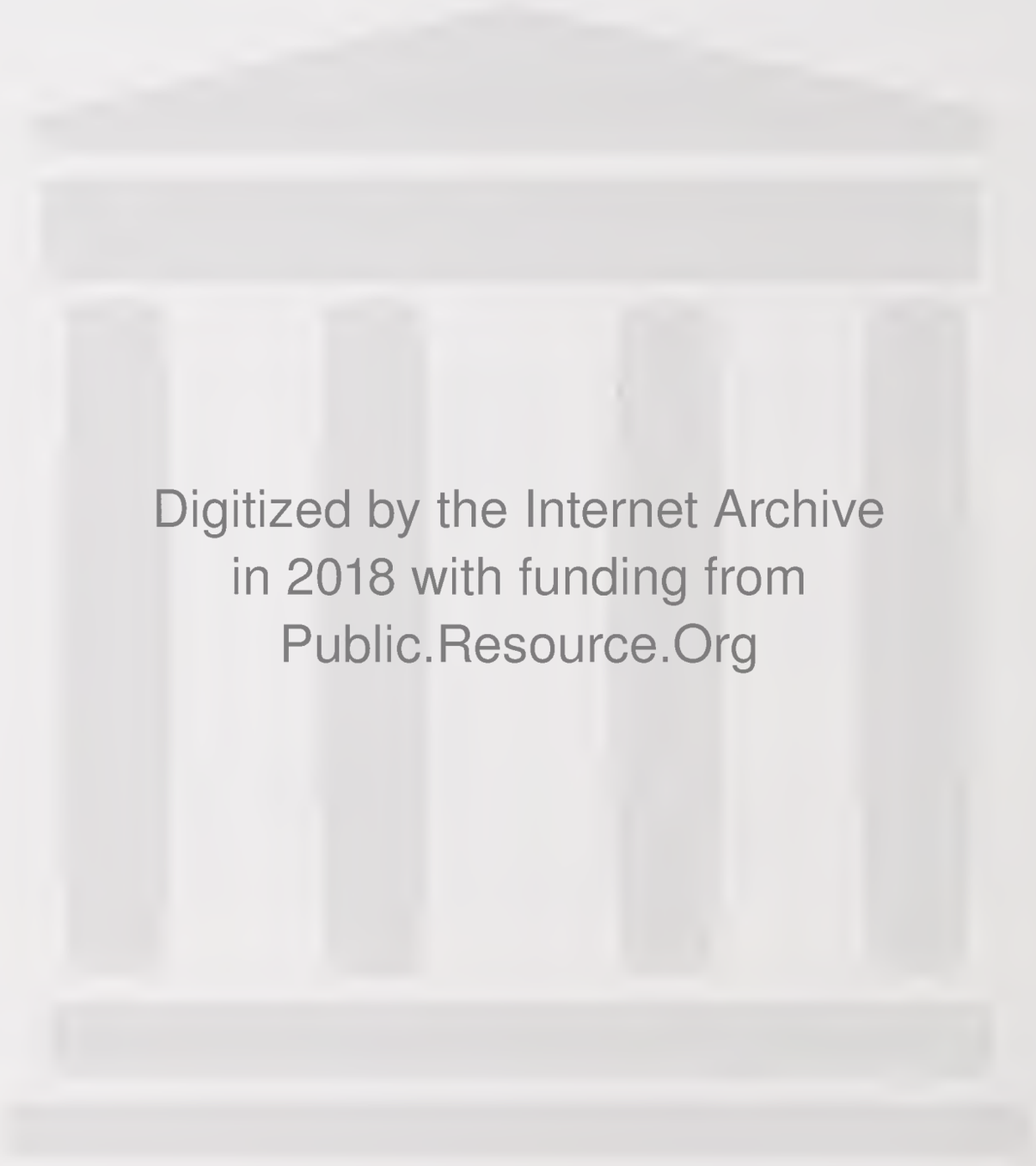
The excavation of a temple constructed of brick and the process of its conversion to stone now provide an insight into a new area of investigation. The fact that this is the earliest reported temple dedicated exclusively to Subrahmanya is another aspect of importance.

I, congratulate Smt. Sathyabhama Badhreenath and her team for bringing out this report and placing the facts of the excavation for the scholars to research upon in future.

R. Tewari

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Director General
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February 23, 2015
New Delhi


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PREFACE

The receding waves of the tsunami of 2004 brought to light new lines of an inscription of Rāshṭrakūṭa King Kṛishṇa III (939 – 967 CE) on a rock boulder adjacent to the earlier copied inscription of Kulottunga III (1178 - 1218 CE). Both inscriptions incidentally mention about a Subrahmaṇya temple. There was no temple anywhere close to the vicinity of this site and an enquiry led to a closer look at the site which fortunately resulted in exposing through systematic excavations one of the earliest temples dedicated to Subrahmaṇya assigned to the Early Pallava period.

The discovery was significant since, it contributed to information which was hitherto unknown.The evolution of a small brick temple to a full complex probably due to the patronage that it received was revealed here. The complete layout of the temple complex was exposed, which revealed three phases of structural activity ranging from Early Pallava to late Chōḷa period. It was for the first time that the inscription of the Rāshṭrakūṭa King Kṛishṇa III was found in the Māmallapuram area. It was perhaps for the first time that the spear (vēl), the attribute (āyudha) of Subrahmaṇya, a popular deity in this part of the region even to this date, has been found in stone.

The story of this episode of excavation will not be complete without mentioning the rich haul of stone inscriptions and antiquities including Roman coins and terracotta plaque with dancing girls that was unearthed from the site. A multi-disciplinary approach through a geological study to understand the sea surge activities in this area was attempted and has been incorporated in this report.

Finally, I only wish to add that a holistic study of the site through this excavation has added a new dimension to the history of the famous world heritage site of Māmallapuram.

Chennai
27-5-2010

Sathyabhama Badhreenath

ACKNOWLEDGEMENTS

The excavation at Śāḷuvaṉkuppam was started as a fall out of the tsunami which in all its fury exposed an inscription of Rāshṭrākuṭa Kṛishṇa III. The excavation was initially started in 2005 by Dr. T. Satyamurthy, then Superintending Archaeologist, Archaeological Survey of India, Chennai Circle and continued by me for the next two seasons in 2006 and 2007.

I am grateful to the Director General, Archaeological Survey of India, New Delhi for having reposed faith in me to continue the excavation. This has given me an opportunity to study and analyze the architectural exploits of the Pallavas who were trendsetters in South Indian temple architecture. During the course of my report writing on this excavation, I had the opportunity to interact and seek the opinion of scholars like Dr. R. Nagaswamy, Prof. R. Champakalaksmi, Prof. A. Sundara, Dr. P.Shanmugham, and the late Dr. B. Narasimaiah.

The structure at the site had been repeatedly reconstructed and therefore posed a great challenge in identifying the different phases of development. The loose sandy soil was also another deterrent during the excavation.

Efforts have been taken to include all the details noticed at the site as this is a unique brick and stone temple of the Pallava period.

S/Sri K.P. Mohandas and G.Thirumurthy, both Asst. Suptdg. Archaeologists were actively involved in the excavation and in finalizing the report. The writing of this report would not have been possible without the co-operation of my colleagues Sri R.S.Mulimani, Dy.Suptdg. Archaeologist; Smt. C.R. Gayathri, Asst. Archaeologist, Sri S. Jayakaran, Sr. C.A.; P.V.Rajan, Librarian and Information Assistant, H. Raghavendra, V. Sarangadharan, Draughstmen; R. Vinayagam, J. Kuppusamy, Surveyors; Smt. Girly Suguna, Photographer; Sri Veeraraghavan, Ms. Nithya, Sri M.Prasanna, Sri R.Ramesh, Research scholars; Dr.S. Rajavelu, Asst.Suptdg. Epigraphist, Epigraphy Branch, Sri Samuel Joshue, Photographer, Temple Survey Project.

I would also like to thank the Publication team of the Survey, especially its Director Sri A.K.Sinha, Dr. P. K. Trivedi, Director (Retd) Dr. Madhulika Samanta, Superintending Archaeologist and Sri Hoshiar Singh, Production Officer.

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ABBREVIATIONS

AI	: *Ancient India*
ARE	: *Annual Report on Indian Epigraphy*
EI	: *Epigraphia Indica*
IAR	: *Indian Archaeology- A Review*
SII	: *South Indian Inscriptions*
SITI	: *South Indian Temple Inscriptions*

Location of Śāḷuvaṇkuppam

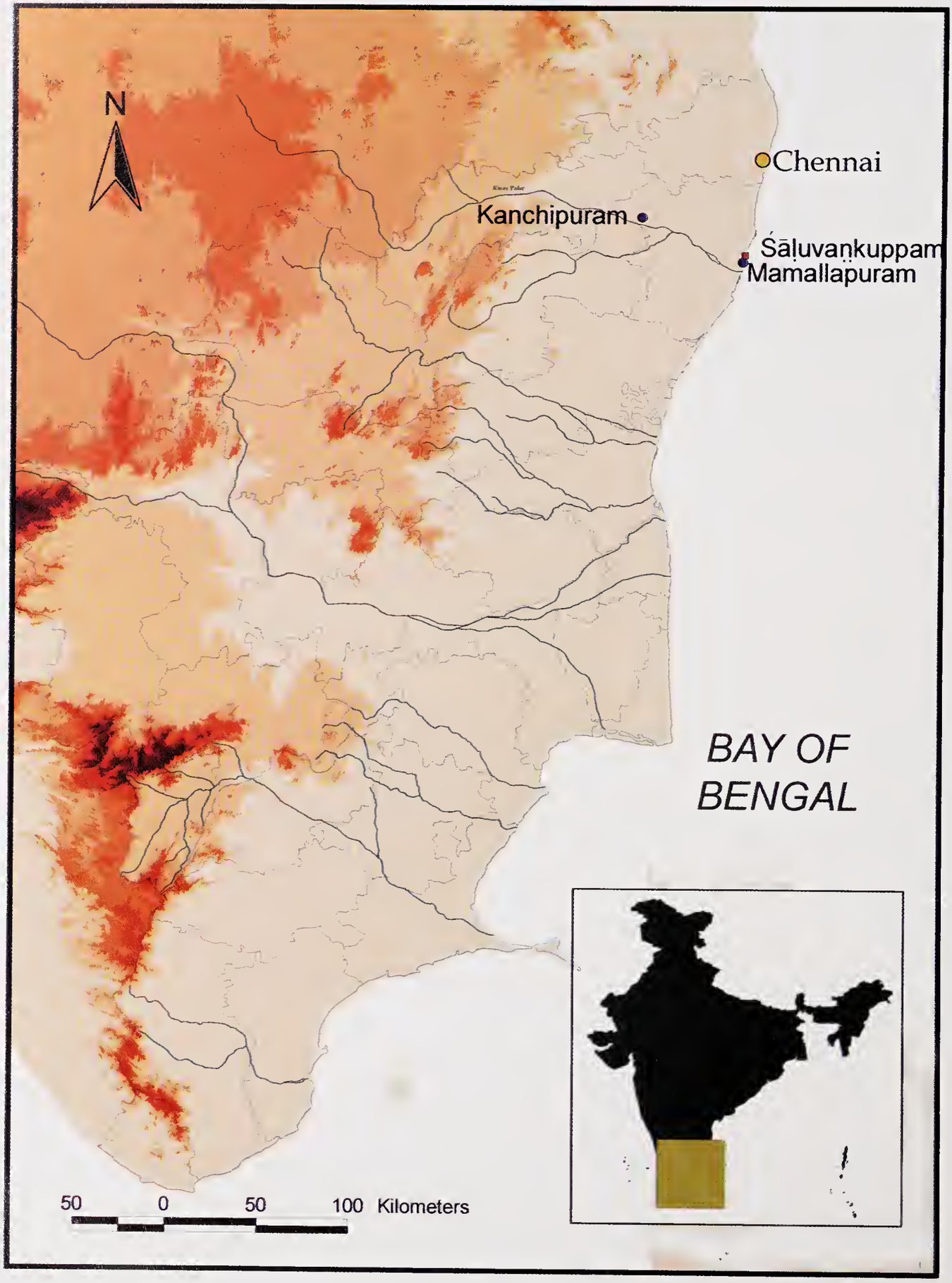


Fig 2 (see p. 1)

CHAPTER I

INTRODUCTION

The shoreline off the coast of Māmallapuram has a large number of stone outcrops, which were judiciously utilized by Pallava craftsmen for their awe-inspiring creations. These works of art have been extensively described in European accounts like those of B.G. Babington, J.F.Fleet, John Goldingham, Charles Gubbins, Alexander Rea, J.Ph. Vogel etc. However, the main problem European travellers encountered was the drifting sand dunes that were continuously burying these places. The Archaeological Survey of India over the last century has been involved in the task of exposing the monuments by removing the drifting sand. Even two decades back during clearing operations in the Shore temple complex, an apsidal temple, a bathing ghat like structure and some brick structures were unearthed. This type of scientific clearance is an ongoing process, which is being continuously adopted by the Survey.

Incidentally the tsunami of 2004 exposed some structures and sculptured rocky outcrops alongside the coast to the north and south of the Shore temple.

The fluorescence of Pallava architecture can be best studied at Māmallapuram, which is studded with rock cut caves, bas reliefs, monoliths and structural temples. What was missing was the brick structural temple. This was corrected with the discovery and excavations of a brick temple complex at Śāḷuvaṇkuppam which must have once formed the periphery of the port town of Māmallapuram (pl. 1 A).

A sporadic archaeological investigation for evaluating the damage caused during the flury of the tsunami that hit the coast of Tamil Nadu on December 26, 2004 resulted in the discovery of new evidence by way of a three-line inscription engraved on the eastern face of a boulder at Śāḷuvaṇkuppam, five kms north of Māmallapuram(Fig.1).

The receding tsunami waves had eroded about 60 cm of sand around the boulder to the north of Tiger's cave, on which is engraved a Chōḷa inscription of Kulōttuṅga III (1178–1218 CE) recording a donation made towards a Subrahmaṇya temple. Incidentally, the newly exposed inscription of Rāshṭrakūṭa Kṛishṇa III (939-967 CE) engraved on the eastern side of the same boulder, too speaks of a Subrahmaṇya temple (Pl. I B).

What puzzled the archaeologist was the non – existence of such a temple in the vicinity. A survey in the region near the boulder (Pl. II A) bore fruit, resulting in the discovery of a Chōḷa coin. Brick bats were also seen strewn around. On a closer examination, the area seemed to be on a higher level than the surroundings giving the appearence of a mound (Pl. II B). A trial trench was laid on this mound (just north of the boulder) which was

otherwise thought to have been a sterile sand dune (Pl. III A). The discovery of two more inscriptions also referring to a Subrahmaṇya temple led to a systematic excavation. The excavations at the north of the boulder taken up in 2005 exposed a brick temple underlying a stone one (Pl. III B). A cloister *maṇḍapa* enclosed the temple.

The excavations were continued in 2006 and an extension to the cloister *maṇḍapa* on the east and west and a *mukhamaṇḍapa* was noticed. The excavations in 2007 led to the unraveling of a *mukhamaṇḍapa* of a later phase. Since no inscriptions pertaining to this temple are available after Kulōttuṅga III and as later inscriptions of the Pāṇḍyas, Telugu Chōḷas and Vijayanagara kings are noticed at Māmallapuram, Tiruvidandai, Tirukkaḷukkuṇṛam and Vāyalūr, all close to Śāḷuvaṇkuppam,it could be presumed that the temple might not have been in existence even during the times of the Pāṇḍyas. Moreover as the British travelogues also do not make mention of any such temple remains, but describe other structures, like the ones at Māmallapuram and also state that the Atiraṇachaṇḍa cave is more than half buried, it is presumed that the temple might have already been in ruins and probably well buried.

The excavation brought to light structural activities of three phases,beginning with the early medieval to the late medieval period. The signifcance of this discovery lies in its being the earliest record of a temple dedicated to Subrahmaṇya in Tamilnadu.[1] The legendary achievements of Muruga (identified with Subrahmaṇya) are often alluded to in the classics of Tamil literature. The worship of this deity is very popular in this part of South India and its antiquity can be traced to 2nd century CE, for there are references to this in the *Sangam* texts like *Puṛanāṇūṛu, Paripāḍal, Ahanāṇūṛu, Pathiṛṛupattu, Perumpāṇāṛṛupaḍai, Śilappadikaram* and *Tirumurugāṛṛupaḍai.*

The site of Śāḷuvaṇkuppam referred to as *Tiruviḷichil* in the inscriptions is located in the Toṇḍaimaṇḍalam region. King Mahēndravarmaṇ I (c 600 – 630 CE) of the Pallava dynasty gave a new impetus to religious architecture in the Tamil country by excavating a number of imperishable rock cut temples. From his inscription at the cave temple of Maṇḍagapaṭṭu it is inferred that brick, timber, stucco and metal were in use for the construction of edifces. Alongside the use of stone for the building of temples, brick also was used in some of the Pallava temples like the Śiva temple at Kūram, the Sundaravarada Perumāl and Kailāsanātha temples at Uttiramērūr to name a few (Fig. 2).

The temple at Śāḷuvaṇkuppam was initially built of brick and had witnessed a change in the medium of construction from brick to stone at a later point of time. Hence, it becomes even more signifcant, as the early phase of this structure in brick is the earliest known temple and could be the earliest edifice in Māmallapuram[2] (as Śāḷuvaṇkuppam is in all probability a peripheral area) before the advent of structures in stone medium.

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1. Velanjeri copper plates of Pallava *Aparājita* (9th regnal year i.e. 884 CE) refers to the temple of Subrahmaṇya atop the hill at Tiruttani as already in existence when he made some donations to it. However the earliest inscription in the *Kaṅdasami* (Subrahmaṇya) temple at Tiruppōrūr is that of Pallava Narasimhavarman II.

2. The elliptical temple complex with miniature shrine to the north of Shore temple is attributed to Narasiṁhavarman I Māmalla.

Further, it is to be recorded here that the conclusion reached from this study is purely based on the evidence available through the excavations. The plan of the temple as envisaged through the study during each phase was quite diffcult to arrive at, due to disturbances and changes that it had encountered at every stage. Finally, it is difficult to say, if in the final point in time the temple as envisaged in stone was ever completed as stone, architectural members, particularly the members above the wall portion,were lying scattered.

The other major problem that was encountered during the excavations was the surrounding loose, sandy soil and hence, recognizing and maintaining stratigraphy was a formidable task. However, all efforts were made to document the finds systematically.

The excavation of this temple site proved to be of immense signifcance in the study of Pallava architectural development because, it is here for the first time that a temple constructed entirely of brick and subsequently converted into stone was exposed.

Plate I



A. General view of the excavated temple complex -Śāḷuvaṇkuppam (see p.1)



B. Inscribed Rāshṭrakūṭa inscription on the rock boulder (see p. 1)

Plate II



A. A close view of the mound with boulder (see p.1)



B. General view of the site – before excavation (see p. 1)

Plate III



A. Stone architectural members and bricks structure exposed during the initial stage of excavation (see p. 2)



B. Stone pillars found in excavation (see p. 2)

Important Pallava Temples in Toṇḍai-Maṇḍalam

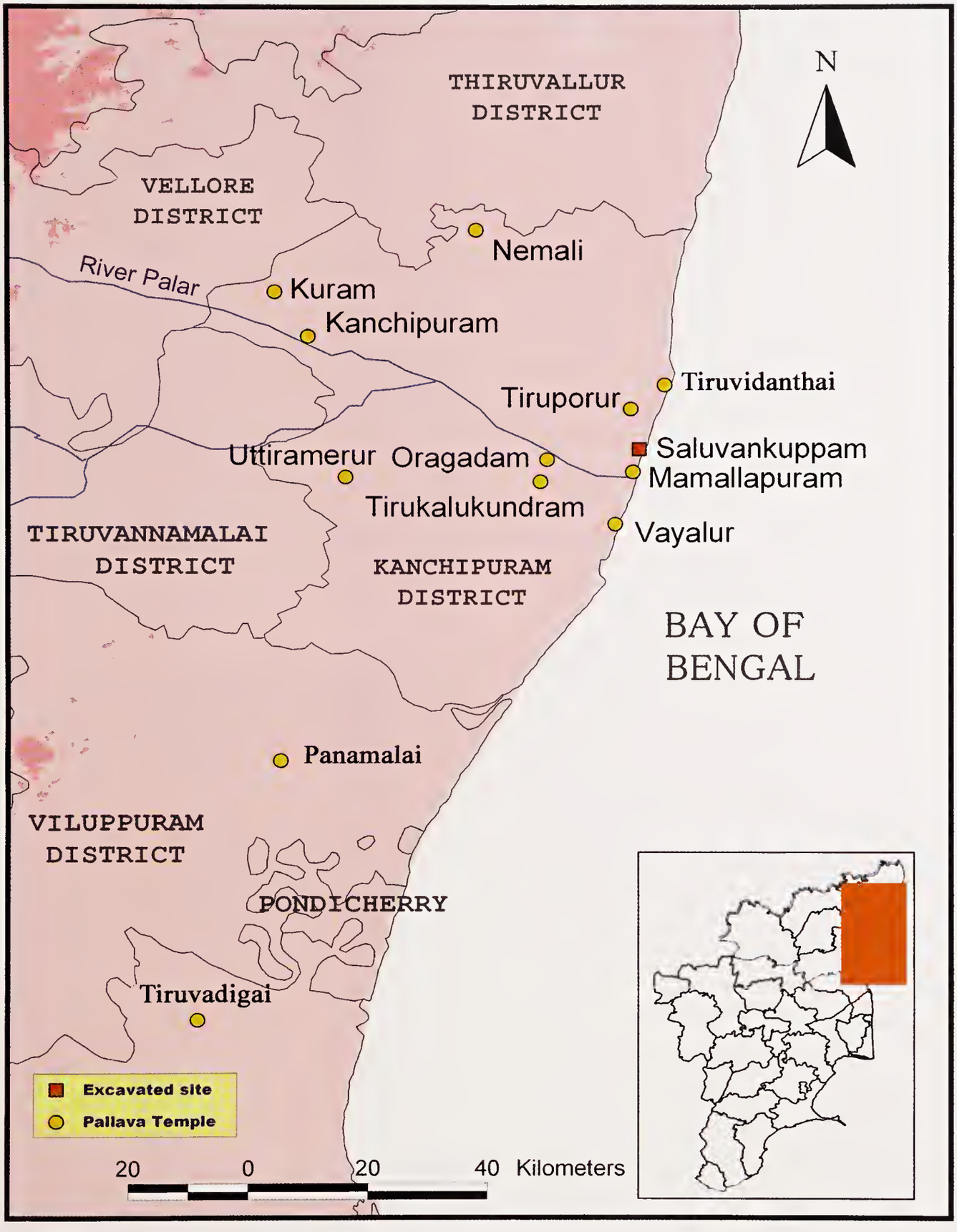


Fig 2 (see p. 2)

CHAPTER II

THE SITE AND ITS ENVIRONS

THE SITE

Śāḷuvaṉkuppam (Lat.12°39'25" N; Long. 80°12'52" E), a coastal village on the fringes of the Bay of Bengal in the district of Kāñchīpuram, is located 53 km south of Chennai, the capital city of Tamilnadu state and 5 km north of Māmallapuram. Probably a peripheral area of the port city of Māmallapuram, the site is best known for its twin Pallava cave temples - the Atiranachanda and Yāḷi *maṇḍapa* attributed to Narasimhavarman II Rājasiṁha (Fig. 3). It is pertinent to note that as the most conducive rocky outcrops at Māmallapuram were exhausted by his time, Rājasiṁha might have exploited those outcrops in the suburbs of the port city where the recently excavated brick temple was unearthed. The discovery and excavations of this temple complex add to the site's importance and has contributed to the study of brick structural temples of the period of the Pallavas (Fig. 4).

PHYSIOMORPHOLOGY OF THE SITE

The district of Kāñchīpuram has a long coastline of 87 km and is interspersed with low hills in Chingleput and Madurantakam taluks. The sedimentary rock types like sandstone and the crystalline formations like charnockite and granitic gneiss are the rock formations in this region.

The coastal belt is flat and sloping seaward and is characterized by several strand lines, lagoons, barrier dunes, spits, beach terraces etc. The Buckingham canal, a later intervention runs parallel to the coast line and not very long ago this canal was also used for ferrying people and merchandise. River Palar, the largest one flowing in this area debouches into the sea near Sadras, about 22 kms to the south of Śāḷuvaṉkuppam.

The coastal town of Māmallapuram is dotted with innumerable rocky outcrops jutting out of the sand dunes. Māmallapuram finds mention in the early period as *Mallai*, *Kadalmallai* referring to the small hillocks that are seen even today at Māmallapuram

The rocky outcrops served as the proper platform for the Pallavas to unleash their skills on stone. Sandstone, granitic gneiss and charnockite were the raw material, which ideally suited the Pallava craftsmen to experiment with different types of architectural activity.

The temple site, which was excavated is situated adjacent to a rocky outcrop on which the most important inscription was engraved giving information about a Subrahmaṇya temple. As the rock boulder might have had some religious significance

and as clay (for preparing bricks) was easily available in the vicinity along with shell lime, the temple might have been conceived at this site. Possibly due to the incessant action of the sea and saline wind activity, the temple was converted into stone subsequently.

It is noteworthy that for the first time in the annals of Pallava architectural study in Tamilnadu, that a temple evolving from brick to stone has been excavated. The mode of construction and its adaptability to developing trends has been effectively studied here for the first time through material data.

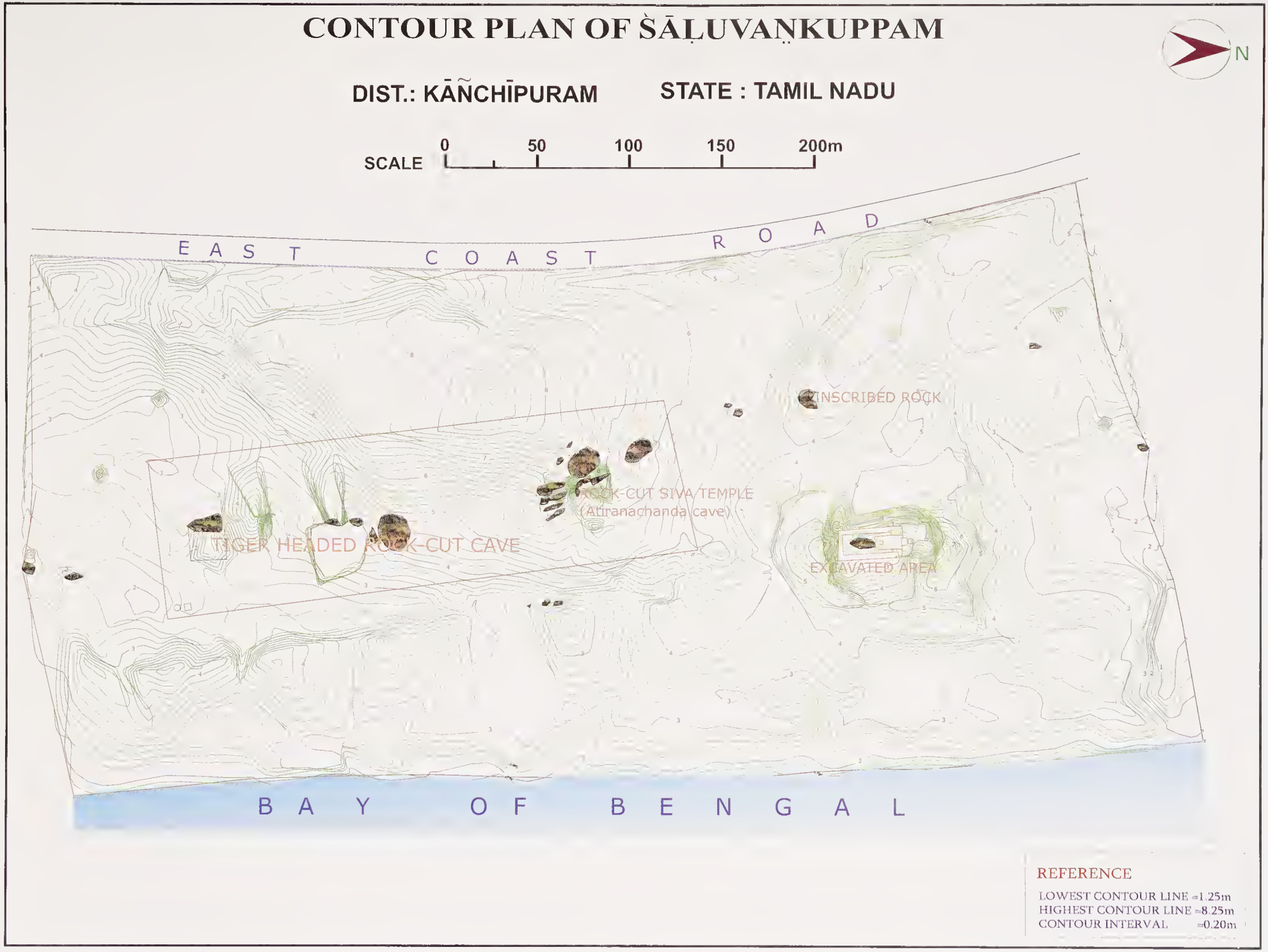

CONTOUR PLAN OF ŚĀḶUVAṆKUPPAM
DIST.: KĀÑCHĪPURAM
STATE : TAMIL NADU
SCALE 0 50 100 150 200m
N
EAST COAST ROAD
INSCRIBED ROCK
ROCK-CUT SIVA TEMPLE
(Atiranachanda cave)
TIGER HEADED ROCK-CUT CAVE
EXCAVATED AREA
BAY OF BENGAL
REFERENCE
LOWEST CONTOUR LINE =1.25m
HIGHEST CONTOUR LINE =8.25m
CONTOUR INTERVAL =0.20m

Fig 3 (see p. 9)

SITE PLAN OF ŚĀḶUVAṆKUPPAM , EXCAVATED STRUCTURE AND SURROUNDINGS



Fig 4 (see p. 9)

CHAPTER III

HISTORICAL BACKGROUND

LITERARY REFERENCES

Śāḷuvaṇkuppam is located to the north of the ancient port city of Māmallapuram. Śāḷuvaṇkuppam referred to as Tiruviḷichil in the inscriptions must have once formed the northern periphery of the port city. Hence, the historical background of Śaḷuvaṇkuppam is to be gauged in the backdrop of the historical scenario unveiled at Māmallapuram. The historical and archaeological evidence prove beyond doubt the existence of Māmallapuram as one of the important early port cities since the beginning of the Common Era and also as a centre for the development of art and architecture during the Pallava period.

The port city of Māmallapuram finds mention in several foreign and indigenous texts as early as the first two centuries of the Common Era. The *Periplus of the Erythraean Sea* of the first century CE makes the earliest reference to Māmallapuram. The Roman geographer Ptolemy of the 2nd century CE refers to it as '*Malange*'.

Some of the scholars like Dr. Rajamanikkanar and R.Nagasamy identify a port town near Kāñchīpuram called *Nirpeyarru*[1] as Māmallapuram, where there was a lighthouse in the days of Toṇḍaimaṇ Ilanthiraiyan (Sangam Chōḷa). M.Raghava Iyengar[2] suggests that *Nirpeyarru* might be *Kadalmallai*, and he also provides the literary meaning of the term, the place which receives the showers of the sea *(nir-peyal-tu)* i.e. "the city which abounds in sea-borne goods."

One of the Vaishṇava saints *Buttat Āḻwār* (5th/6th century CE) refers to Māmallapuram as *Mallai* and *Māmallai*, a celebrated pilgrimage centre of Lord Vishnu. Another Vaishṇava saint *Tirumangai Āḻwār* (c 8th century CE) describes the port activities of *Kadalmallai* in his *Periya Tirumozhi*[3]. *Dandin*, the great Sanskrit poet of the Pallava period (c 8th century CE), in his famous Sanskrit work *Avantisundari katha* mentions Māmallapuram as Māhamallapuram[4]. *Nandikalambakam*[5] a Tamil composition of Pallava Nandivarmaṇ III (846-869 CE) mentions *Mallai* as amongst the prominent ports on the *Toṇḍaimaṇḍalam* coast.

Māmallapuram is also referred to in the works of *Jayakondar, Sekkilar* and *Ottakuttar* during the Chōḷa period.[6]

None of the literary works refer to Śāḷuvaṇkuppam or Tiruviḷichil exclusively.

EPIGRAPHICAL SOURCES

While literary sources clearly attest to the fact that Māmallapuram was a flourishing

1 N.S. Ramaswami, *2000 years of Māmallapuram,* Vol. I (Text), pp. 19-21.
2 Ibid., p.186.
3 Ibid., pp..33-34.
4 Ibid., pp.28-31.
5 Ibid., pp.34-35.
6 Ibid., p. 38.

port city, the epigraphic references indicate the architectural and religious developments and also the role played by the administrative hierarchy in its growth and expansion.

The earliest inscriptions at Māmallapuram of the Pallava rulers like Narasimhavarmaṉ I, Paramēsvaravarmaṉ I and Narasimhavarmaṉ II are mostly enumerating the various epithets of the kings and do not reveal much about any other activity. Chōḷa inscriptions of Rājarāja I, Rājēndra II, Kulōttuṅga I and a couple of Telugu Chōḷa inscriptions refer to the donations made to the temples, and also ment in the revenue divisions that prevailed at that point of time.

Some epigraphic sources also refer to the place as Māmallapuram. Nandivarmaṉ II Pallavamalla in an inscription dated in his 65th regnal year refers to the place for the first time at Māmallapuram itself. Subsequent Chōḷa inscriptions also mention the name and even mention the revenue divisions. But, an inscription of Nandivarmaṉ II Pallavamalla dated in his 37th regnal year found at Paiyanur, about 7km north of Māmallapuram gives a reference to Māmallapuram and hence this is even earlier to his Māmallapuram record. An inscription of Chōḷa Rājēndra II in the Varahāswami temple at Māmallapuram refers to donations made to it from the village of Tiruveḷichil and hitherto this was the earliest reference to Tiruviḷichil.

Inscriptions at Śāḷuvaṇkuppam

The rock cut excavations at Śāḷuvaṉkuppam attributed to Narasiṁhavarmaṉ II Rājasiṁha have some inscriptions engraved on them which speak of a temple dedicated to Siva called *Atiraṇachaṇḍēśvara*. The other inscription on the boulder to the north of the cave temple refers to the offering of gold for the deity *Chattaperumanar*. The lengthy inscription of Chōḷa Kulōttuṅga III engraved on the boulder around which excavations were carried out spells out the strategy for saving the Subrahmaṇya temple by the sale of temple lands.

The excavation at Śāḷuvaṇkuppam has revealed an inscription of Nandivarmaṉ II Pallavamalla dated in his 12th regnal year and has become the earliest inscription so far found, which refers to the port city as Māmallapuram. The other inscriptions record the donations made to the temple and the role of the village administrative bodies. The importance of the temple and its popularity during the Pallava period can be gleaned from innumerable donations that had been made even by the later Pallava rulers like Nripatuṅga, Kāmpavarmaṉ(whose inscriptions have been found here in the Māmallapuram area for the first time) and also of Nandivarmaṉ II Pallavamalla.

The status of Śāḷuvaṇkuppam through epigraphical/literary sources is not known after Kulōttuṅga III and until it is mentioned by Mrs. Maria Graham in the 'Journal of a Residence in India[7]. Even in 1844 Braddock only speaks of the Tiger's Cave[8]. A. Rea in 'Pallava Architecture' mentions that numerous mounds exist both at Māmallpuram and Śāḷuvaṇkuppam, where inspite of the heavy mass of sand; several pottery, bricks and other older relics are found strewn about in many directions[9]. The monuments at Śāḷuvaṇkuppam including the inscription of Kulōttuṅga III on this boulder were cleared of the sand dunes and were protected by the Archaeological Survey of India in 1941.

7 Ibid., p. 71.
8 Ibid., p. 96.
9 Ibid., p. 136.

CHAPTER IV

EXCAVATIONS

Lay-out Plan

The inscription of Kulōttuṅga III of the Chōḷa dynasty inscribed on the rear side of the boulder assumed a new implication with the discovery of an inscription of Rāshṭrakūṭa Krishna III. The possibility that a temple dedicated to Subrahmaṇya could be in the immediate environs assumed importance as already stated and the area was probed. With the aim of exposing a temple site as referred to in the two inscriptions, the excavations were started. The archaeological excavation which primarily aimed at tracing the remains of a Subrahmaṇya temple was initiated in 2005-06 in an area [70m (north-south) and 40m (east-west)] to the north of the protected rock boulder bearing the inscriptions and about 130m from the coast line. The most contentious issue during the entire period of the excavation was the loose, sandy soil which often caved in.

The removal of accumulated sand dunes upto a depth of 1.5 m brought to light structural remains of a temple built with dressed granite stone as veneer; brick and stone rubble filling as core. The stone alignment formed a temple plan consisting of a *garbhagṛiha, ardhamaṇḍapa, mahāmaṇḍapa and mukhamaṇḍapa* facing north, axially oriented in north-south direction. Further in 2006-07, the excavation work was continued with a view to tracing out all the possible structural remains of the temple, especially its complete plan. The excavation exposed a second and earlier level consisting of *garbhagṛiha, ardhamaṇḍapa* and *mukhamaṇḍapa* enclosed by a cloister *prākāra* measuring 36.50m (north-south) and 18.30m (east-west). A *māhamaṇḍapa* (of the last phase), *balipīṭha* and ring wells further north of the *mukhamaṇḍapa* were also noticed.

The sanctum (*garbhagṛiha* and *ardhamaṇḍapa*) is rectangular on plan and measures 7.50m (east-west) and 8.00m (north-south). It is built with bricks abutting the natural rock boulder on the southern side. The brick walls of the *garbhagṛiha* are 2.65m thick on the east and west, whereas its thickness varies on south and north as they measure 2.75m and 3.60m respectively. The brick walls of the *garbhagṛiha* has 23 courses of brick and rises to a height of 2.45m. The inner hollow portion of the *garbhagṛiha* was filled with fine sand, brickbats, laterite stone chips and clay and was provided with thin granite slabs at the top plinth level. The bricks measure on an average 40x21x7cm. The mortar used as binding medium for the brick structure is clay. The side walls of this unit were covered with fine lime plaster on all sides. The inner face of the northern wall has two layers of plaster with the outer surface having a very fine lime plaster. This surface might have been painted as fragmentary pieces of painted lime plaster were found.

The brick courses rest on four courses of dressed laterite blocks and beneath this is a flooring of brick (one layer only). A cushion of fine river sand was seen below the last level of the brick course of the *garbhagṛiha* at a depth of 3.65m indicating that such a practice was already in vogue. The *ardhamaṇḍapa* (later phase) measures 2.50m (north-south) and 3.50m (east-west) and is raised along with the *garbhagṛiha* on the same foundation as a single unit.

The sanctum is approached by stone steps of 1.40m width situated on the lateral sides of the *ardhamaṇḍapa*, which has been subsequently made defunct by the construction of a stone *ardhamaṇḍapa* at a later date. The extant *mahāmaṇḍapa* measures 7.70m (north-south) and 7.80m (east-west). It is built with cut stones and the core is filled with stone rubble. The existence of four sockets for pillar bases at the centre of this *maṇḍapa* reveal that an *ardhamaṇḍapa* of an earlier phase once existed here. The sockets which are rectangular in shape and measure about 30x17 cm must have been intended for the erection of stone pillars. Also a portion of the *maṇḍapa* at the centre is preserved with polished stone flooring.

The brick sanctum and the elliptical rock are enclosed by a cloister *prākāra* 36.50m (north-south) and 18.30m (east-west). It has a width of about 3.20m while the outer and inner veneer walls measure 0.65m in thickness. Like the foundation of the *garbhagṛiha*, here also river sand has been used as a cushion. There is a granite stone insertion at the first phase level, which demarcates the two phases of the cloister. The cloister *maṇḍapa* shows two levels of lime plastered floor and also at the south eastern corner were found few hearths which could have been used as lime kilns. The walls were covered with fine lime plastering. Stone pillars, many of which are broken or fallen and a few standing on a nearly square stone base is found placed at a regular interval of about 2.0m. One of the pillars (fallen down) on the south western corner has the inscription of the Rāshṭrakūṭa king Kṛishṇa III (939-967 CE). The cloister *maṇḍapa* has been enclosed by a *prākāra* wall on the east, south and west, of which only a portion has survived. The extant damaged *prākāra* wall on the eastern side was leaning towards the inner side i.e. the western side which was probably due to lateral pressure caused by the sand dunes. At the centre of the southern *prākāra* wall is found a stone boulder packing which probably was a rear entrance. It was closed at a later date. The second level (raised) of the cloister exhibits the *adhiṣṭhāna* moulding like *tripaṭṭa kumuda* on the northern side. The remaining portions are not visible due to later activity. An extension to the structure has been made beyond the cloister on the northern side on both east and west. The eastern side has provision at three points probably for pillars, and appears as a rectangular pillared *maṇḍapa*. The western side has the mouldings of an *adhiṣṭhāna* like *jagati* and *tripaṭṭa kumuda*. As a good quantity of fresh shell and clay was noticed here, it might have been a store house.

Loose granite stones and broken pillars of an earlier phase were utilized for the creation of an entrance platform on the north to the *mukhamaṇḍapa*. Adjacent to this on the eastern side were found traces of sun baked brick flooring.

A completely ruined *mukhamaṇḍapa* has been traced just in front of the *mahāmaṇḍapa* but outside the cloister *maṇḍapa*. This must have been constructed during the last phase of this temple. It measures [10.50m (N-S) and 10.38(E-W)] on an elevated area of about 1.15m height. The stone pillars of the earlier phase, which have been obviously reused, and cut stones were arranged in a wall-like alignment on the western and northern sides. One of the broken pillars on the northern side bears an inscription in characters of the 8th century CE. The *mahāmaṇḍapa* has been raised using rammed brick bats and terracotta tiles. In the centre of this *maṇḍapa* and in the alignment of the *garbhagṛiha*, a stone spear (*vēl*) of 1.75 m height is placed. Further north of this in the same alignment, a stone platform circular in shape was encountered at a depth of 1.15 m. The semi-circular stone in two pieces and placed together, measures 1.25m in diameter. A rectangular stone slab (97x3x7cm) is found to have been placed over a sand cushion above the circular stone slab. The circular stone slab is lined with two courses of brick at its edges. Below this is found an intact square brick *balipīṭha* plastered with lime and erected on a granite stone platform, which obviously served as a base stone. The brick *balipīṭha* has the usual *adhiṣṭhāna* moulding of the *pādabandha* type having a *tripaṭṭa kumuda* (Fig 5).

A rectangular stone platform was found at a distance of 0.80m from the ruined *mahāmaṇḍapa* in the north western corner. It was found at a depth of 1.15m. It measures [1.50m (north-south) and 0.95m (east-west)]. The stone platform was lined with two courses of bricks at its edges and it was sloping towards the north.

Two ring wells of terracotta were exposed in the north eastern and north western corners and another just adjoining the *balipīṭha* towards the east. A fallen brick wall running in north-south orientation was encountered at a depth of 1.00m on the north eastern corner of the temple. It is about 0.70m wide. The bricks were laid with clayey earth as binding material. The wall extends towards north to a length of 16m and seems to have been displaced on the eastern side. It is made of differently-sized bricks and is not in proper alignment. Bricks, brickbats, brickbats with lime plaster, broken stucco and terracotta figurines were found dumped to the east of this wall. At its southern end were found buried two saddle querns in inverted position. These querns were made of granite and sandstone respectively.

The temple dedicated to Subrahmaṇya had undergone numerous changes evidently due to the constant threat from the sea; progressive elaborations of the temple occurred alternatively due to religious and ritual needs and finally because of the growth of the village. All these factors undoubtely contributed to the expansion of the temple, but have made the work of the excavator an uphill task as it has been very difficult to identify the various stages of development without destroying the present ruined structure. Accordingly, only wherever possible and required, the structure was disturbed to understand the progressive development of the complex. Added to this was the loose sandy soil, which ensured that stratigraphy occupied a backseat in the excavation. The different phases of structural activity has been determined on the basis of their sequence, based on evidence within

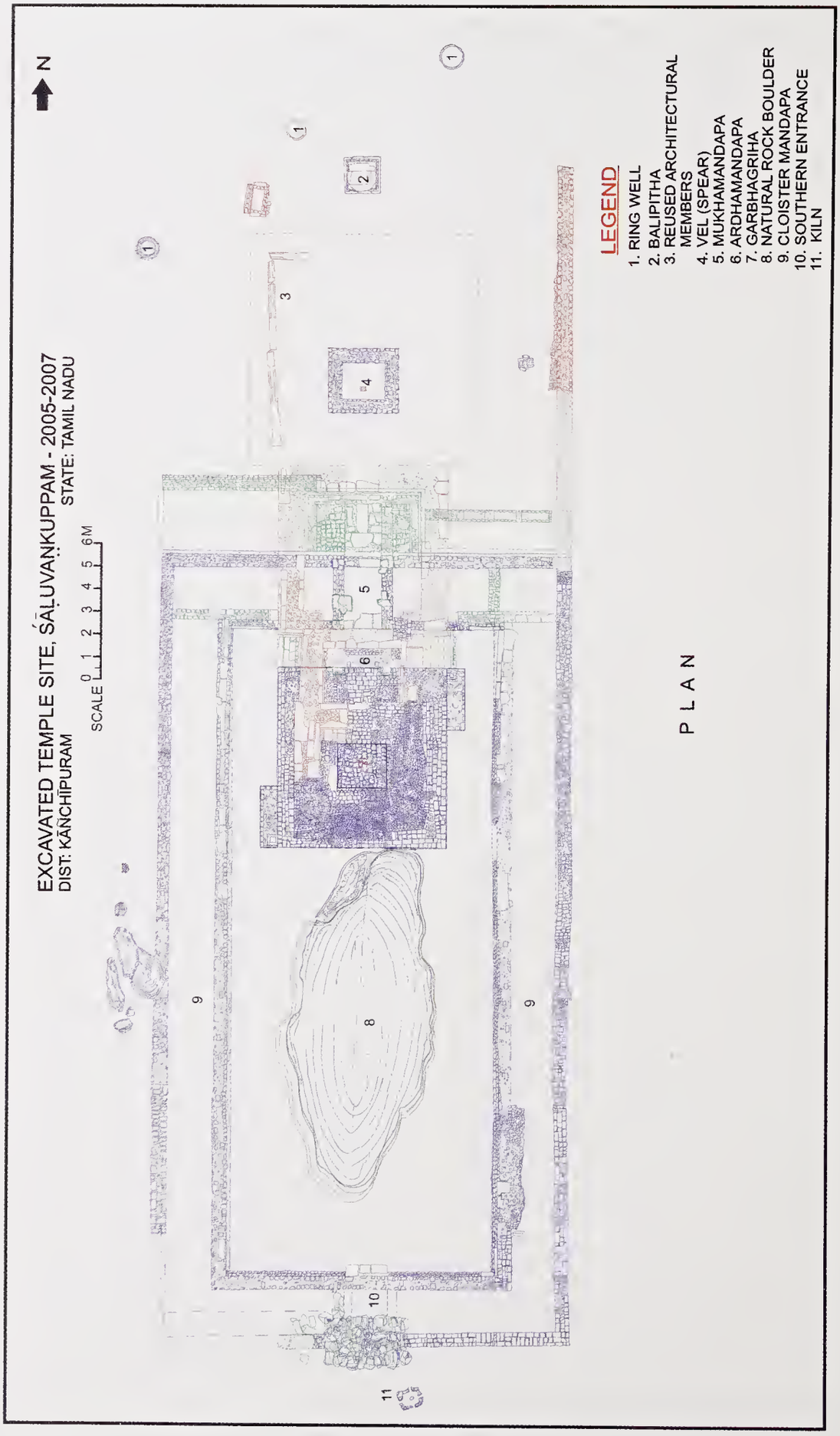
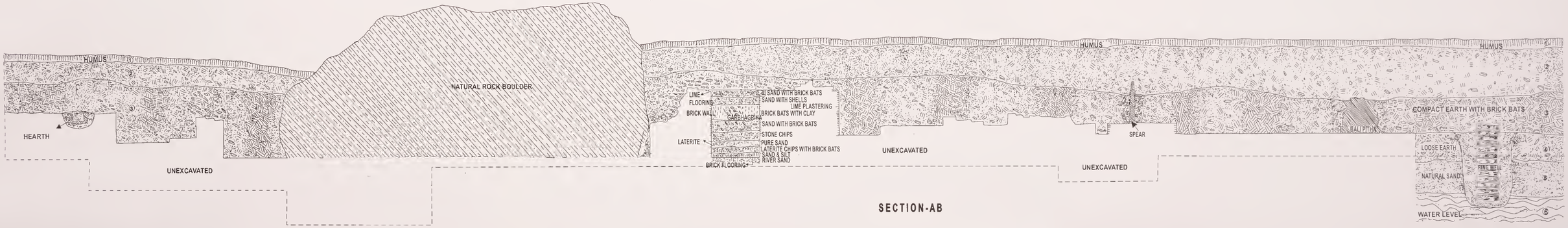

N
EXCAVATED TEMPLE SITE, ŚĀḶUVAṆKUPPAM - 2005-2007
DIST: KĀÑCHĪPURAM
STATE: TAMIL NADU
SCALE 0 1 2 3 4 5 6M
LEGEND
1. RING WELL
2. BALIPITHA
3. REUSED ARCHITECTURAL MEMBERS
4. VEL (SPEAR)
5. MUKHAMANDAPA
6. ARDHAMANDAPA
7. GARBHAGRIHA
8. NATURAL ROCK BOULDER
9. CLOISTER MANDAPA
10. SOUTHERN ENTRANCE
11. KILN
P L A N

Fig 5 (see p. 17)

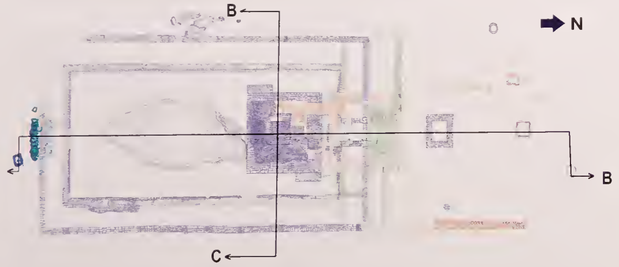
EXCAVATED TEMPLE SITE, ŚĀḶUVAṆKUPPAM - 2005-2007

DIST: KĀÑCHĪPURAM STATE: TAMIL NADU

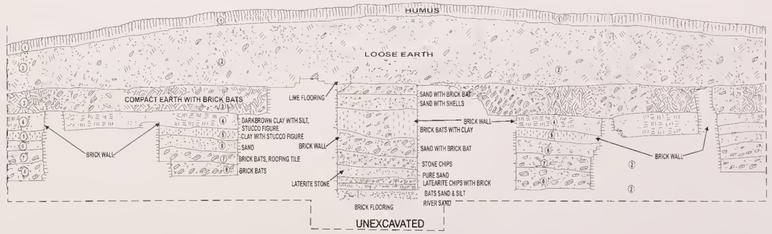
SCALE 0 1 2 3 4 5 6M



SECTION-AB



KEY PLAN



SECTION-BC

Fig.6 (see p. 22)

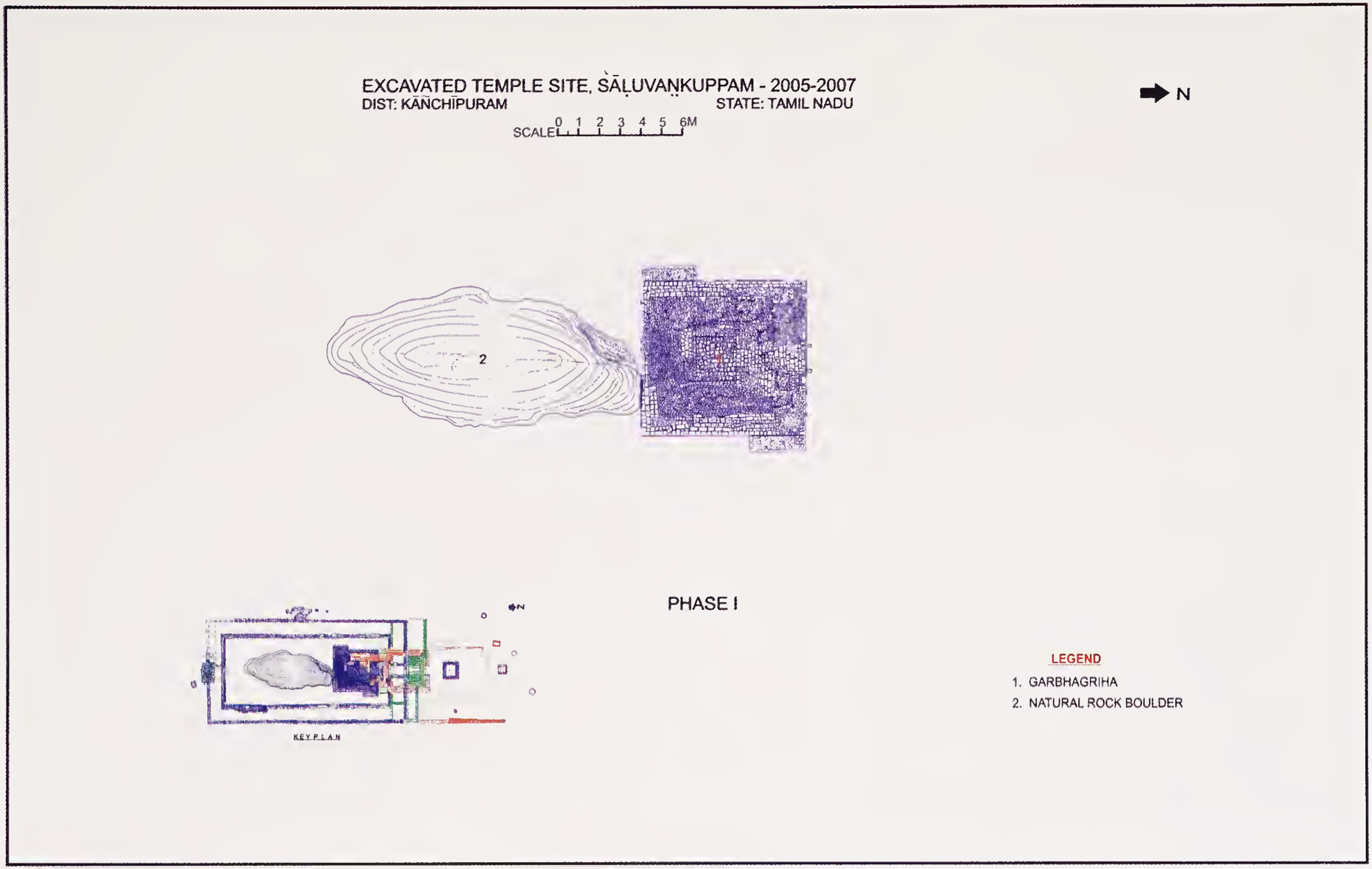

EXCAVATED TEMPLE SITE, SĀḷUVAṆKUPPAM - 2005-2007
DIST: KĀÑCHĪPURAM
STATE: TAMIL NADU
SCALE 0 1 2 3 4 5 6M
N
2
1
PHASE I
KEY PLAN
LEGEND
1. GARBHAGRIHA
2. NATURAL ROCK BOULDER

Fig 7 (see p. 22)

the framework of construction activity in the temple site.

It is to be noted that this is for the first time a temple complex initiated from the rudimentary brick structure evolving into a stone one has been unearthed through excavations in Tamil Nadu.

PHASES OF STRUCTURAL ACTIVITIES:

The excavation has exposed multiple occupational levels showing different phases of structural activity. The nature and evidence of the exposed structure reveal that the temple had witnessed basically three phases of construction and they are as follows: (Fig.6)

Phase I : Early Pallava period – pre-6th century CE.

Phase II : **Sub phase A** - Pallava period c. 6th – 8th century CE

Sub phase B - Late Pallava period c.8th – 10th century CE

Phase III : Chōḷa period c. 10th -13th century CE.

Phase I

The earliest structural activity in this temple complex has been noticed adjoining the rock boulder, which, when seen after the excavation, appears to resemble a seated elephant. Since the southern portion of the boulder looks like the hind part, the structure has been constructed to the north of the boulder. The complex as it should have been in the early stages was a modest one and comprised only a rectangular structure using bricks of the size of 40 x 21x 7cms. on an average. This is the characteristic of early structures.

In this phase it is evident that only brick had been used with dressed laterite for the foundation. Probably above the laterite foundation(0.85m in height) only four courses of bricks were laid, as a layer of lime plaster was observed just above these brick courses. The laterite foundation has a base of a single course of brick laid on a cushion of river sand (pl. IVA). It was a very moderate structure, having a simple rectangular plan [(7.50m (east-west) and 8.00m (north-south)] with an entrance from the north (pl. IVB) (Fig 7) .

The base of the structure was river sand over which a layer of bricks of different size, was laid some of which were also broken ones. The core of the structure was filled with layers of fine sand and silt, chips of laterite and finally pure sand over which the floor must have been laid. Four courses of dressed laterite blocks were laid as inner and outer veneers. The intervening core was filled with fine river sand. The outer surface of the laterite wall - the last course and half of the third course was plastered with lime along with four courses of bricks above it. The binding medium for the lower courses of the laterite as also for the inner walls of the structure was clay. Two pillar bases were seen placed perpendicular to the laterite wall on the north and in alignment to the centre of the structure. It is to be surmised that this was the entrance to the structure which was facing north (pl. V A).

A redeeming feature was the discovery of a conical vase in the foundation level on the western side just abutting the laterite (pl. V B). A Roman coin alongside terracotta lamps were found outside the structure on the west in line with the upper

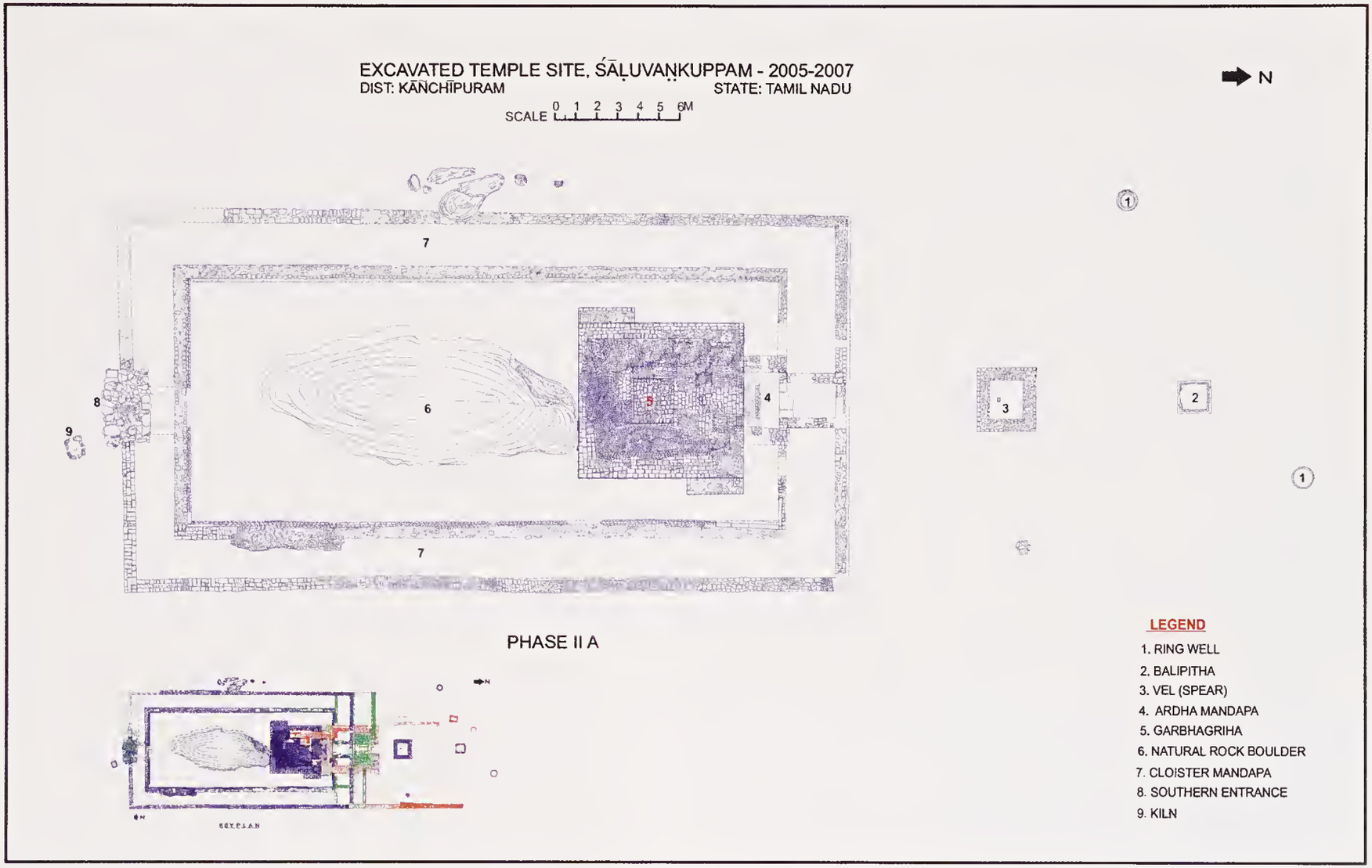

EXCAVATED TEMPLE SITE, ŚĀḶUVAṄKUPPAM - 2005-2007
DIST: KĀÑCHĪPURAM
STATE: TAMIL NADU
SCALE 0 1 2 3 4 5 6M
N
PHASE II A
LEGEND
1. RING WELL
2. BALIPITHA
3. VEL (SPEAR)
4. ARDHA MANDAPA
5. GARBHAGRIHA
6. NATURAL ROCK BOULDER
7. CLOISTER MANDAPA
8. SOUTHERN ENTRANCE
9. KILN

Fig 8 (see p. 26)

Plate IV



A. North-west wall of the temple showing plastered brick wall with blocks of laterite in the foundation and a layer of brick at the bottom. (see p. 22)



B. Aerial view of the garbhagṛiha with the ardha maṇḍapa (see p. 22)

Plate V



A. Two pillar bases in front of the plastered wall on the north (see p. 22)



B. Conical vase found in the western side of the laterite foundation (see p. 22)

most course of laterite. The use of both a layer of brick and laterite for the foundation is noteworthy as the porosity of these materials prevent capillary action especially in coastal areas.

The first phase of activity in this complex as stated earlier was very modest and did not exhibit any of the customary features usually associated with a temple. The inner and outer veneer walls of laterite were plain and merely plastered above the foundational levels probably more as a protective measure in saline conditions. There are no other indications to establish the nature of this structure. It can be merely surmised that this might have been a temple based on the inscriptions of the later period. The fact that the temple faces north does not give it Pre *Āgamic* date as temples dedicated to Subrahmaṇya, can be north facing as per the Āgamas[1].

Phase II

The temple reached its climax during this phase in both constructional and cultural activity. The temple might have received royal patronage and with the progression in respect of religious sanctions other subsidiary structural elements added to its simple plan. This level is found at a depth of about 1.90 m. Again, in this phase two sub–phases viz. A and B could be seen wherein some additions and alterations had been carried out to the already existing structure.

Sub-phase A

The level of the structure was probably raised to a height of 2m by adding another ten courses of bricks (pl. VI A). The brick size continues to be the same i.e. 40x20x7cms. A *prākāra* with stone pillared cloister was constructed enclosing the *garbhagṛiha* and the large boulder (pl. VI B). A *mukhamaṇḍapa* had been added in the north with four pillars. To accommodate the pillars near the sanctum, the first course laterite was cut into on either side, ensuring that the alignment of the entrance was not disturbed. Lateral entrances from the east and the west were introduced though the initial approach was still from the north. Sockets for pillars noticed on the eastern side might have been meant to support a small *mukhamaṇḍapa* (Fig 8).

On plan the temple at this stage, comprised a *garbhagṛiha*, and a *mukhamaṇḍapa* which was enclosed by a cloister *maṇḍapa* and a *prākāra* wall (pl. VII A). All these structures seem to have been plastered with a fine coat of lime plaster as indicated by patches of plaster (pl. VII B). The rise in the level of the temple is directly proportional to the rise in the level of the *garbhagṛiha* as seen in the core filling with an addition of levels – comprising chips of laterite, a thin layer of sand mixed with brickbats and a layer of sand mixed with clay and brickbats.

The floor of the cloister *maṇḍapa* was plastered with lime. The inner side of the floor of the cloister *maṇḍapa* was lined with granite. Traces of plaster on the outer surface of the wall of the cloister (seen in the east) signify that the *prākāra* wall was a later addition. The inner portion of the cloister was lined with granite pillars. These pillars along with the *prākāra* wall must have supported a tiled roof. The cloister *maṇḍapa* on the southern side had been pierced with

1 T.A. Gopinath Rao, *Elements of Hindu Iconography*, Vol II, Part II p.487.

an entrance whose lower layer consisted of small boulders (pl. VIII A).

In front of the temple on the same axial line of the *garbhagṛiha*, (north-south) there is a stone spear (*vēl*) (pl. VIII B) similar to the one at Arvalem, Goa. As an attribute of Subrahmaṇya i.e. (*āyudha*) it has been placed on a square brick platform constructed with broken bricks at a distance of 3.40m north of the *mukhamaṇḍapa*. The pillar, shaped like the spear of Subrahmaṇya, rises to a height of 1.60 m and is fixed at the point where the *dhvajastaṁbha* of the temple is generally located. It is made of granite, rectangular in section and similar in profile to the pillars in the cloister *maṇḍapa*. The uppermost portion of this unique piece is shaped like a spearhead to a height of 0.35m out of the total height of 1.60m and has two lines of *padma*[2] and *ūrdhapadma*[3] respectively. This is followed by a flat band and a bulbous, curved portion, which culminates in a flat rectangular base. The entire pillar is faceted hexagonally with small facets at the sides and a broad facet at the centre. The lower most part, which is buried in the brick platform, is not dressed. Such a type of stone spear has not been reported from any temple in Tamilnadu. This stone spear might also have been a *dhvajastaṁbha* as it is located outside the *mukhamaṇḍapa* but before the *balipīṭha* in alignment.

Further north and again on the same axis as the main temple is a *balipīṭha* constructed of brick having a *pādabandha adhiṣṭhāna* with mouldings like *jagati, tripaṭṭa kumuda, kaṇṭha* and *paṭṭika* and plastered with lime. A granite stone base has been provided to this structure (pl. IX A).

In this phase two ring wells with thirteen courses of terracotta rings and running to a depth of 2.40 m at a distance of 4.2m north - west of the ruined *mahāmaṇḍapa* and another at a distance of 4.25 m to the northern side of the ruined *māhamaṇḍapa* and adjacent to the *balipīṭha* with twenty two rings running to a depth of 5.5m were exposed. Each ring measures 0.88m in diameter with a height of 0.20m Sweet potable water oozed out from both the wells (pl. IX B & C).

The older inscriptional records datable to the middle of the eight century of the Pallava period found at this temple can be associated with this phase. Both of them refer to the gift of gold for the celebration of the festival of *kārtigai* at the temple.

The earlier structure was meticulously modified with bricks of similar size and now exhibits all the features of a temple. The construction of a cloister around the boulder also seems to indicate its sanctity. As it appears like a seated elephant one of the *vāhanas* of Subrahmaṇya, the cloister *maṇḍapa* might have been constructed around it. Lateral entrances, the erection of a *balipīṭha* and the stone spear might indicate that the temple has now started adhering to other Āgamic injunctions. The reference to a temple dedicated to Subrahmaṇya in the inscribed records has lent credence to its religious affiliation. The fact that donations were made for the celebration of *kārtigai* festival indicates that the temple dedicated to Subrahmaṇya might have already been in existence and had gained sufficient popularity.

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2 Inverted cyma recta moulding with lotus petal pattern.

3 Minor inverted cyma recta with lotus petal pattern.

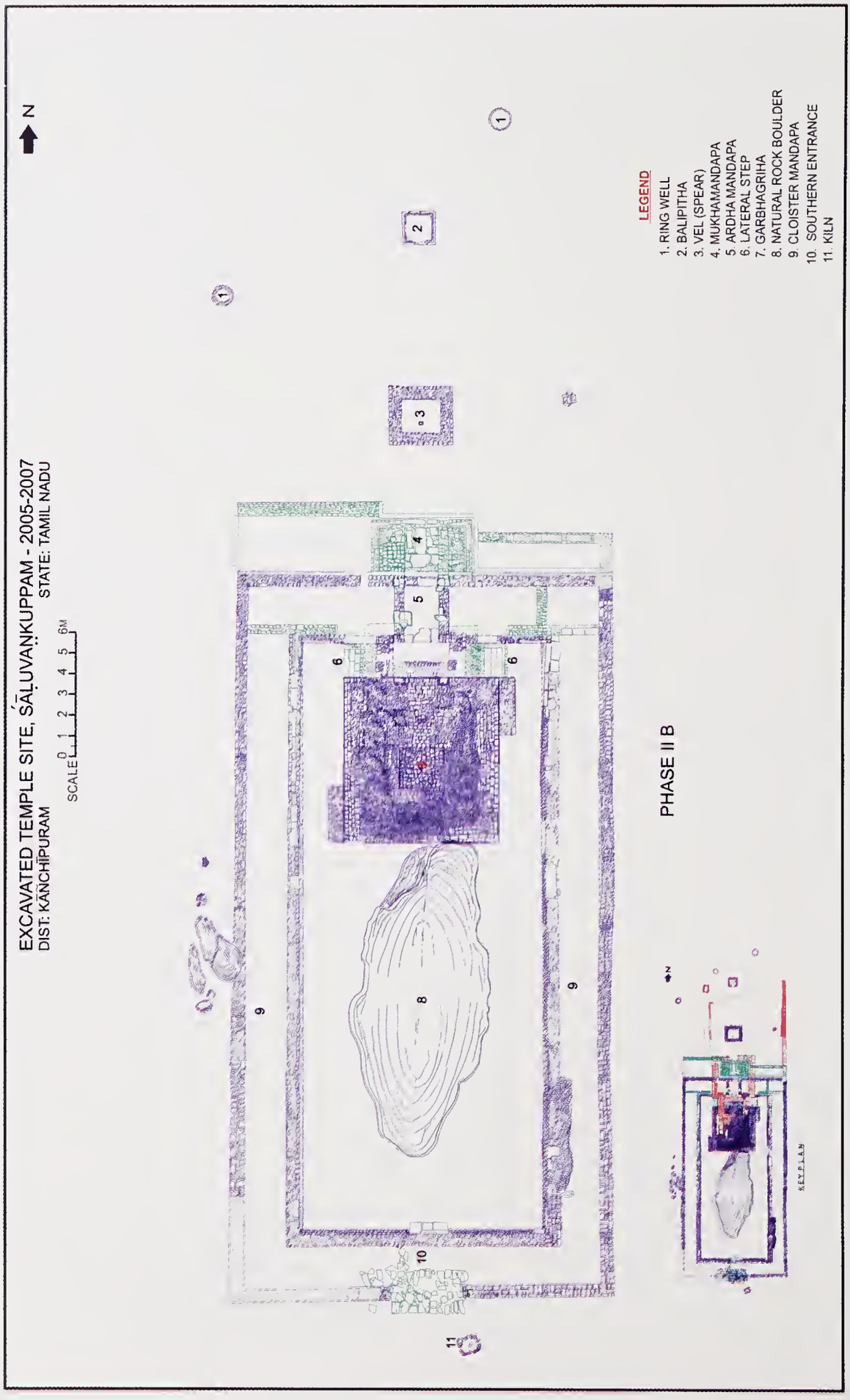

EXCAVATED TEMPLE SITE, ŚĀḶUVAṆKUPPAM - 2005-2007
DIST: KĀÑCHĪPURAM
STATE: TAMIL NADU
SCALE 0 1 2 3 4 5 6M
N
LEGEND
1. RING WELL
2. BALIPITHA
3. VEL (SPEAR)
4. MUKHAMANDAPA
5 ARDHA MANDAPA
6. LATERAL STEP
7. GARBHAGRIHA
8. NATURAL ROCK BOULDER
9. CLOISTER MANDAPA
10. SOUTHERN ENTRANCE
11. KILN
PHASE II B
KEY PLAN

Fig.9 (see p. 29)

Sub-phase B

The next phase also records a spurt in structural activity. A general rise in the working level is noticed all round the temple (Fig 9). In this period, the cloister level was raised to a height of 2.50 m. The lateral entrance with three steps in stone were added on either side of the main temple (pl. X A). A few broken pillars were converted into steps as seen on the western side. Further, the front wall of the older rectangular structure (*garbhagṛiha*) had been plastered once again. The floor level of the cloister was raised and carried a series of stone pillars at regular intervals of approx. 2 m. The pillars are hexagonal in section and in shape with a flat centre. Only a few pillars are extant while most of them are broken at the base level itself and are only about 30cm high (pl. X B).

The cloister *maṇḍapa* floor had been plastered with lime and has a paving of granite on the sides facing the boulder. The floor has on its surface shallow depressions both on the granite and lime portions. The southern entrance to the cloister is found to have been closed by packing stone boulders and the construction of a brick wall (pl. XI). Wherever the earlier cloister *maṇḍapa* had collapsed, the portion has been strengthened with the use of stone members. Behind the *prākāra* wall on the southern side a hearth(?) (probably a lime kiln) was found. Rectangular in shape it has four openings, probably for insertion of wood (pl. XII A).

On the northern side, the *mukhamaṇḍapa* of the earlier phase (Phase II A) made way for a new *mukhamaṇḍapa* with four pillars about 6m away from the *garbhagṛiha*. A proper offset has been provided at the two sides and has provisions for the erection of pilasters. Patches of plaster are noticed here also. Moulded bricks of 40 x 20 x7cm dimension have also been retrieved. The northern cloister has been pierced in the east and west to facilitate accessibility through the lateral steps. The steps have been supported by four courses of brick. On either side of the *mukhamaṇḍapa*, two rectangular chambers have been added. The chamber on the west has *adhiṣṭhāna* mouldings like *jagati* and *tripaṭṭa kumuda*, which has been finely plastered. This chamber in all probability served as a storehouse,as layers of sea shells and virgin clay were found. No proper passageway has been provided to this structure. The chamber to the east has three slots presumably for pillars. The binding medium is unbaked shell lime and the plastering is coarse. Lower in level than the other chamber, this has a step providing easy access to the *mukhamaṇḍapa*.

A floor rammed with earth and tiles was seen to the north, at the centre of which the spear has been placed on a brick platform. A raised platform utilizing broken pillars and stone members has been laid as an entrance to the *ardhamaṇḍapa* of this phase. Unbaked bricks also form a part of this platform (pl. XII B).

Most inscriptions of the Pallava, Rāshṭrakūṭa and Chōḷa periods are associated with this sub-phase justifying the fact that the temple received the maximum patronage. Inscriptions convey donations endowed to the temple and are mainly centered on the gift of gold and land for the burning of perpetual lamps.

The temple structure has not undergone any radical change in this phase, and only reflects certain improvements

Plate VI



A. Inner view of the garbhagṛiha (see p. 26)



B. View of the excavated temple complex from the north east (see p. 26)

Plate VII



A. Cloister maṇḍapa on the east (see p. 26)



B. Cloister maṇḍapa on the west (see p. 26)

Plate VIII



A. Cloister maṇḍapa in the south with a stone paved entrance (see p. 27)



B. View of the Vēl (Spear) (see p. 27)

Plate IX



A. Balipīṭha (see p. 27)



B. Ring well on the north west (see p. 27)



Ring well on the north east (see p. 27)

Plate X



A. Lateral entrance on the east with stone steps (see p. 29)



B. View of the broken pillars in the western cloister maṇḍapa (see p. 29)

Plate XI



Southern entrance closed by stone boulders in later phase (see p. 29)

Plate XII



A. Hearth (?) on the south (see p. 29)



B. Unbaked brick platform (see p. 29)

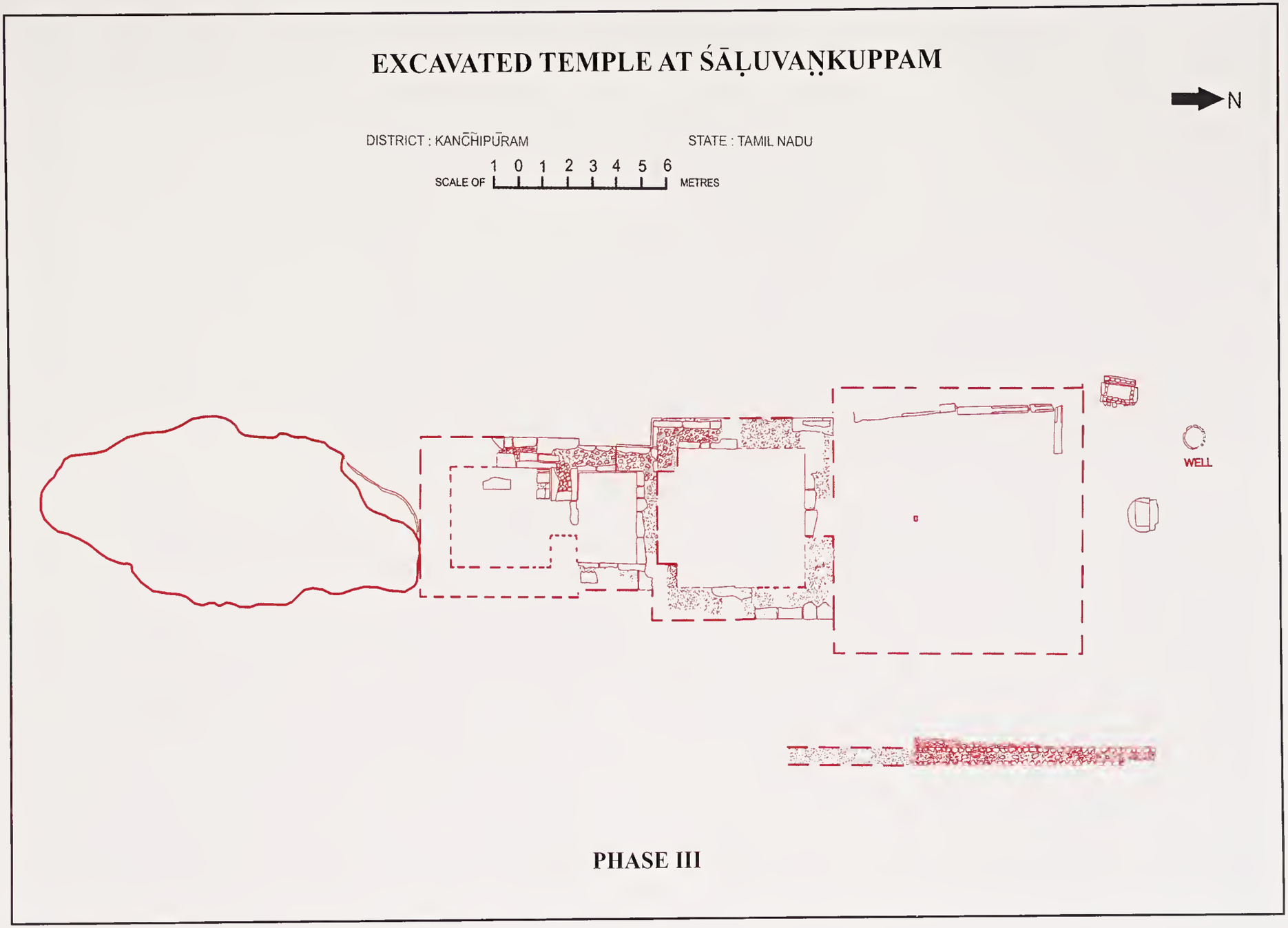


Fig. 10 (see p. 38)

like the provision of stone steps and the extension of the cloister on the northern side. The shallow depressions encountered in the floor of the cloister *maṇḍapa* and the discovery of saddle querns, a rectangular hearth with openings on the four sides, large quantities of shell lime and clay (pl. XIII A), in all probability establish the fact that all activities of the construction of the temple were confined to its precincts. This could be because the temple and its environs might have become important entities due to the growth of the village structure, the temple *per se* and the religious cult. The *garbhagṛiha* continued to be a simple brick structure. The simple structure obviously continued to be popular even when structural temples built of stone were experimented upon. The process of experimentation in different medium of construction was in vogue as stone structural temples vie for place with brick temples in the Toṇḍai-maṇḍalam area itself. The southern entrance of the cloister *maṇḍapa* was closed by filling it up with large boulder stones. The perseverance to adhere to this medium of construction despite the pressure of the sea is a point to be debated.

Phase III:

The last phase of temple activity heralded a new mode of construction and the temple has been converted into a stone one. Dressed stone has been used and is extant on the western side only (pl. XIII B). This level of activity is found at a depth of 1.20 m. The earlier brick structure had been modified both by strengthening the earlier brick activity and also by raising the floor level of the temple.(Fig 10)

The plan of the temple at this stage confirms to the conventional one and consists of a *garbhagṛiha, ardhamaṇḍapa, mahāmaṇḍapa* and *mukhamaṇḍapa*. The floor of the main temple has been paved with stone as seen through the evidence present on the western side. The floor of the *garbhagṛiha* had been plastered with lime after the core area had been raised systematically with clay and finally with sand and shell. The stone wall of the sanctum has a brick core (pl. XIV A).

To accommodate the *mahāmaṇḍapa*, a stone wall running east–west and parallel to the wall of the *garbhagṛiha* was constructed, which abutted the *mukhamaṇḍapa* offsets of phase II A. The intervening void was filled using brick bats, sand, small chunks of lime plaster etc. In this cavity a number of potsherds like bowls, pieces of grooved tiles, brickbats, stucco pieces, a few painted stucco pieces, and a stone inscription of the13th century were found. The entry points of phase II were filled up with earth and brickbats. The inner stone veneer of the extant *mahāmaṇḍapa* on the western side was supported by broken pillars of the earlier phase at the bottom (pl. XIV B). As seen through the extant wall of this stage the construction pattern seems to be that, plain dressed stone members were used as veneering and brickbats and stone chips as filling for the core. The available material like pillars, terracotta tiles, bricks were freely recycled to raise the *mukhamaṇḍapa* of the temple. Terracotta tiles have been rammed on the eastern and western sides. Intact stone pillars arranged in north-south and west orientation have been exposed on the

western and partially on the northern side. A number of pillars were also traced in the central filling and they included the inscribed pillars of the Pallava rulers Kamphavarman, Nṛpatuṅga; and Chōḷa Rājarāja I. The last two were broken and placed on either side of the stone spear. All of them were found uniformly at the same level (1.60m). A Chōḷa coin of Rājarāja also finds place. The spear, given prime importance in Phase II is now seen buried upto the neck portion, though its position and orientation has not been disturbed (pl. XV A).

The *balipīṭha* has also been buried and a semi-circular and rectangular stone slab was laid over it probably signifying the new *balipīṭha*. The total height of the *balipīṭha* is 1.25m. A brick revetment wall of very poor construction is noticed running in north-south orientation for 10m.

It is also to be noted that during this phase the *prākāra* wall and cloister *maṇḍapa* were completely buried. The entrance and approach to the temple was restored to its original direction i.e., north. The lateral entrances of phase II have been blocked by the walls of the *mahāmaṇḍapa* of this phase. Further, north of the *balipīṭha* was seen a ring well having thirteen courses. On the outer sides i.e., on the eastern side, from the accumulated black soil containing dumps, full bricks (40x20x8cms and 40x19x6cms), bricks with lime plaster, broken stucco figurines, broken terracotta pieces and roofing tiles, terracotta *naṇdi*, lamps, and votive Śiva *liṅga* were retrieved.

Even in this phase of development no image has been encountered and no ritualistically aligned development is seen in the *garbhagṛiha*. The noteworthy feature of this temple is the remarkable modifications that were carried out in the *garbhagṛiha*, which conceal all earlier constructional processes.

A number of stray finds like stone architectural members along with inscribed stone pillars were lying scattered amidst the sandy accumulation. Since all the architectural members have not been recovered and also some of the stone members found seem to be unfinished, it can be reasonably concluded that the conversion of the temple into stone may not have been completed. (pl. XV B)

The irony of what type of idol was worshipped is left to imagination as no sculpture has been found. The *garbhagṛiha* also does not betray any religious characteristics. A palm portion of a right hand made of sandstone was found. This broken palm apparently can be symmetrically fixed to a damaged sandstone sculpture found in front of the Atiraṇachaṇḍa cave at Śāḷuvaṇkuppam (pl. XVI A). Whether this could have been the presiding deity is a topic for further research.

Plate XIII



A. Chamber on the north – storehouse for clay and lime (see p. 38)

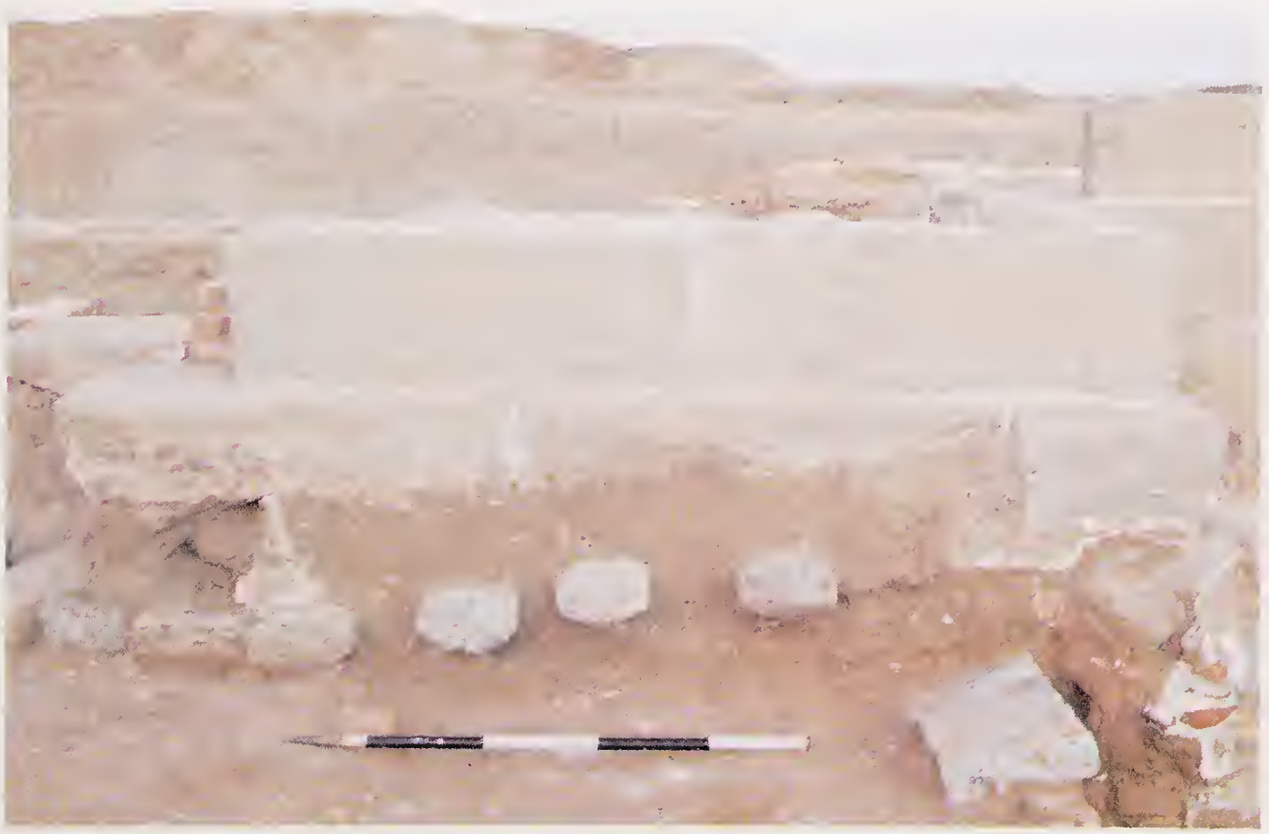


B. A view of the final phase of the temple with stone construction (see p 38)

Plate XIV



A. A stone wall of the garbhagṛiha with brick core (see p 38)



B. Old pillars reused for raising the plinth of the succeeding phase *(see p. 38)*

Plate XV



A. A view of the vēl in the fnal phase projecting only the upper portion (see p 39)



B. Scattered architectural members on the surface (see p 39)

Plate XVI



A. Palm retrieved from the excavation matching the damaged sculpture preserved near the Atiraṇachaṇḍa cave (see p. 39 and 73)



B. A conical Vase (see p 47)

CHAPTER V

CERAMIC ASSEMBLAGE

The excavated pottery from the temple site of Śāḷuvaṇkuppam is fairly large in collection and uniformly distributed all over the site. The pottery is essentially red ware of both slipped and unslipped varieties along with few other shards of celadon and glazed variety.

The pottery from Śāḷuvaṇkuppam, essentially a temple site comprises of lids, plates, dishes, bowls, basins, vases, miniature jars, storage jars, conical jars, dish-on-stand and lamp stand. Such a large quantity of pottery reveals that the temple was a very popular one in which the whole village was actively participating in all activities of the temple. From the inscriptions it is known that annual festivals and the *Kārtigai* festival in particular were being regularly celebrated. The fabric in slipped variety is usually medium to fine in texture with normal degrassiant like sand and grit but with husk or straw occasionally present and well fired in full oxidized conditions.

SLIP

Potteries were made non absorbent by the application of a thick or thin coat of red slip both externally and internally up to the rim. More than 30% of the vessels of the red ware variety have been treated with mostly hematite slips variously of red, brown, pale red, from bright to the light shades. No salt glazing is seen.

COMMON TYPES

The types that generally occur in the red ware are dishes, bowls, lids, lid-cum-dishes, lid-cum-bowls, plates, basins, dish-on-stand, pots and storage jars in a wide range as well as carinated vases of different sizes. Decorations can be seen either on the outer rim portion or on the shoulder. They are mostly incised slashes, oblique strokes, wavy lines, herring bone pattern etc.

The sophisticated level of the ceramic production here is indicated by the use of high necked pitcher (*Kuja*), variety of spouts and few conical jars. Besides these, stemmed stands and lamps are worthy of mention.

Majority of the pottery pieces were collected from the dump on the north-eastern corner of the temple complex. Lamps were mostly found in the south eastern corner at a depth of 1.80m. Conical jars were found at the same level as the lateritic foundation on the western side and on the south-eastern and south-western corner of the cloister *prākāra* at a depth of 2.40m. Both grooved and "L" shaped tiles were found at different levels. Most of them have come from north-eastern side and north-western side of the cloister *prākāra* at a depth of 3.40m and 2.90m respectively. Storage jars and bowls were collected on the southern side of the brick wall at a depth of 2.40m.

Spouts were collected from the dump on the northeastern corner of the temple complex at a depth of 0.94m.

Vase, Dishes and Lids

Most of the dishes, lids and lid-cum-dishes or lid-cum bowl types appear to have been used as cooking utensils as is indicated by the soot stains on the exterior and at rims. Very often they carry a light red slip, which is peeled off or affected by soot stains.

Dishes (Fig. 11)

Generally they have an incurved or externally grooved, thickened rim with flat or sagger base. They have a grooved rim with incised oblique designs on the exterior. Both the interior and exterior have a slip with soot marks.

Dish-on-stand (Fig. 11)

The dish-on-stand is available both in handmade and wheel made variety. In case of wheel made variety both the base and dish are wheel made and are luted together. They are of different sizes and shapes. They have either rounded or flattened base. An almost intact dish-on-stand was recovered from the excavation which shows a splayed outer rim base and the dish has a blunted rim and flanged waist. They have short or big concave waist. Some of them have corrugated interior with grooves on the exterior of the stand. Bottom stands are thicker and the diameter ranges from 10 to 12cms. They are devoid of any slip.

Carinated Vases (Fig. 12)

They have splayed out or both externally and internally grooved rim with sharp or mild carination at the waist. They have internally ridged neck with either flat or sagger base. Decorations can be seen in the form of multiple grooves, incised oblique lines and herring bone pattern executed either on the neck or shoulder.

Plates (Fig. 13)

Plates were found in limited quantity. They show a variety of rims of both plain and decorated types including horizontally pulled out and out-turned rim, centrally grooved rim along with incised decorations like rope at the centre of the rim. Multiple grooves can be seen both on the neck and shoulder.

Lids (Fig. 13)

The lids were used either as lid-cum dish or lid-cum bowl. The fact that they were used as lids is evident from the slip applied on one surface only, the other being intended to remain as the inner portion not exposed. Examples of both medium and coarse fabrics possess no slip but their surfaces are smooth obviously because of the application of a self-slip or something like an ochre wash. They have blunted rim, carinated waist culminating in a rounded base. Some of them have corrugated exterior with flat base.

Basins (Fig. 14)

The basins are mostly devoid of any slip, however in a few cases, slip is visible on the exterior. They show a variety of rims such as beaded, clubbed, internally thickened, cut and centrally grooved rim. Some of the deep basins are decorated in the form of incised oblique lines and are grooved on the neck. The fabric is medium to coarse with finger impressions in some cases.

Bowls (Fig. 15)

Both medium sized and deep bowls were found from the excavation. The fabric is medium and devoid of any slip, but their surfaces have been made smooth apparently by a self-slip or something like an ochre wash. The rims are very simple like featureless, knife-edged, sharpened or in-turned rim with a steep body profile, mildly corrugated both on the interior and exterior. Only one specimen of a convex sided bowl was recovered.

Storage jars (Fig. 16)

The storage jars in particular are applied with a thick coat of slip on the exterior as well as interior up to the rim. They are of medium fabric and the core ranges from medium to thick section. Almost all of them are well fired with a few exceptions having unoxidized smoky core. A variety of rims are seen in storage jars such as internally clubbed rim, externally thickened rim, or flattened sharpened rim, or heavy rolled rims or closing rim. Some of them are without neck or with incipient neck and sloppy shoulders. They show a variety of design repertoire comprising multiple grooves, incised designs like oblique slashes, dots, wavy lines and vertical lines on the shoulder.

Pots (Fig. 17, 18, 19)

The pots range in size from small to big and varying in shapes. The fabric is mostly medium and occasionally coarse. The texture of the clay ranges from medium to coarse without the presence of much sand and grit. Generally the pots do not show any slip and only in a few cases the pots are slipped both externally and internally up to the rim. Some of them show weathered surface. Different types of rim can be seen in this category. They include pots with both internally and externally grooved rim, obliquely cut rim, externally sharpened and grooved rim, externally vertically cut rim and splayed rim and externally thickened collared rim. Some of them have globular body with short concave neck. A large water vessel with a straight featureless rim, internally ridged neck and externally grooved rim with oblique incised slashes on it is worthy of mention. Some of the pots have their inner surface coated with lime. Decorations are available in the form of multiple grooves on the neck and shoulder. Besides this, incised oblique lines with dots at the end and herring bone pattern are important.

Conical jars (Fig. 20)

Conical jars have tapering sides with an average thickness of the side walls being 1.5cm. They have either pointed or flat bottoms with a diameter of 3cm and solid base. In some of the cases the base is very heavy and solid to a height of 6cm. They are hand-made, of well levigated clay and well burnt. Mostly, both the interior and exterior has smoothened surface except a few, which shows a grooved interior. The colour of the core ranges from red to smoky black (pl XVI B).

SPECIAL POTTERY

Chinese celadon ware

The bottom portion of a pale green bowl with ring footed base and well polished sides was found. The diameter at the base is 9cm and the thickness is 0.8cm.

GLAZED WARE

Two shards of glazed ware were recovered from the excavation - one glazed green on the exterior with a thickness of 2.3cm with the core rough and well fired. The other smaller shard is glazed green on coarse red ware of thickness 0.7cm.

High necked Pitcher *(kuja)* (Fig. 20)

A broken upper portion of a sprinkler of polished red ware, measuring 9cm in height with a diameter of 7cm at the top and a thickness of 0.8cm was found. It has a spout of 2.5cm diameter at the top. Another was a fragment of a sprinkler with the nozzle portion alone intact.

Potshard with mat impression

A fragment of pottery, white in colour was recovered. The colour has been lost due to weathering. It is roughly rectangular in shape and measures 6 × 4cm. Its outer surface has mat impressions.

Spouted vessels

Several decorated terracotta spouted knobs were recovered at Śāḷuvaṇkuppam excavations from different levels. Most of the spouted vessels are found in coarse red ware and red slipped ware variety. Spouted vessels occur in two forms.

In the first category the plain spouts with grooved exterior would be made along with the pot itself. In the case of the second category, decorated spouts were separately made and fixed at the shoulder portion of the vessels. This type of spout is more abundantly found than the previous one.

Following varieties of spouted knobs are noticed in the second type of spouted vessels-

1. Plain tubular channel with beaded mouth and a thin perforation at the centre.
2. Crudely modelled tubular spout with a small perforation slightly raised collar in the edge.
3. Exquisitely modelled tubular spout with collared mouth decorated with grooved design.

Red ware is the dominant pottery at this site. The shards have not been found in a stratified context as already explained due to the sandy soil of the site. The other significant factor is that the level of the temple site has been raised continuously to accommodate alterations and in doing so earth was dumped along with brickbats and pottery. The pottery so found in a peculiar context is basically an indicator of the pottery of the times all mixed up and not exhibiting any chronological aspects. However, of consequence is the discovery of conical jars which is also found at Vasavasamudram and Kāñchīpuram and the sprinklers as found at Kavēripaṭṭinam.

Dishes & Dish-on-stand

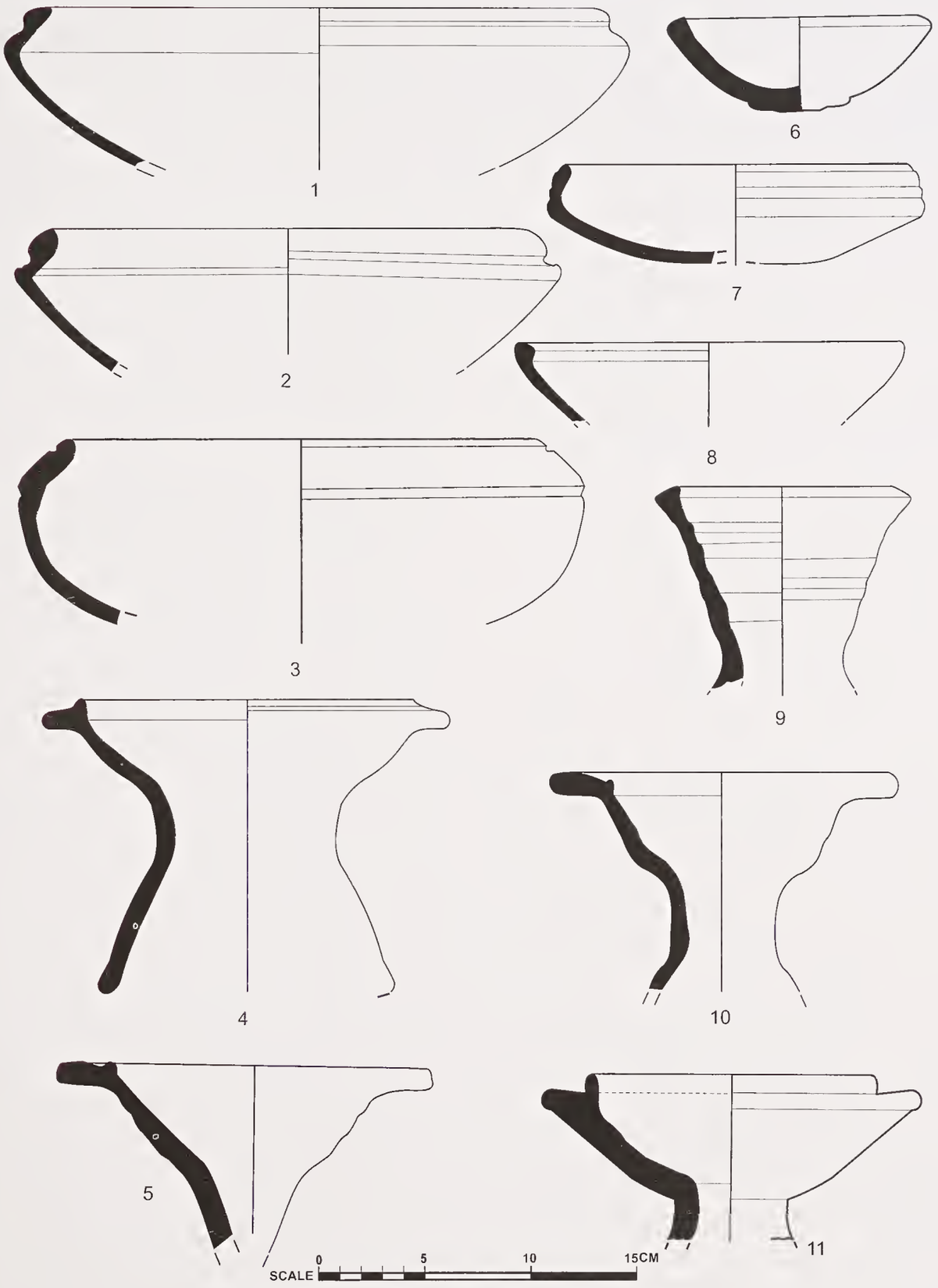


Fig.11 Dishes & Dish-on-stand (see p. 46)

Fig. 11 Dishes & Dish-on-stand

1. A dish with internally projecting wide beaked rim, bluntly carinated with contiguous, rounded base.
2. Fragment of a dish with an incurved and externally thickened rim with a sharp shoulder and contiguous, rounded base.
3. A dish with an incurved featureless rim with prominent bulbous profile. A groove just below the rim and a prominent groove cut at the waist.
4. A dish-on-stand with thick rounded base and short waist. The dish with a very short blunt rim prominently projecting a slightly curved flange.
5. Upper portion of a dish-on-stand with a flattened and very short blunt rim base. It has a corrugated exterior and thin waist.
6. A shallow bowl with an externally cut rim and a small flat base.
7. Dish with an inturned rim and corrugated exterior with rounded base.
8. Fragment of a small dish with internally obliquely cut rim.
9. Portion of a dish-on-stand with slightly obliquely cut base with corrugations both on interior and exterior. It has a short waist or stem.
10. Bottom portion of a dish on stand with flat base and luted at the waist.
11. Upper portion of a dish-on stand of red ware with vertical, blunted rim and broad collared waist and short luted neck.

Fig. 12 - Carinated vases

1. A carinated vase with a flaring externally cordoned rim and externally ledged neck with corrugated exterior. It has blunt carination with a sagger base.
2. Fragment of a carinated cooking vessel with a grooved rim. It has multiple series of grooves on the exterior.
3. Fragment of a vase with an internally chamfered and externally square cut rim with multiple grooves on the neck and shoulder. It has a bluntly carinated waist.
4. A carinated vase with deep outwardly tapering body, a horizontally thickened and pulled out grooved rim with multiple grooves on the exterior.
5. Fragment of a vase with an internally thickened, grooved and externally cordoned rim. It has prominent multiple grooves on the shoulder.
6. Fragment of a cooking vessel. It has a prominent flanged rim, which afforded an easy grip. A series of ridges at the shoulder just above the point of carination with incised decoration in between.
7. Carinated vessel with thick internally grooved rim and ridges on the exterior. A sharp carination at the waist with a shallow sagger base.
8. A vase with thickened grooved rim, internally ledged neck, bluntly carinated to shallow sagger base.
9. Fragment of a cooking vessel with a prominent grooved rim. A series of grooves on the exterior with a flat base.
10. Fragment of a carinated vase with an out-turned internally grooved rim and internally ridged neck. It has multiple grooves on the neck with bluntly carinated waist.

Carinated Vases

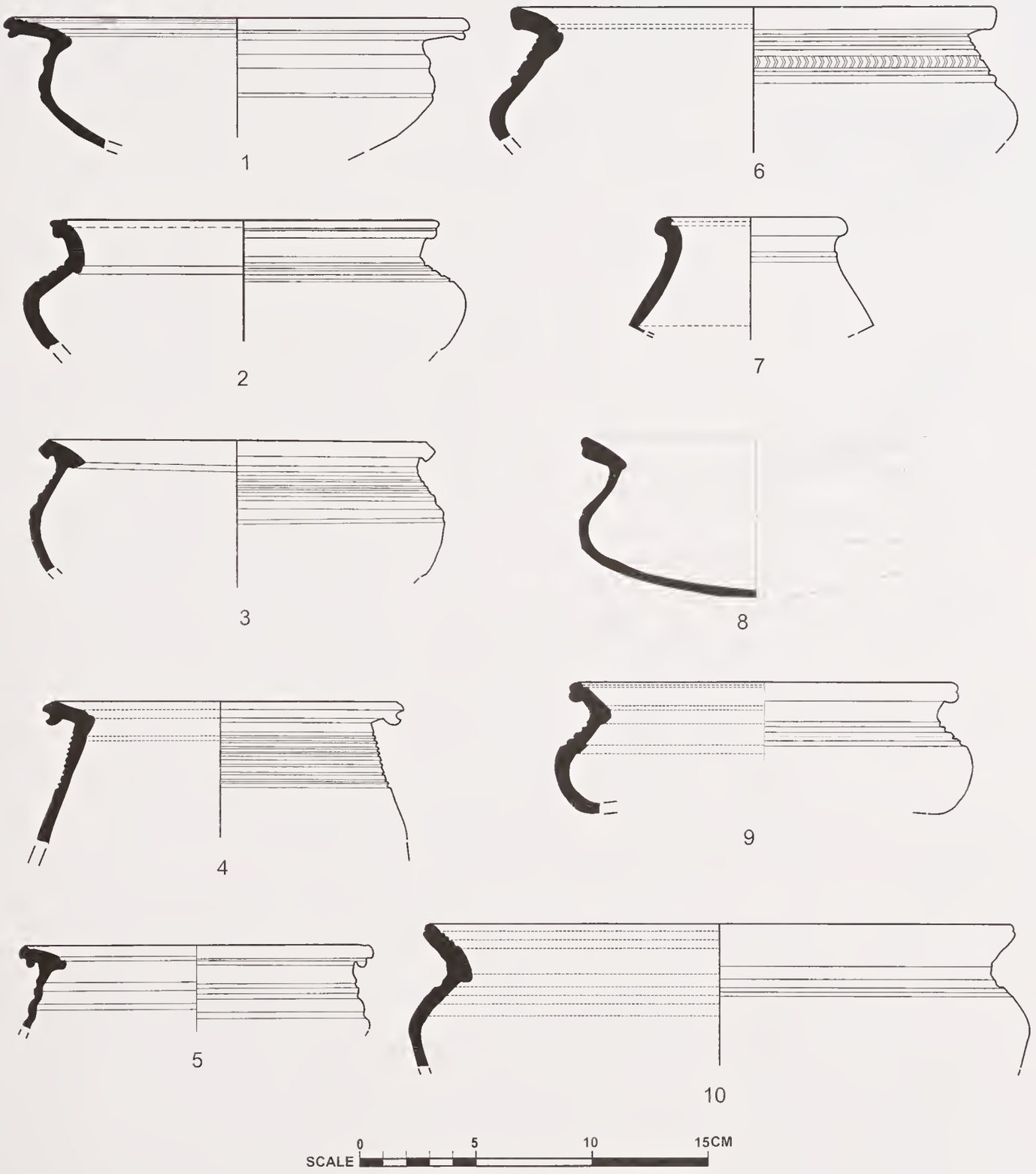


Fig.12 Carinated Vases (see p. 46)

Plates & Lids



Fig.13 Plates & Lids (see p. 46)

Fig.13 Plates & Lids

1. Fragment of a plate with flanged externally curved and thickened rim, internally ledged neck. It has incised strokes on the rim with multiple grooves right from the neck up to the shoulder, and shallow sagger base.
2. A plate with flaring and an out-curved rim grooved exterior and a shallow base.
3. Fragment of a plate with an inturned rounded edge and ridged neck.
4. A red ware lid with a flaring rim and a tapering profile with a thick base and correspondingly thick interior.
5. Plate with splayed out externally thickened and grooved rim, ledged shoulder bluntly carinated to a shallow base. Interior has concentric circle decorations and in between are rope and bead like incised decorations.
6. A lid with horizontally thick ended and a short blunted rim with tapering profle and a thick base.
7. A plate with horizontally out turned rim and grooved neck with rounded base.
8. A sturdy red ware lid with slightly everted prominent rim and fanged grooved waist.
9. Fragment of a small lid with a rounded base and prominent internally ridged and grooved interior.
10. Fragment of a plate with horizontally pulled out and out-turned rim. It has a groove on the shoulder with probably a thick base.

Fig. 14 Basins

1. Fragment of a basin with a centrally grooved and internally thickened rim.
2. Fragment of a basin with a beaded rim and internally grooved.
3. Fragment of a basin with an internally thickened rim.
4. Basin with incurved and beaded rim, corrugated exterior and somewhat deep profle.
5. Fragment of a deep basin of red ware with internally rounded rim and externally grooved just below the rim.
6. Fragment of a deep basin with a rolled rim. It has a groove on the exterior.
7. Basin of red ware with an externally thickened rim and grooved body.
8. Fragment of a basin with an inturned, externally thickened rim, grooved neck with smooth rounded shoulder.
9. Fragment of deep basin with an inturned featureless rim and a ridged neck with slashes just below it. A series of thin grooves just below the neck.

Basin

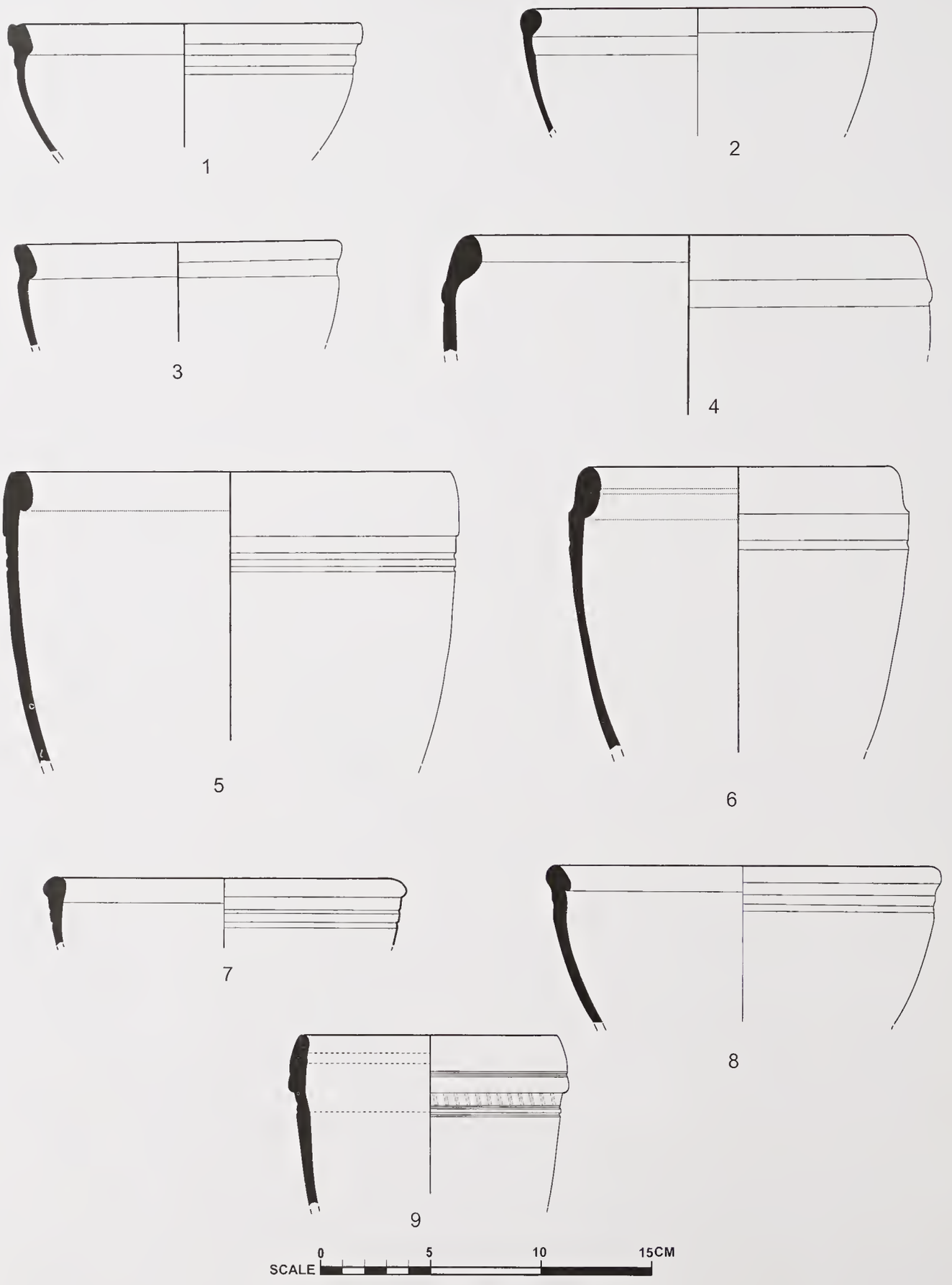


Fig.14 Basin **(see p. 46)**

Fig. 15 Bowls & Lid-cum-bowl

1. Fragment of a thick carinated lid cum bowl with a short, incurved and externally thickened rim, with insignifcantly fanged waist, and externally grooved neck with a sagger base.
2. A lid-cum dish with a very short blunt rim prominently projecting curved fange with corrugated exterior. It has a thick base.
3. A red ware lid with a short projecting rim and thick tapering sides.
4. Bowl of dull red ware with inturned obliquely cut rim or with closing rim, corrugated both internally and externally and a thick base.
5. A complete bowl with sharp edged rim and tapering profile. It has corrugated interior and footed base.
6. Fragment of a deep bowl with thickened top rim grooved internally and with convex sides weakly ribbed at the exterior neck region.
7. A carinated deep bowl with a thick featureless rim and multiple grooves just below the rim. It has a slanting ridged waist with smooth tapering base.
8. Fragment of a red ware bowl with an inturned featureless rim. It has a convex profile.
9. Fragment of a deep bowl with a beaded or rounded rim and an almost vertical side.

Fig. 16- Storage jars

1. A large storage vessel with blunt and thick straight rim and heavy bulging body.
2. Fragment of a storage jar with a grooved rim, hardly any neck. It has prominent multiple grooves on the shoulder.
3. Fragment of a wide mouthed storage jar with a flattened top and sharpened rim. A series of prominent grooves start just below the rim and continue up to the shoulder with hardly any neck.
4. Fragment of a storage jar with a clubbed rim and a grooved shoulder.
5. A decorated fragment of a red ware storage jar with a rounded rim, internally ledged neck, multi - grooved shoulder having vertical wavy lines between incised strokes.
6. Fragment of a storage jar with a splayed out externally thickened and internally grooved rim. It has a slightly widening shoulder with decorations, which include a series of grooves, incised dots and wavy lines.
7. Fragment of a wide-mouthed storage jar with an inturned internally thickened rim and multiple grooves both on the interior and exterior.
8. Fragment of a jar with clubbed rim and hardly any neck. It is of medium fabric, treated with red slip outside. It has two grooves on the shoulder.
9. A large storage vessel with blunt and thick straight rim and heavy bulging body.
10. Fragment of a wide-mouthed jar with a rolled or beaded rim. It has a single ridge on the shoulder with widening body.
11. Fragment of a wide-mouthed storage jar with an internally clubbed rim and hardly any neck. It has multiple grooves both on the exterior and interior.
12. Fragment of a wide-mouthed pot with an externally grooved rim. It has internally ridged short neck with multiple grooves on the shoulder.

Bowls & Lid-cum Bowls

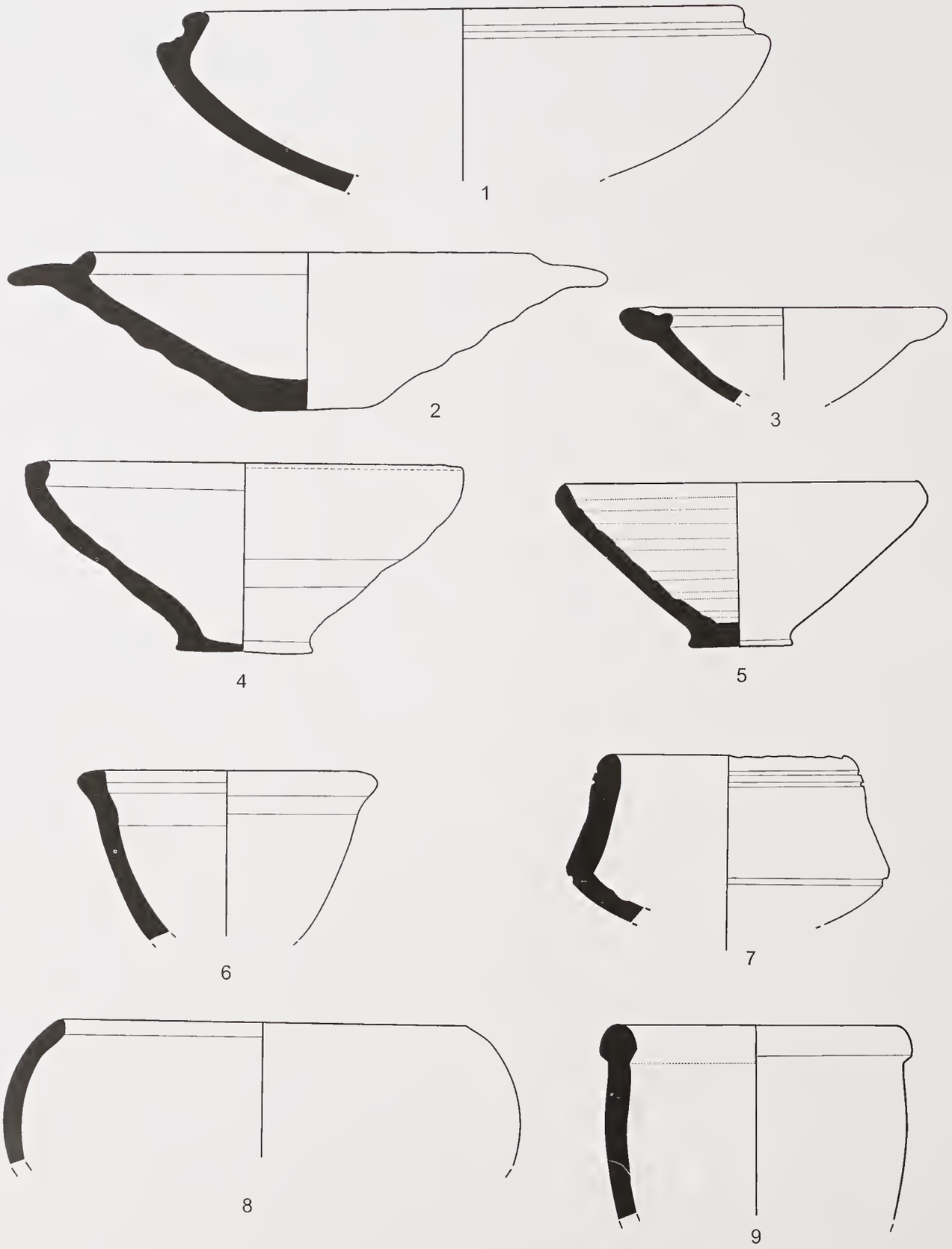


Fig.15 Bowls & Lid-cum Bowls (see p. 47)

Storage Jars

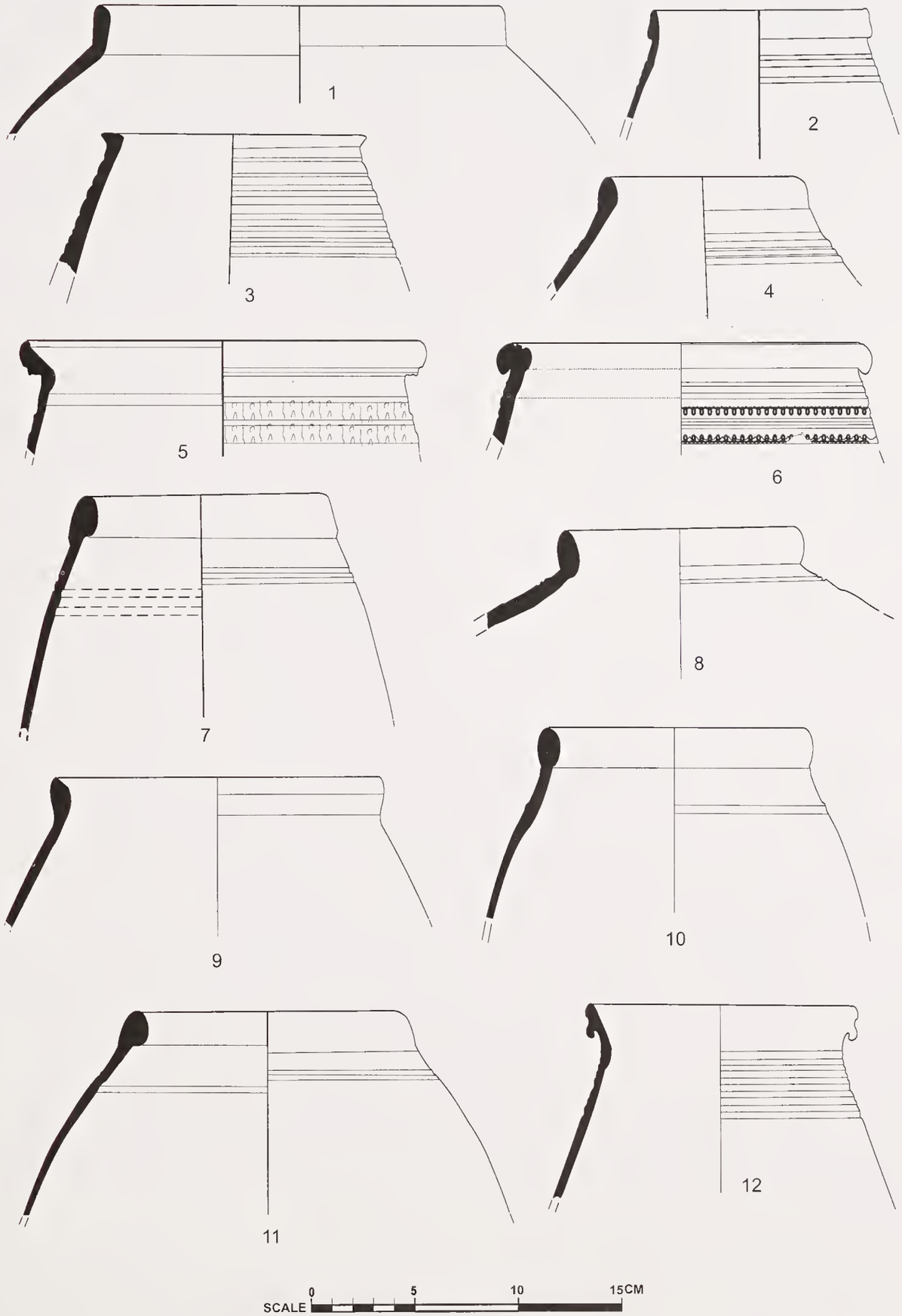


Fig.16 Storage Jars (see p. 47)

Pots - 1

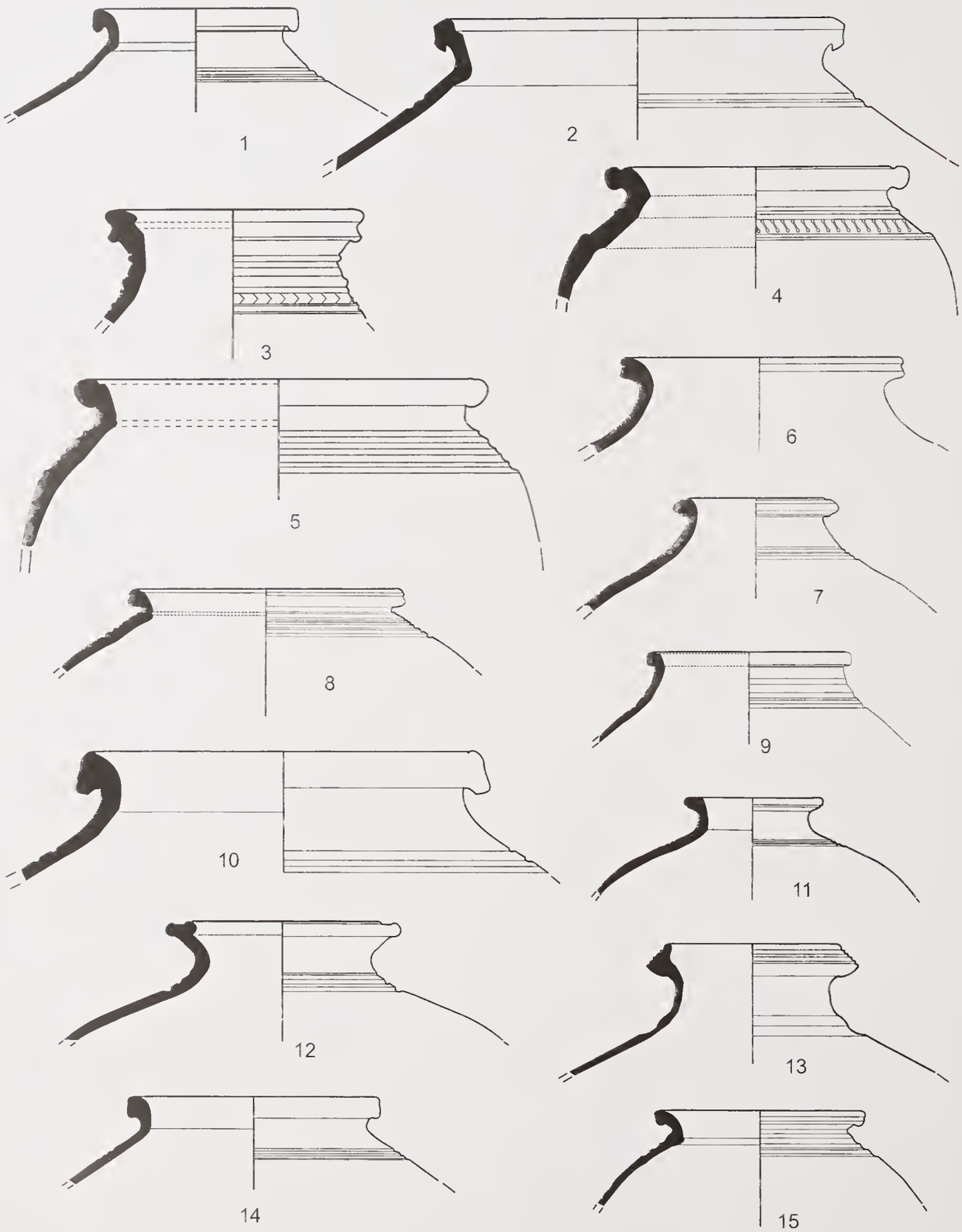


Fig.17 Pots - 1 (see p. 61)

Pots - 2



Fig 18 Pots - 2 (see p. 61)

Pots - 3

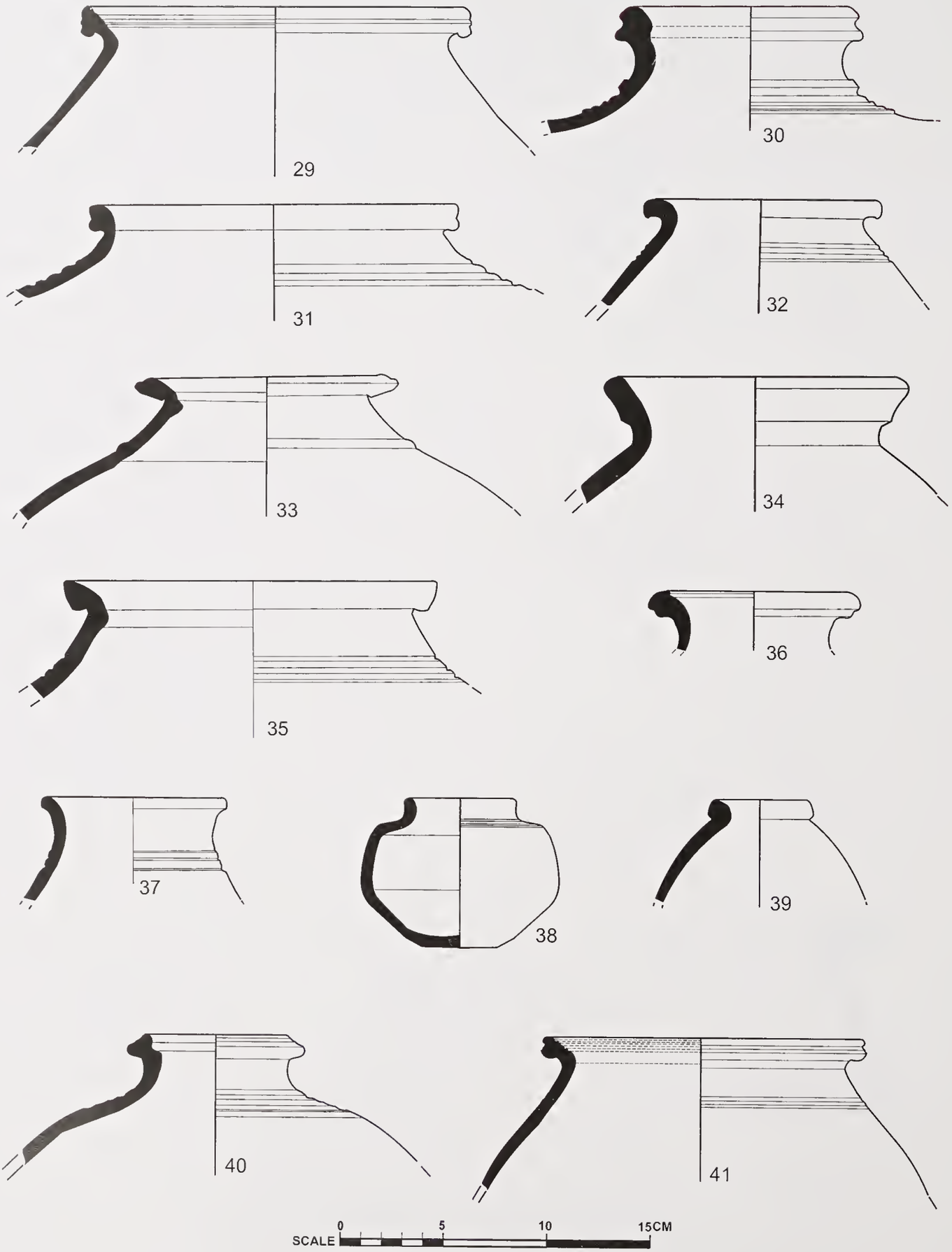


Fig.19 Pots - 3 (see p. 61)

Fig. 17, 18, 19 Pots

1. Fragment of a water vessel with an out-turned vertically cut and undercut rim. It has a short internally ridged neck with multiple grooves on the shoulder.
2. Fragment of a pot internally grooved with a vertical collared sharp undercut rim in the interior.
3. Fragment of a pot with a vertical externally thickened and internally undercut rim. The exterior portion is decorated with a series of grooves and herring bone decorations.
4. Fragment of a pot with faring, grooved and rounded rim. It has a prominently ridged shoulder with vertically slanting design in-between.
5. Fragment of a pot with an externally thickened and internally grooved rim having external grooves on the shoulder.
6. Fragment of a pot with an out-turned externally grooved rim. It has concave neck with a widening body.
7. Fragment of a pot with a centrally grooved and externally thickened rim. It has multiple grooves on the shoulder with widening body.
8. Fragment of a wide mouthed pot with a grooved rim. It has series of prominent grooves on the shoulder with widening body.
9. Fragment of a water vessel with a centrally grooved and externally flanked rim, ridged neck and shoulder.
10. Fragment of a water vessel with an externally chamfered undercut rim and short neck. It has multiple grooves on the neck with a widening body.
11. Fragment of a short necked globular pot with flattened top rim with grooves on the exterior shoulder.
12. Fragment of a larger water vessel with a slight projected rim, short neck and globular body. A series of grooves on the shoulder.
13. Fragment of a water vessel with a splayed out, thickened and multi grooved rim. It has prominent ledged shoulder with widening body.
14. Fragment of a water vessel or pot with an out-turned fattened top and vertically cut rim. It has a small concave neck with multiple grooved shoulder and widening body.
15. Fragment of a globular pot with an out-turned externally grooved rim. It has internally ledged neck with ribbed neck and shoulder.
16. Fragment of wide-mouthed storage jar with an out-turned grooved rim and internally pointed neck. It has multiple grooves both on the interior rim and exterior shoulder with a widening body.
17. Fragment of a pot with an out-turned obliquely cut rim with a short neck and multiple grooves on the shoulder.
18. Fragment of a pot with flaring featureless rim and internally ridged neck. It has a prominently ledged shoulder developing into an elliptical profile.
19. Fragment of a pot with splayed out rim, internally ridged neck with multiple grooves on the interior.
20. Fragment of a pot with an externally thickened and internally ledged neck.
21. Fragment of a water vessel with an out-turned rounded rim, short neck and multiple grooves on the shoulder with widening body.

22. Fragment of a wide-mouthed pot with a grooved rim. It has a series of prominent grooves on the shoulder with a widening body.
23. Fragment of a narrow-necked vessel of dull red ware with an out-turned beaded rim. It has a ridged shoulder.
24. Fragment of a water vessel with a centrally grooved and externally vertical cut rim ridged neck and shoulder.
25. Fragment of a pot with an internally thickened and externally collared rim.
26. Fragment of a pot with an externally thickened projecting rim and internally grooved rim.
27. A carinated vase with deep outwardly tapering body, a horizontally thickened and pulled out grooved rim with multiple grooves on the exterior.
28. Fragment of a small pot with fanged rim and a short neck. It has multiple grooves on the neck.
29. Fragment of a wide-mouthed pot with vertical externally thickened grooved rim a little undercut. It has a short neck with widening body.
30. Fragment of a pot with an out-curved externally thickened and internally collared rim and ribbed below the short concave neck.
31. Fragment of a storage jar with an externally grooved rim and a series of prominent grooves on the shoulder.
32. Fragment of a medium-sized pot with splayed out rim. It has a short neck with multiple grooves on the shoulder.
33. Fragment of a pot with fanged, grooved rim with internally ridged neck. It has a single ridge on the shoulder with globular body.
34. Medium-sized round bodied pot of red ware with a flanked collared rim. It has a short neck with a wide shoulder.
35. Fragment of a water pot with an out-turned obliquely cut rim. It has multiple grooves on the shoulder with deep widening body.
36. Fragment of a pot with an externally thickened and internally grooved rim with a short concave neck.
37. Fragment of a small pot with an out-turned featureless rim. It has two prominent grooves on the shoulder.
38. Miniature pot of dull red ware with almost straight rim, spherical sides and a thick string cut base.
39. Fragment of a small pot with an everted thickened rim, hardly any neck with globular body.
40. Fragment of a globular water pot with an internally grooved and externally thickened rim. It has a short neck with multiple grooves on the shoulder.
41. Fragment of a wide-mouthed jar with an out-turned rim, internally and externally grooved. It has multiple grooves on the shoulder with widening body.

Fig. 20 High necked vessels, sprinklers and Conical jars

1. Fragment of a mouth of a high-necked jar or water pitcher with a closed rim and flanged exterior with straight sides having a short neck.
2. Upper part of a high-necked kuja or water pitcher with a sharpened rim and flanged exterior with grooved shoulder. Of fine fabric, it is treated externally with bright red slip.

High necked vessels, Sprinklers & Conical jars



Fig.20 High necked vessels, sprinklers & conical jars
(see p. 62)

3. Upper part (mouth) of a pot with closing featureless rim, tapering sides and grooved stem or waist.
4. Tapering end of conical jar with pointed base. It is of thick coarse fabric.
5. Lower part of a tall conical vessel tapering down to a point at the base. It has a thick, coarse fabric with grey core.
6. Body portion of tall conical vessel or jar of red ware with tapering sides. It has three grooves on the interior. It is of thick and coarse fabric. It is badly potted with coarse interior but smooth exterior. It is devoid of slip and is burnt black in sections.
7. Fragment of a lower part of a tall conical vessel or jar tapering down with a thick fat base.

CHAPTER VI

OTHER FINDS

The antiquities of this site primarily confirm the existence of a temple complex. The excavation yielded a total number of seventy three antiquities from different levels and comprise forty four terracotta objects, three stone objects, four bronze objects, fourteen copper objects (including thirteen coins), one iron object, six stucco objects, four shell objects and five glass beads and paste and other miscellaneous items

Some of the articles found have a close association with the temple, like the stucco, terracotta objects and tiles, which most probably embellished the temple structure in its various stages of existence. The medium of construction preferred in this temple was brick and stucco. The stone objects found in association with a large quantity of lime and the occurrence of shallow receptacles seen in the cloister *maṇḍapa* indicate that structural activities pertaining to the temple were mostly confined to the precincts of the temple.

The religious leanings of the temple and the ritualistic activities involved can be perceived from the rich repertoire of votive terracotta objects, terracotta lamps, a human palm made of sandstone and bronze pieces.

Apart from all these the occurrence of copper coins assist in the dating of the temple complex.

As already stated, though, most of the antiquities were not found in proper stratified contexts owing to the nature of the site; yet when studied in association with the structural phases, they are reflective of the chronological sequence of the site.

The site being unique in the excavated annals of Tamil Nadu, most of the antiquities except the damaged ones have been described in detail.

A) TERRACOTTA OBJECTS (Fig 21)

Introduction

The terracotta figurines from Śāḷuvaṇkuppam can be classified into four categories. They are human and animal figurines, cult objects and miscellaneous items. Terracotta animal figurines are dominant. A number of elephant legs, trunks and heads mostly fragmentary in nature were found. The retrieved elephant faces are shown with trunks either in normal, dropped or curved position. A terracotta bull (Naṇdi) also finds place.

Human figurines include head of a male figure, pedestal with legs and plaque with dancing women. The cult figurines are those of Gaṇēśa and Naṇdi. Besides these a number of architecturally related items like finials, decorated pieces, tiles; ritualistic objects like lamps, lamp stand, and miscellaneous objects like hop scotches, and a spindle whorl were also found.

In terms of technique, majority of the figurines are hand modelled with few moulded ones. Further, while the bigger ones

are hollow, the smaller ones are solid. The clay used in the case of the bigger ones is usually medium to coarse-grained and mixed with sand particles whereas in the case of the smaller ones it is usually medium-grained, but sometimes even fine and well levigated.

The firing, by and large, is good, though there are a few pieces, which had not been well fired. For obvious reasons, the core of the bigger specimens remained smoky, even though the exterior is well-fired. Some of them show either flat or concave surfaces on one side with finger impressions.

In a few cases a thin red slip was applied and often only a wash of the same material was provided. This would further go to show that they were intended for the ordinary citizens mostly.

Human figurines

Human head (Fig. 25.1)

A solid head of a male figurine in round was recovered to the west of the main shrine. It is externally well fired with smoky black core. The eyes and eyebrows indicated by deep grooves are broad. The pupils are indicated by holes and the nostrils are prominent. The ears are elongated, shaped like a fan and the chin is broad and slightly bulging. The mouth is shown open with a thick lower lip. The back portion of the head is roughly finished and without any workmanship.

Terracotta plaque (Fig. 21.2)

The most outstanding find from the site is a solid terracotta plaque probably depicting the folk dance referred to in the Sañgam literature as *Kuravai Koothu* (pl. XVII A). It is made out of well levigated clay and rectangular in shape. The terracotta plaque contains a row of five women dancing by holding their hands at the shoulder level. The long hands of the women standing at the extreme ends are shown hanging down. Though the feet are not prominent, it is noticed that the feet are shown turned to the left and right side respectively except the extreme right figure, whose feet are shown turned to the left. The facial features are prominent. A head band is clearly seen. The mouth is slightly open. The neck portion is adorned with a thick plain *hāra*. All the features are modelled with appliqué decoration. The figures are shown with prominent breasts and wearing a short lower garment. This type of dance is commonly associated with Subrahmaṇya and is reflected in Sañgam literature like *Puṛaṇāṅūṛu, Śilapadikāram* and *Tirumurukāṛṛuppadai*.

Pedestal with broken legs (Fig. 21.4)

It is rectangular in shape of height 8cm, of which the pedestal alone is 3cm. The pedestal has a broad base. The top of the pedestal has a pinched decoration in the front and sides. The back portion is plain. The toes are prominently shown by incised lines and the insteps have a circular punched design at the centre. Punched design is also seen on either side. It is difficult to assess whether it is a religious symbol or a sculptor's mark. Anklets are shown in the form of incised dots in two rows. The figure was observably handmade as can be seen from the finger impressions at the bottom. The length and width of the pedestal at the bottom measured 10cm and 8cm respectively.

Animal figurines:

Bull figurine (Fig. 21.8)

A broken portion of a standing bull figurine, with a prominent hump and the right leg intact was found. The neck portion is decorated with a beautiful *hāra* of bells. It shows that the bull figurine must have been a sacred motif. The hump and the leg are artistically made.

Bull Head (Fig. 21.5)

Head portion of a bull, with eyes shown prominently by projected eyeballs and eyelids. Small holes pierced in the face depict the nostrils and the mouth is shown by a horizontal slit. The ears are partly broken. It is made of coarse clay and the exterior surface is worn out. As the neck of the bull is unusually elongated it must have been made separately and then luted with the main body.

Elephant figurine

A hollow broken elephant figurine (pl. XVII B) includes the head, trunk, tusk, ears and the right leg. The physical parts are prominently shown. The forehead is broad and quite prominent. It has an internally curved broken trunk along with the tusk. The outer surface is worn out due to weathering. It is made of coarse clay mixed with sand.

Elephant trunk (Fig. 21.6)

A handmade lower part of the trunk of an elephant, measures 9cm in length and has a diameter of 3cm. The mouth portion is shown prominently. It is solid and made of well-levigated clay and is well burnt.

Elephant trunk

Broken trunk of an elephant with flat frontal part measuring 7cm in length. The lower portion of the trunk is more natural. It is coated with red slip.

Fragment of an elephant leg

This could be a part of a broken leg of an elephant. It is cylindrical, with a hole at the centre.

Elephant trunk

Fragmentary piece of a broken trunk of an elephant with its mouth kept open and with rough surface treatment. It is a solid one and made of well-levigated clay.

Elephant leg

A cylindrically shaped and solid broken part of an elephant leg. It must have been the hind leg of the elephant as seen through the heavy thigh portion at the top. The top of this leg portion is plain, smooth and well finished. It is made of well-levigated clay.

Elephant leg

A featureless, solid broken portion of a leg of an elephant. A fluted broken leg with a circular foot portion. The surface of the latter is very rough with pitted marks.

Fragment of an elephant figurine

Fragmentary head portion of an elephant figurine with a typically large and intact ear. The edges of the ear are beautifully curved with a hole at the centre. It has smoothened surface, well fired and devoid of any slip.

Cult figurines

Gaṇēśa (Fig. 21.3)

A solitary moulded stylised figure of Gaṇēśa in the round is an interesting find. He is shown seated on a pedestal and wears *Karaṇḍa makuṭa*. He has four arms holding probably the *pāsa, mōdaka* and *aṅkuśa*.The trunk gently turns to the left. The right hand rests on the knee and holds *mōdaka*. The

belly is prominently shown. The forehead bears a round auspicious mark. The back of this figurine show various details such as *yajñōpavīta* across the torso, *udarabandha*, other ornaments, etc. There is a mortise hole at the underside of the base extending to a depth of 2cm. This is meant for placing the deity over a pedestal or upright on a wooden stand. It is devoid of any slip. The elaborate ornamentation of the figure shows that it may belong to the medieval period (pl. XVIII A).

Naṇdi (Fig. 21.1)

One of the most important findings from the site was a hollow Nandi found on the eastern side of the extended brick wall on the north-eastern corner of the temple complex.

The Naṇdi figurine is depicted as sitting elegantly on a high three-tiered pedestal. The Naṇdi is resting on its forelegs bent backward and the hind legs bent forward. The hooves are clearly seen. The well proportioned Nandi has the head facing front. The details of the Nandi such as horn, sharp ears and big eyes are clearly shown. The mouth is shown with a horizontal slit opening, and the nostril is shown with pierced holes. It has a semi-circled prominent hump and tail. The neck is adorned with a *hāra* marked by cut lines, with a conch as a *pendant*. The back has been deliberately flattened to accommodate a semicircular hump and a small pinched *liṅga* with a circular *pīṭha* behind it. The bottom of the pedestal is hollow whereas the main body is solid, indicating that this might have been mounted on some structure (pl. XVIII B).

Finials (Fig. 22)

The temple at *Śāḷuvaṇkuppam* obviously had a superstructure of brick and terracotta as can be seen from the terracotta architectural members like finials and decorated shards, moulded bricks, tiles and stucco objects.

The excavation at *Śāḷuvaṇkuppam* yielded five finials, of which three are almost intact. Among them two have red slip on the exterior while the rest have lost their slip, may be due to weathering. Besides these, a few fragmentary finials of coarse fabric were also collected from the site. The finials represented here bear resemblance to those recovered from other early historical sites like *Sālihundam, Kolhāpur*, etc. These finials might have been once a part of the super structure of the temple.

Terracotta finial (Fig. 22.1&3)

These finials have circular flattened base and tapering neck, decorated with four successively diminishing sharpened flanges and ribs. The inner hollow portion of the finial is filled with fine lime mortar. They have a circular perforation at the top. The top of the finials are broken and it has a diameter of 6cm. The diameter of the base of the finial is 10.5cm.

Terracotta finial (Fig. 22.3)

The terracotta finial is of medium fabric and wheel-made. The upper part is bulbous with a bud-like projection at the top and two small perforations on the sides while the bottom portion has tapering sides with a light thumb impression.

Stand for lamp

A stand probably used for holding the lamp is a unique find from the site. It is made of well-levigated clay with red slip on the exterior. It is roughly conical in shape and the top is broken. The bottom is circular with thick out-turned blunt rim and the diameter is 16cm. It has a small perforation at the top with prominent multiple grooves

at the bottom. It is pertinent to mention that donation of lamp stand is also mentioned in the inscriptions. (pl. XIX A)

Spindle whorl (Fig. 28.1)

A highly decorated rectangular shaped spindle whorl with incised decoration on the top and the bottom surfaces is another unique object of this site. One side is decorated with eleven oblique lines at regular intervals with five dots in between, while the other side is decorated with three segmented floral designs having four petals in each, enclosed within two marginal ends with dots. The sides are concave and polished. It could have been used for weaving nets used by fishermen.

Ear ornament (Fig. 28.5)

The ear ornament is black in color, having concave sides with circular top and bottom. It has a central hole with a diameter of 4mm. This ear ornament is a little worn out at the bottom portion.

Decorated shard

A broken terracotta piece with semi-circular incised decorations having fish scale motifs on the exterior surface could have been part of the body of a mythical animal motif that decorates the exterior portion of the temple. It bears red slip on the upper surface.

Miscellaneous–Terracotta Objects (Fig. 23)

A section of terracotta objects with prominent mouldings were probably part of the brick superstructure of the temple.

Terracotta lamps (Fig. 24)

The category of pottery-cum antiquity type is represented by lamps of both decorated and plain variety. A total number of seventy four terracotta lamps were recovered from the excavation. Of this 30 lamps were found intact, while the rest were broken. It is interesting to note that a set of 26 lamps was found *in situ* near the lateral entrance steps on the western side to the main temple placed vertically one above the other. The numerous references to festivals and donation of lamps at the temple underscore the rich collection of terracotta lamps (pl. XIX B).

In plain variety, there are two types-

Type A: This is a common variety found in large numbers. The fabric is fine to medium grained and well-fired. It has disc or flat base, flaring outward with either obliquely cut or featureless rim. The inner diameter ranges between 6 and 10cm.

Type B: In this variety the base is either flat or irregular with a thick bottom. The rim is incurved and featureless. The inner diameter is about 8cm. In general the above variety has complete circular rim without groove for wick.

Wick lamp

A broken portion of a terracotta wick lamp with a flat base is the only one of its kind found here. It is made out of fine clay with beaded rim and three sharp and pinched lips of which two are intact while the third one is half broken. Two rows of incised strokes are visible in between the lips. It has black stains or oil incrustations on the interior. Similar type of lamp has been reported from Kavēripaṭṭinam and it is dated between 4th and 5th Century CE.

Hop scotches

A total number of seven hopscotches fashioned out of potshards were recovered. Two are intact while the rest are broken. They are of various sizes. They are of red ware having smooth and round edges. Though hopscotches are generally used for games, the hopscotches recovered in this temple site might have been used as measuring weights. Similiar hopscotch specimens were reported at the temple site of *Peddavegi* and *Salihundam* in Andhra Pradesh.

a) A small hopscotch fashioned out of a potshard of coarse red ware. It measured 1.50cm diameter with a thickness of 0.30cm.

b) A well finished hopscotch, fashioned out of a shard of fine red slipped ware. It has well finished and smoothened edges with lime patches.

B) STUCCO OBJECTS (Fig.25)

The excavation at *Śāḷuvaṇkuppam* has yielded a total number of six stucco objects (pl. XX A). These were probably used to embellish the superstructure. They are fragmentary in nature and belong to human, animal figurines and mythical animals. The human figurines consist of torso, head, and hand portions. Lime with fine to medium sand granules were used in the preparation of stucco figurines. Though the stucco objects are fragmentary they show high degree of craftsmanship especially in the animal figurines. The human heads are round with prominent eyes and nose, prominent cheeks with receding or flat head. The hands are round and heavy. Some of them were probably used as appliqué decorations for the brick structures, as some of the specimens show rough finish with brick pieces at the underside.

Torso of a male figurine (Fig. 25.6)

This solid bust is the standing figure of a young male figurine, preserved from the neck up to the abdomen portion with two broken arms. The right arm is raised upwards while the left is hanging down. The chest portion, neck, and stomach are carefully delineated. The back portion of the figurine is partly damaged. The ornamental on work of this figurine is lost due to weathering.

Male head (Fig. 25.4)

A solid defaced head probably of a male figurine with an elongated face and the right part of the mouth alone visible. The back of the figure is flat and clearly shows that it was fixed on to a flat surface. It is made of lime mixed with coarse sand.

Female head (Fig. 25.2)

A solid head probably of a female figurine, with an oval face and a long sharp nose was found. The eyes are shown by holes but it is not clear and the left ear is broken. The chin is slightly bulging and the lips are thick. The head dress is also attempted in the form of a protruding hair do. The back of the figure is a little shallow and without any details. The length and breadth of the head is 10 and 11cm respectively with a thickness of 7cm.

***Vyāla* head (Fig. 25.5)**

This mythical animal, partly lion and partly crocodile is exquisitely carved with details including prominent eyes with protruding eyeballs and the typical nostrils of a lion. The back portion of the stucco figurine is plain and unfinished and indicates that the figurine must have decorated the outer walls of the temples. It measures 7 × 6cm. This *vyāla* also appears to be the earliest example

so far reported and might be a precedent to the stone *vyālas* of the later period.

Mythical head (Fig. 25.7)

Head portion (*kīrtimukha*) probably of a lion or mythical animal. It has prominent eyeball, well-modelled ear and prominent nose. The figure has a wide open mouth with sharp teeth, and measures 10.50 × 10.50cm with a thickness of 6.50cm. The facial features are naturally shown and well proportioned. The back side of the figurine is plain and indicates that it was probably fixed on the brick wall.

Half of the fore arm (Fig. 25.3)

A fragment of a solid hand portion of a human figurine, measuring 10 × 5cm. The hand is round with broken fingers. Two thick bangles are visible. It has traces of ochre colour painting. It is made out of fine clay mixed with fine lime and the exterior surface is worn out.

C) COPPER OBJECTS

Copper bangle

Fragment of a copper bangle. It is plano-convex in section and decorated with a grooved design on the top (Pl. XXB).

Coins

A total of thirteen copper coins were recovered from the site and of significance is the discovery of two Roman coins in the stratified context. The two Roman coins are of the late period. One can be identified as that of Arcadius with the reverse depicting two standing figures and a legend. The find spot just outside the *garbhagṛiha* and closer to the upper levels of the laterite can probably help in dating the earliest structure to c. 5th Century CE.

The most significant find is a Pallava coin with bull and with the inscription in Pallava Grantha which reads as "*lakshita*". "Lakshita" on the obverse is a title of Pallava *Mahēndravarmaṇ* I (600 – 630 CE). The reverse of this coin has a central depression, which could be due to corrosion. The other Pallava coins (4 coins) with bull on the obverse and a not-so-clear reverse appear to be of the late Pallava period. The Chōḷa coins (2) are of Rājarāja I. The remaining 4 coins are badly corroded and could not be identified.

Roman coins have been reported from the *Māmallapuram* area as surface finds and they also belong to the late Roman period. This reiterates the view that the port was flourishing in the early medieval period.

D) BRONZE OBJECTS (Fig. 26)

Four bronze objects viz. a bell, lamp, tripod stand and spoon were found. The objects seem to be ritualistic in nature (pl. XX B).

1) Bell (Fig. 26.4)

A small conical shaped bell with ring, probably used as a neck ornament for animals was found. It has a slightly everted beaded rim and the clapper is missing. It measures 3.50cm in height with a diameter of 2.50cm at the base.

2) Bronze lamp (Fig. 26.3)

A bronze lamp was recovered at a depth of 3.45m. The diameter is 11.5cm and the height is 2cm. It is circular in outline with a flat bottom. The lamp is well designed with a projected single wick holder. The die struck lamp has a beading in the outer edges to enable it to be hung as a hanging lamp.

The most important feature of the lamp is the presence of a cock at the opposite side of the wick in the same orientation. There are two big and two small hook-like projections in the top of the lamp. The cock is intricately carved with its circular tail. The beak and feathers of the cock are shown prominently.

3) Tripod stand (Fig. 26.2)

A tripod stand was found at a depth of 2.60m on the western side of the cloister *maṇḍapa*. It has three curved legs luted through a prominent triangular projection to the circular ring at the top. The legs have been provided with a circular base. Each leg is flanked by a small projection on either side. It is about 3.6cm in height with a width of 6cm. The technique of manufacture was probably in a nascent stage and this is the only bronze object of the site, which has not been made in the mould technique.

4) Bronze spoon (Fig. 26.1)

This bronze spoon was found 3metres to the north-east of the stone spear at a depth of 2.90m. It is roughly circular in outline, has a flattened handle, broken at the end. There are six horizontal lines in two groups of three each in the middle part of the handle. Its length is about 20cm.

E) IRON OBJECTS

Only a single iron object was retrieved from the excavation. It is a fragment probably of a knife and highly rusted. It measures 22x4.5cm (pl. XXI A).

F) STONE OBJECTS

The minor objects of stone are a miscellaneous collection comprising types such as saddle querns, pestles, polished stones, palm portion and a votive *liṅga*.

1) Saddle querns and pestles

They were found on the eastern side of the *prākāra* wall at a depth of 2.50m (pl. XXI B).

Querns

The excavation has yielded three sets of saddle querns. Two of the saddle querns were made of rectangular granite (dressed stone) and one was made of dressed sandstone. The upper surface was either flat or had turned concave on account of constant use. They have rectangular outline and trapezoid cross section. The sides slope inwards ending in a small square or rectangular base.

Pestles

The pestles are circular in section with proper handle. As far as their use is concerned it may be presumed that they were used for grinding lime as is evident from the lime paste and other receptacles found *in situ* in the cloister *maṇḍapa*.

1) The saddle quern is made out of black granite, rectangular in shape with proper base. Its top measures 40 × 21cm with a height of 10cm. The smooth flat surface has turned slightly concave as though it was regularly used as the pestle stone and measures a length of 32cm with a diameter of 12cm. The pestle stone is fresh without any damage.

2) The small black granite saddle quern is rectangular in shape and the top measures 25 × 17cm and has a height of 7cm. The upper surface is worn out due to constant use. The pestle stone associated with this quern measures a length of 25cm with a diameter of 8.50cm. Unlike the previous one this has a rough surface and is also damaged at the handle portions.

3) The saddle quern is made of sandstone and rectangular in shape. The top of the saddle quern measures 42 × 22cm and has a height of 13cm. The upper surface is smooth probably due to constant use. The pestle stone has a length of 32cm with a diameter of 14cm (pl. XXI B).

Votive *Śiva liṅga* (Fig. 28.2)

A small monolithic votive Śiva *liṅga* was recovered in the north eastern corner of the temple complex at a depth of 1m. It is made of green stone and well polished. The Śiva *liṅga* consists of a square *pīṭha* with a cylindrical *liṅga*. It is 4cm in height. The *pīṭha* measures about 3cm on each side at its base with a height of 2cm. The cylindrical *liṅga* has height of 2cm and diameter of 1.70cm. It has incised lines on the surface of the *pīṭha*.

Palm portion

One of the most important findings from the excavation is the broken palm portion of a human hand made out of sandstone. The hand (i.e. right) is in *abhaya mudrā*, in typical Pallava style. Though the tips of the fingers are broken, the palm and the wrist portion are proportionately made. It has rounded fingers with shallow depression in the palm. It is important to note that this palm may be a portion of the broken sand stone torso image of a deity kept in front of the Atiraṇachaṇḍa *maṇḍapa* (cave temple) which is 200m away from the excavated site. The circumference of the wrist of the palm is 8cm and it matches with the wrist size of the deity figure. This might have been the deity worshipped in the temple. (pl XVI A)

G) BEADS

The excavations yielded a total number of six beads. They can be classified into three groups according to the materials used for making them namely, (a) Paste (b) Glass and (c) Terracotta. Four of them are paste beads and the remaining is one each of glass and terracotta (pl. XXIIA).

Glass bead

A standard circular shaped bead, green in colour.

Paste beads

Four beads were collected from the north eastern corner of the temple.

a) A medium-sized bead, circular in shape, dark green in colour.

b) Slightly bigger than the above one, circular in shape, green in colour, slightly worn out.

c) Short, tubular in shape, light green in colour, edges are not perfect.

d) Circular, dark green in colour found at a depth of 0.40m

Terracotta bead (Fig. 28.4)

A complete bead having a bulbous, round bottom and tapering at the top was the only terracotta bead found from the site. It has a perforation in the middle. A similar type of bead has been reported from *Kavēripaṭṭinam–Pallavaneśvaram* site.

H) SHELL OBJECTS

The shell objects from Śāluvaṇkuppam may be divided into two groups viz.

(i) Objects of cut shells. (ii) Uncut shells.

1) Shell bangle

Fragment of a bangle, plain, thin, flat, rectangular in section. It has a length of 3cm

with a thickness of 0.3cm. The diameter of the bangle measures 2.50cm (pl. XX A).

2) Uncut shells

Uncut shells from Śāluvaṇkuppam are mainly conch shells. A group of three medium sized conch shells were recovered from the excavation. The outer surface of the conches is worn out due to weathering. They might have formed a part of the temple paraphernalia.

I) MISCELLANEOUS OBJECTS

Antler (Fig. 28.3)

The excavation has yielded a piece of antler. It measures 16.50cm in height with a maximum diameter of 2cm. The top end of the antler is very sharp and smooth and the lower is broken and rough.

Tiles (Fig. 27)

Considerable number of tiles were found from the excavations. They have been found on three sides –between the main temple and cloister *maṇḍapa* on the east, west and north. Tiles have been rammed to raise the floor of the *mukhamaṇḍapa* of Phase III. This testifies to the fact that probably in the initial phase, the temple had a tiled superstructure (pl. XXII B).

They can be classified into two main categories. viz, Grooved tiles, and Plain 'L' shaped tiles.

a) Grooved roofing tiles

They are light red in colour, thick and in general rectangular in shape. They have longitudinal deep grooves on the upper surface with a circular perforation at the end. The grooves vary from 2 to 5 in number. In some there is a single groove in length so as to provide grip for the overlapping tile. These appear to have been used in the construction of inclined roofs. These tiles are slightly convex so that the rain water can pass through the grooves rapidly without giving chance for the water to stagnate. The maximum length of the extant tiles varied between 10 and 20cm whereas the breadth ranged between 5 and 10cm. The tiles were made of gritty clay mixed with husk as tempering material.

b) 'L' shaped tiles

A few L-shaped rectangular tiles have been collected from the site along with the grooved tiles only on the eastern side of the main temple. They are thinner than the grooved tiles, and are orange in colour. These well burnt tiles are found in different sizes and are made of fine clay. They are having perfect edges and triangular end. These tiles might have been fixed to the reapers running on the rafters. The thickness of the tiles varies from 0.5cm to 1cm.

Terracotta Objects



Fig.21 Terracotta Objects

1. Nandi 2. Plaque 3.Ganesa 4. Pedestal with broken legs
5. Bull head 6. Elephant trunk 7. Torso of a male figurine 8. Bull figurine

Terracota Finials

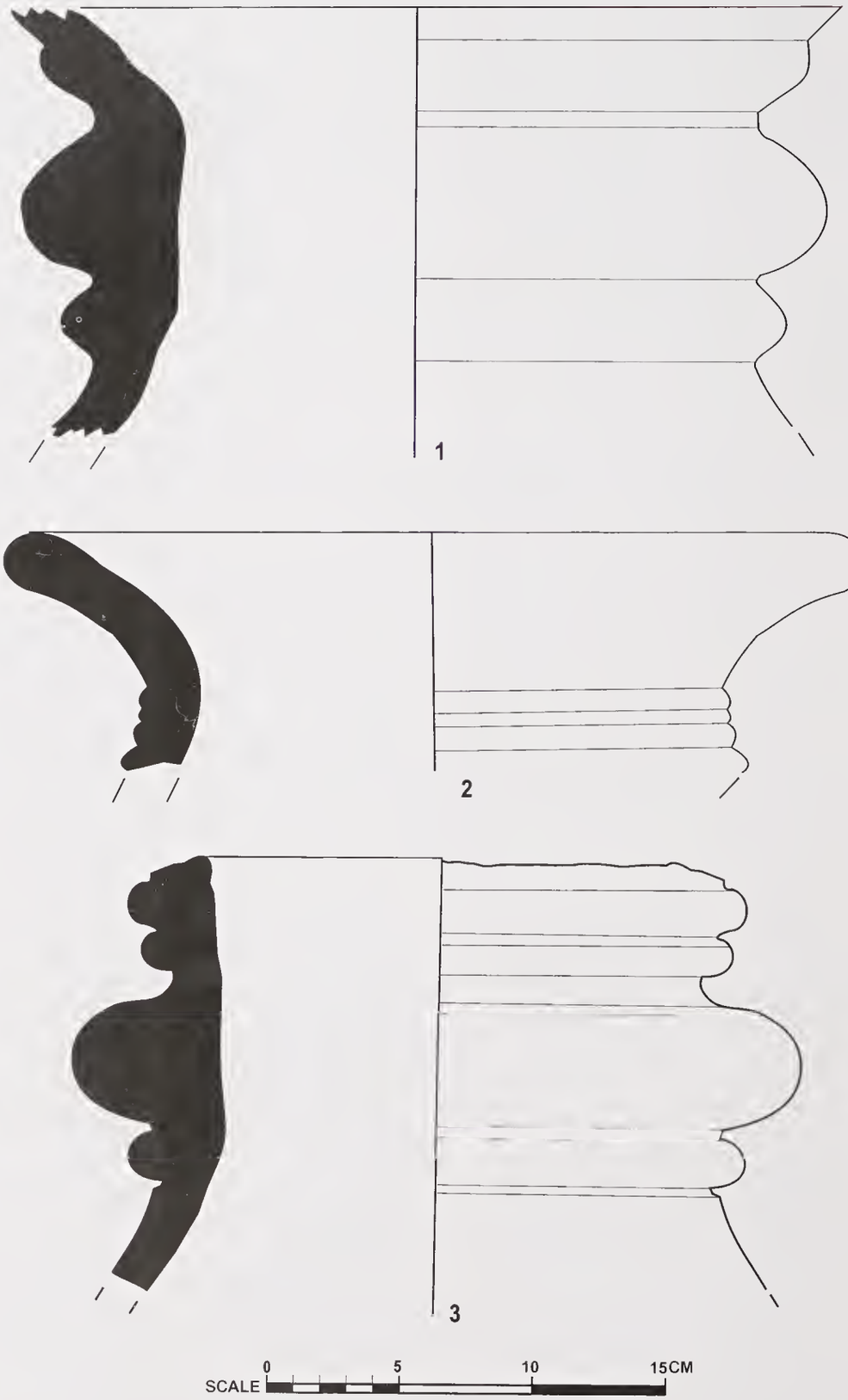


Fig.22 Terracotta Finials

1&3 - Part of finial-decorated with four successively diminishing sharpened flanges and ribs
2 - Part of finial-upper part is bulbous and with two small perforations on the sides.

Miscellaneous - Terracotta Objects

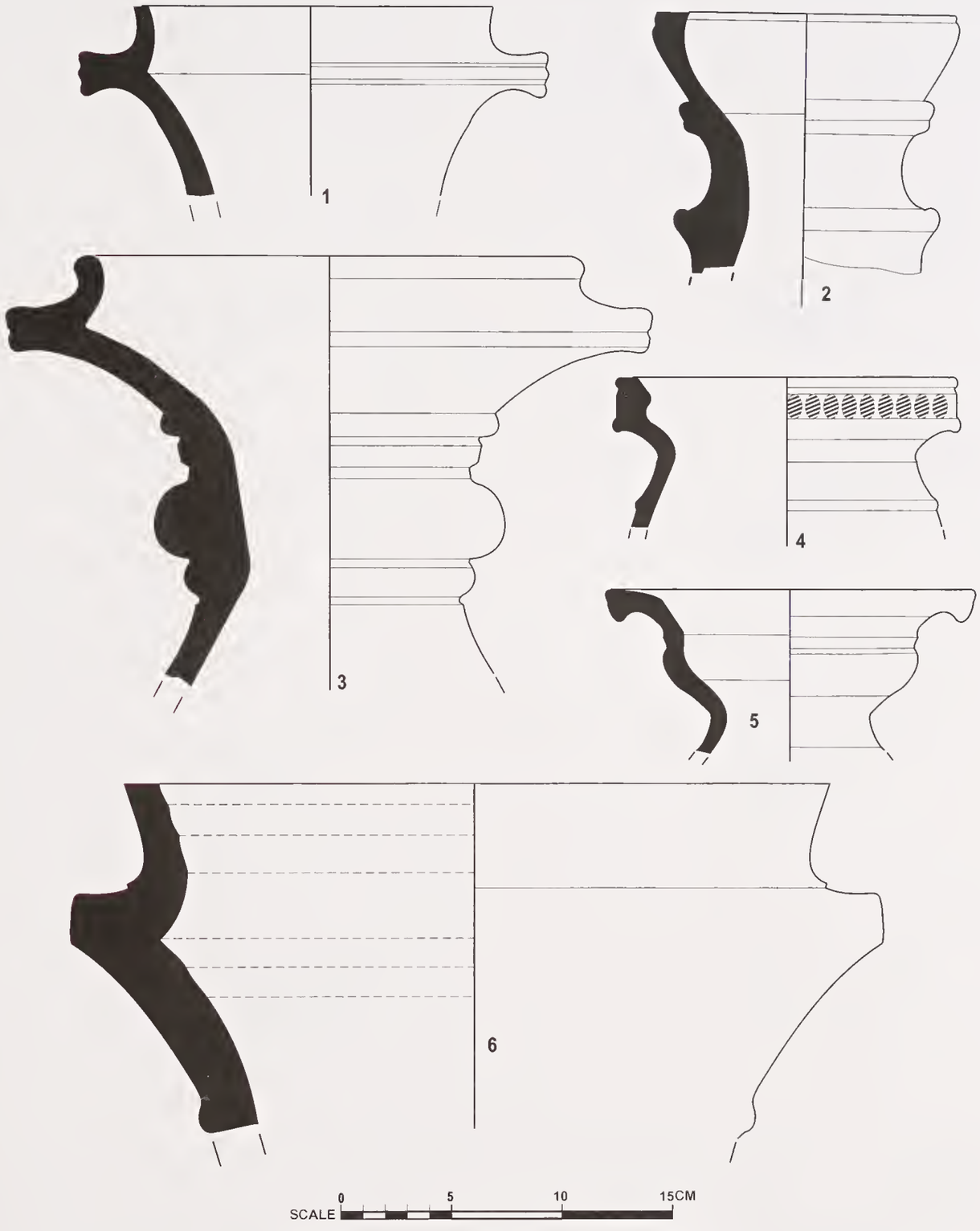


Fig.23 Miscellaneous - Terracotta Objects

1. Fragment with projecting slighty curved top, grooved waist.
2. Lower part of a thick conical object.
3. Fragment portion with a straight blunted edge, having a prominent flange.
4. Fragment with grooved and incised rim like portion with a short neck and a single ridge on the shoulder.
5. Base fragment of a hollow portion with thickened edge.
6. Fragment portion with horizontal flattened top rim and thick collared tapering sides.

Terracotta Lamps

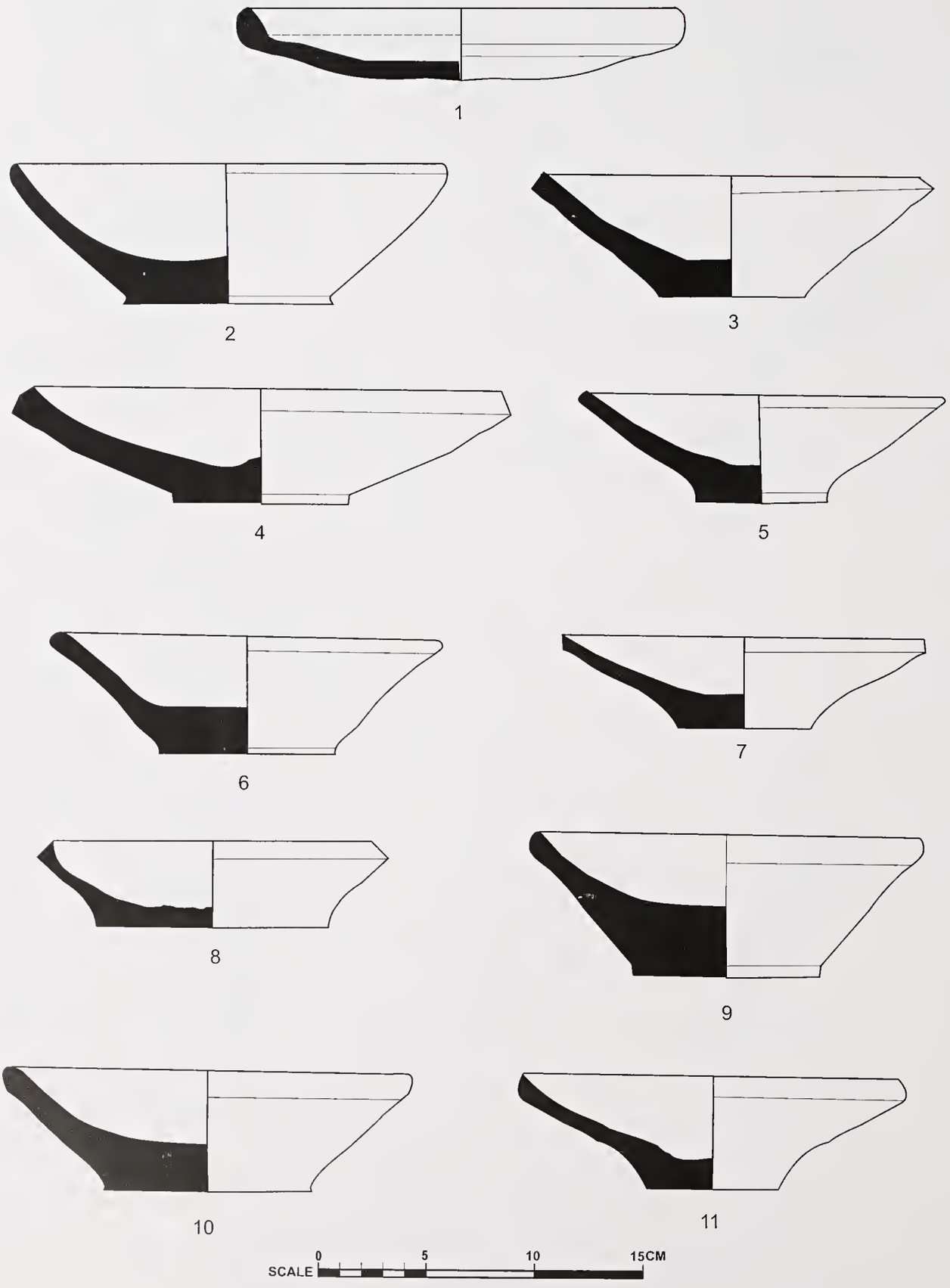


Fig.24 Terracotta Lamps

Fig. 24 Terracotta Lamps:

1. A shallow lamp with thick featureless rim and wide thickish base.
2. A lamp with a sharpened rim and thick disc base.
3. A shallow lamp of dull red ware with an obliquely cut rim and thickish base.
4. A shallow lamp with a thick obliquely cut rim and is shallower and wider than the above. It has a disc base.
5. A shallow lamp of dull red ware with a chamfered rim and thick footed base.
6. A shallow lamp of dull red ware with a featureless rim and disc base.
7. A shallow lamp with a sharpened and externally cut rim with thickish base.
8. A shallow lamp of dull red ware has a sharpened and externally cut rim. It has a wider thickish base.
9. A lamp of dull red ware with featureless rim and thick disc base.
10. A shallow lamp with featureless rim and thick base.
11. A shallow lamp of dull red ware with a sharpened vertical rim and a thickish base.

Terracotta & Stucco Figures



Fig.25 Terracotta & Stucco Figures

1. Terracotta human head
2. Stucco female head
3. Stucco broken part of an arm
4. Stucco male head
5. Stucco vyala head
6. Stucco torso of a male figurine
7. Stucco mythical head

Bronze Objects

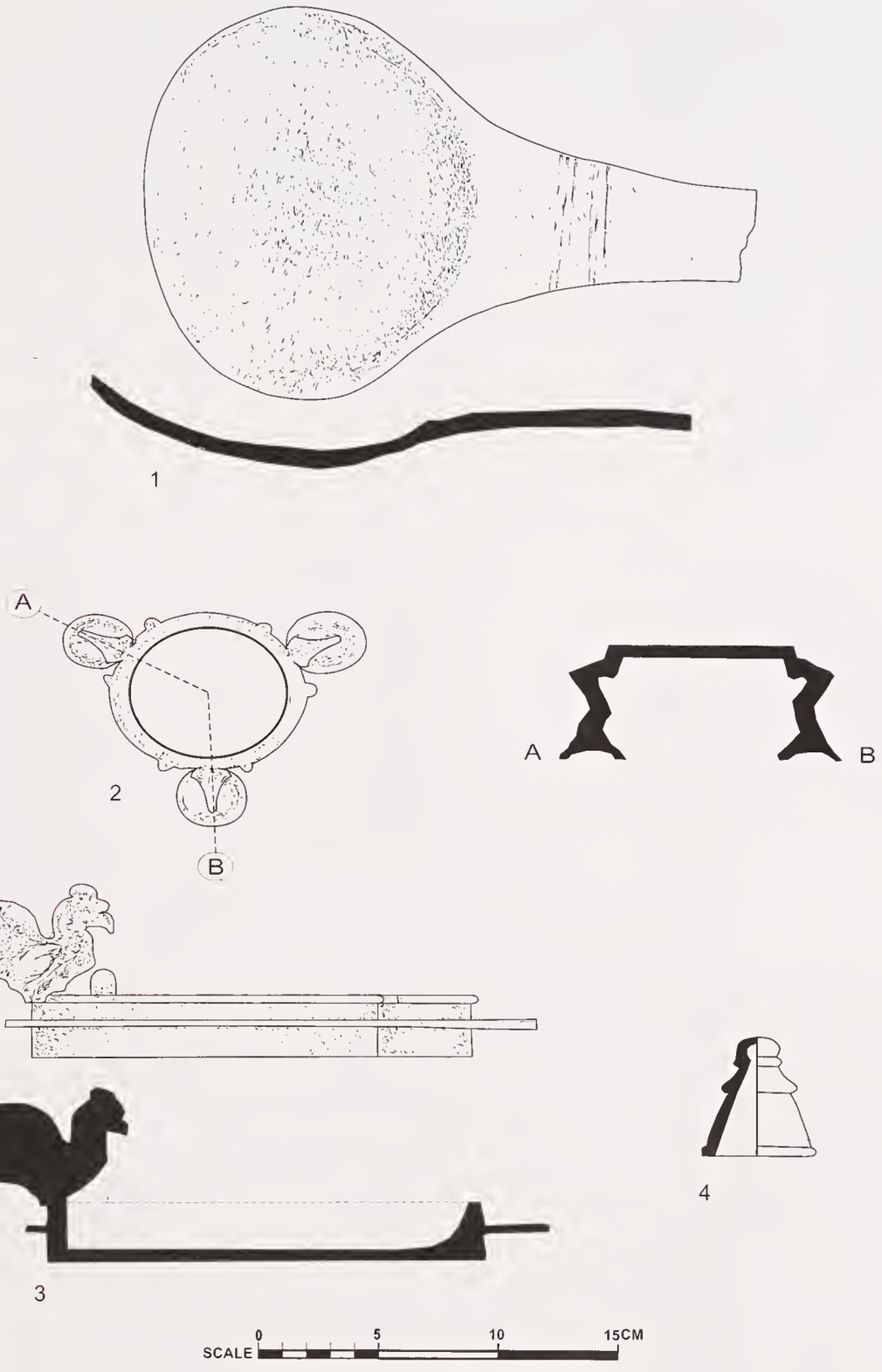


Fig.26 Bronze Objects

1. Spoon 2. Tripod Stand 3. Bronze Lamp 4. Bell

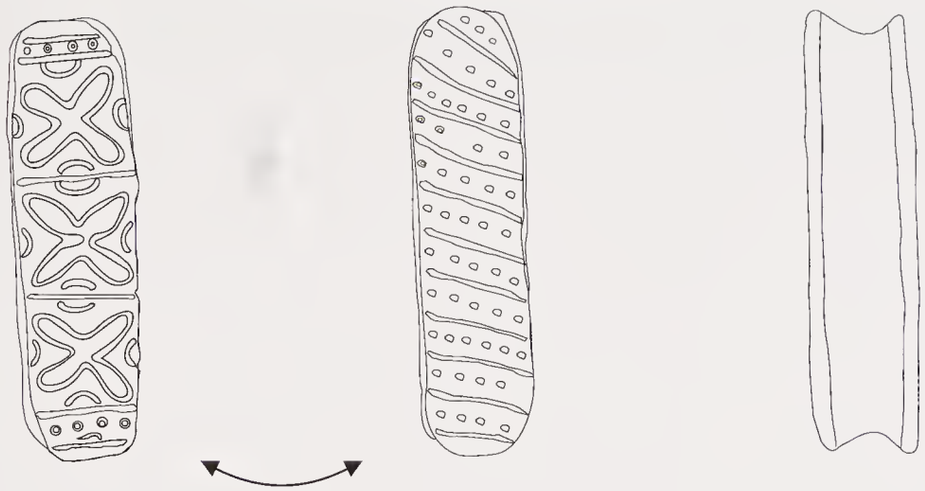
Roofng Tiles



Fig.27 Roofing Tiles

1 to 4 - Grooved tiles
5.to 7 - Flat 'L' shaped tiles

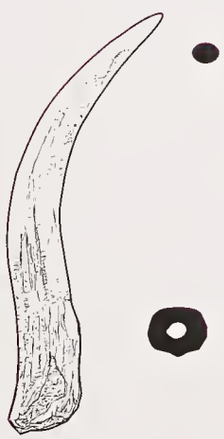
Miscellaneous Objects



1



2



3



4



5

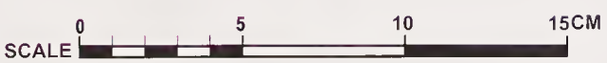


Fig.28 Miscellaneous Objects

1. Spindle whorl - Front incised floral design; Back-dots between oblique lines
2. Votive Siva linga 3. Antler 4. Terracotta bead 5. Terracotta Ear stud

Plate XVII



A. Terracotta plaque (see p. 66)



B. Terracotta elephant head (see p. 67)

Plate XVIII



A. Terracotta Gaṇēśa (see p. 68)



B. Terracotta Nandi (see p. 68)

Plate XIX



A. Terracotta lamp stand (?) (see p. 68)



B. Terracotta lamps (see p. 69)

Plate XX



A. Stucco objects (see p. 70)



B. Bronze & copper objects (see p. 71-72)

Plate XXI

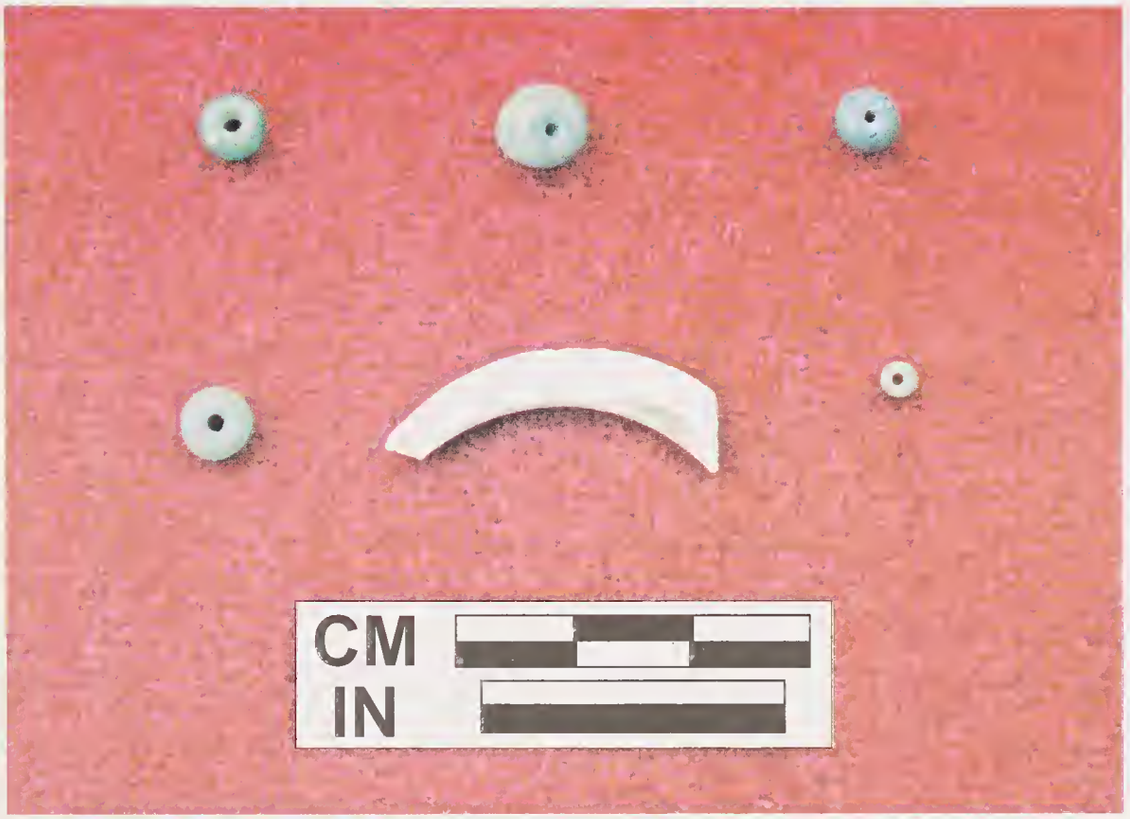


A. Iron object (see p. 72)



B. Saddle querns (see p. 72)

Plate XXII



A. Beads and a shell piece (see p. 73)



B. Terracotta roofing tiles (see p. 69)

CHAPTER VII

INSCRIPTIONS

The Chōḷa inscription of Kulōttuṅga III on a boulder at Śāḷuvaṇkuppam which was copied in 1890[1] was brought alive with the discovery of an inscription of Rāshṭrakūṭa Kṛishṇa III in 2004 on the same boulder. Both the inscriptions are relating to land – the earlier one refers to the sale of temple lands for maintaining the temple while the later one refers to the gift of land.

The brick temple converted subsequently to stone yielded a rich haul of nineteen inscriptions, of which ten are engraved on pillars. Of the ten inscriptions inscribed on the pillars, 5 belong to Pallava period, 1 belongs to Rāṣṭrakūṭa king Kṛishṇa III (936-967 CE), and 2 belong to the Chōḷa period. The remaining two inscriptions on paleographic grounds can be dated to eighth -ninth century CE. Two inscriptions are on the boulder adjacent to the temple complex and they are of Rāshṭrakūṭa Kṛishṇa III and Chōḷa Parāntaka I. The one on the slab can be dated to the 8th century on paleographic grounds. Stone members of this temple have been incorporated in a temple located in the same village – Patavettamman temple and six of them carry fragmentary inscriptions belonging to the Subrahmaṇya temple.

Pillars and inscriptions form an important substratum in the unveiling of this temple complex because both contribute substantially to the presence of the earliest structural temple dedicated to Subrahmaṇya in the history of Tamilnadu. Prior to this discovery, the earliest extant temple dedicated exclusively to *Subrahmaṇya* is the early Pāṇḍya temple of *Bālasubrahmaṇya* at *Kannanur* built prior to 880 CE[2]. The temple of Subrahmaṇya at Tiruttani is known to have been in existence during the 9th regnal year of Pallava *Aparājita* i.e. around 884 CE and takes the history of the temple to an earlier date. Similarly it is to be surmised that the temple at Tiruviḷichil dedicated to Subrahmaṇya was in existence even prior to the 8th century inscription. Festivals like *kārtigai* had obviously gained in significance and were being celebrated in a grand manner. Celebration of festivals in temples might have been an offshoot of the *Bhakti* movement. Quite a few temples were dedicated to Subrahmaṇya, but the continuous patronage this temple received is well discerned from its inscriptions from 8th century to 12th/13th century CE.

Of significance is the fact that this is the only temple in the vicinity of Māmallapuram to have epigraphical evidences of the later Pallavas like Nṛipatuṅga and Kampavarmaṇ. The inscriptions of Nandivarmaṇ II in this temple have set a few precedents. For the first time an epigraphical reference is seen for the celebration of particular festivals like *kārtigai* mentioned in the inscription of Naṇdivarmaṇ

1. ARE, 1890, no 57.
2. Dr.R.Nagaswamy, *Thiruttani and Velanjeri Copper Plates*, Madras, 1979; p.p.5 and 31.

II. The port city of the Pallavas is referred to as Māmallapuram here and is earlier than his Paiyanur record made in his 37th regnal year 768CE[3]. The earliest inscription of *Kṛishṇa* III so far reported from Tamilnadu, has also been found in this temple. It is dated in his 4th regnal year.

It is gleaned from these inscriptions that the Subrahmaṇya temple was located in Tiruviḷichil in *Amūr koṭṭam* and *Amūr nāḍu*. The name Tiruviḷichil[4] can be interpreted to mean a sacred village on the coast. (*tiru* =sacred, a prefix; *vili*=shore/bank/ghat; *il*=village)[5] As to when the name Śāḷuvaṇkuppam was given to this village is not known, but it is pertinent to note that *sāḷuvaṇ* might have been derived from *salam*[6] meaning sand and kuppam meaning fishing hamlet. Since no inscriptions from the temple can be traced after 13th century CE it is probable that it was totally buried in sand and thus earned the name '*Śāḷuvaṇkuppam*'. The administration of the temple from the beginning was entrusted to the *sabhā*. By the Chōḷa period under Rajārāja I it becomes a *Dēvadāna Brahmadēya* in *Amūr koṭṭam* in *Jayakoṇḍachōḷamaṇḍalam*.

All the inscriptions are in Tamil language and script. The inscriptions can be classified into four categories –

1. inscription on the rock boulder
2. inscriptions on pillars
3. inscription on the floor /door jamb ?
4. inscriptions at the Patavettamman temple

INSCRIPTION ON THE ROCK BOULDER

RĀSHTRAKŪṬA KṚISHṆA III

Inscription No 1

The rock boulder was an integral part of the temple complex and two inscriptions have been engraved on them and both are very crucial for the temple. The Rashṭrakūṭa inscription of *Kṛishṇa* III speaks of land donated to the temple for its maintenance and the other inscription of *Chōḷa Kulōttunga* III (already discovered and published) records the sale of temple lands probably for its renovation.

The inscription found on the boulder adjacent to the temple complex belongs to Rashṭrakūṭa Kṛishṇa III, and is dated in his 4th regnal year (943CE.). It refers to the gift of land donated to the temple of *Subrahmaṇya* at *Tiruviḷichil* in *Amūr koṭṭam*. This is the earliest inscription of Kṛishṇa III discovered so far in Tamilnadu. Prior to this an inscription of Kṛishṇa III of his 5th regnal year has been reported from Siddhaliṅgamaḍam[1](appendix p. 156).

CHŌḶA

Inscription No 2

An incomplete one line inscription of Chōla Parāntaka is engraved on the western side of the boulder.

INSCRIPTIONS ON THE PILLAR

The pillars formed the mainstay of this brick temple and most of the inscriptions

3 ARE, 1932-3, No 109.

4 The place where divine or holy processions were conducted or which the king graced by his royal presence (camped). *Cave Temples of the Pallavas*, K.R.Srinivasan P. 182-183.

5 Tamil Lexicon vol 3 part 1, p. 3721.

6 " " vol 6 part 3, p. 1395.

1 ARE, 1909, No. 375, 5th regnal year Vyaghrapadesvara temple, Siddhalingamadam, Tirukoyilur Taluk, Villupuram District.

(ten) have been recorded on these pillars. However, since most of the pillars have been reused, some of the inscriptions are in fragments. None of the inscribed pillars were found *in situ*. Apart from the usual donations made towards burning of perpetual lamps, supply of oil for lamps; donations for celebrating festivals like *kārtigai* find mention here for the frst time.

PALLAVA

Inscription No: 1

The pillar of granite was found on the western side of the cloister *prākarā* at a depth of 1.00m. The pillar square in section measures 0.29 x 0.29 x 3.0m and is inscribed on three sides. The fourth side of the pillar has a *triśūla* symbol. This inscription belongs to the Pallava king Naṇdivarmaṇ II, and is dated in his 12th regnal year (743CE.). It records that 10 *kaḻañju* of gold was donated to the temple at Tiruvi̱ḷichil in *Amūr koṭṭam* by an individual - Kiraippiriyan of Māmallapuram. The gold was entrusted to the members of the *sabhā* of the village who accepted to conduct the *kārtigai* festival *in lieu* of the interest on the gold (Appendix p. 147).

Inscription No: 2

The pillar placed on one side of the stone spear at a depth of 1.60m, measures 0.29 x 0.10 x 0.30m, and is rectangular in section (small broken piece). This inscription belongs to *Nṛipatuṅgavarman* dated in his13th regnal year (882 CE). Only the name and regnal year of the king can be read as the pillar is broken. This is the first inscription of Nṛipatuṅga in the Māmallapuram area (Appendix p. 154).

Inscription No: 3

The pillar found on the western side of the *mahāmaṇḍapa* at a depth of 1.60m, of white granite, square in section, measures 0.25 x 0.19 x 3.0m, with inscriptions engraved on its three sides. This inscription belongs to Pallava king Kampavarman dated in his17th regnal year (887 CE). It refers to the gift of 16 *kaḻañju* of gold for burning the lamp to the deity by a lady named Vasanta piraṭṭiyār wife of Kampabhattan who was the son of *Sāndalya kiḻavan vada kayilaya sarman.* A house site was also donated by which sacred food offering was provided. The *sabhā* agreed to maintain the food offerings annually.

An individual named Tirukkin kilavan Nambhidhaya Bhattan is also mentioned at the end of the inscription. Inscriptions of Kampavarman have not been found in the Māmallapuram area and this is the first one reported so far (Appendix p. 149).

Inscription No: 4

The pillar excavated near the western side of the cloister *prākāra* at a depth of 1.00m, measures 0.27 x 0.18 x 2.65m. It is of granite, rectangular in section, with inscription engraved on its three sides. This inscription also belongs to Pallava king Kampavarman and is dated in his 17th regnal year (888 CE). It refers to the gift of 16 *kaḻañju* of gold by a lady Vasanthanar wife of Siyasarman alias Kampabhattan of Sandalya *gōtra* of *Manaiyir kottam* in *Manaiyir nāḍu* for burning of perpetual lamps in the temple of Subrahmaṇya at Tiruviḻichil. The gold was entrusted to the *sabhā* of the village and they accepted to maintain the lamps *in lieu* of the interest on the gold. She also donated land to the temple for food offering (Appendix p. 150).

Inscription No: 5

The pillar was found near the northern side of the stone spear at a depth of 1.60m. It measures 0.26 x 0.20 x 2.60m. It is of white

granite, square in section, and engraved on three sides. This inscription belongs to Pallava king Kampavarman, dated in his 18th regnal year (888 CE). The inscription records the gift of 10 *kaḷañju* of gold by a lady named *Nangaipperumal* wife of *Siyaśarman* alias *Kambhabhattan* of *Sandalya gotra* who resided at *Manaiyir nāḍu*, a subdivision of *Manayir koṭṭam.* It appears that he is the same person who is mentioned in the above inscription (No: 4).The gold was entrusted to the *sabhā* of the village *Tiruviḷichil* for burning the perpetual lamp in the temple (Appendix p. 152).

Inscription No: 6

The pillar of white granite and square in section found on the western side of the *mahāmaṇḍapa* at a depth of 1.60m, measures 0.25 x 0.19 x 3.0m. The inscription on it records that a certain individual *Tirukkin kilavar nambhi dhaya bhattar* donated 5 *kaḷañju* of gold for burning the perpetual lamps in the temple and *in lieu* of the interest on the gold, the *sabhā* accepted to maintain the lamps every year. This individual may be the same as mentioned in the inscription of Pallava king Kampavarman (inscription No: 2), dated in his 17th regnal year (887 CE). Palaeographically, this inscription is assignable to c 9th century. On the other side of this pillar is engraved the inscription of Kampavarman- vasantapirattiyar. (Inscription No 3) (Appendix p. 154).

Inscription No: 7

This inscription is found on the pillar having Inscription No 6 on one side. This pillar was found on the western side of the *mahāmaṇḍapa* at a depth of 1.60m, it measures 0.25 x 0.19 x 3.0m and is made of white granite and is square in section. It records the gift of land to the temple for the *talippali* festival by an individual *Perunchatti Arrulana Nayana sarman.*Besides another individual named *Kaniyur Korra sarman Catankaviyan* donated land for the same festival. This inscription can also be paleographically dated to c. 8th century (Appendix p. 156).

RĀSHṬRAKUṬĀ

Inscription No: 8

The flat pillar found at the south west corner side of the cloister *maṇḍapa* at a depth of 2.00m, measures 0.32 x 0.8 x 1.76m and is rectangular in section. This badly damaged inscription belongs to Rāshṭrakūṭa Kṛishṇa III, and is dated to his 21st regnal year (960 CE.). It seems to refer to the gift of gold for supplying oil for the lamps to the Subrahmaṇya temple at *Tiruviḷichil in Amūr koṭṭam* (Appendix p. 157).

CHŌḶA

Inscription No: 9

The pillar found on another side of the stone spear (Pillar with Inscription No. 2 was found on one side) at a depth of 1.70m, measures 0.27 x 0.15 x 0.50m and is rectangular in section. This inscription belongs to the Chōḷa King Rājarāja Rājakēsari dated in his 19th regnal year (1004 CE) and mentions the temple by name as Subrahmaṇya deva at *Tiruveḷichil* in *Amūr koṭṭam* in *Amūr nāḍu* and records it to be a *Dēvadāna Brahmadēya.* The remaining portions are missing as the pillar is broken (Appendix p. 159).

Inscription No: 10

This broken pillar inscription was found on the northern side of the main shrine at a depth of 1.70m. It measures 0.63 x 0.36 x 0.08m, is rectangular in section

and inscribed on both sides. It gives the title - Kopparakēsari and could probably be identified with Chōḷa Rājēndra I. Recorded in his 3rd regnal year, it pertains to the gift of land – *Ainnuruva peraiyan kuli* to the deity *Subrahmaṇya* situated in Tiruveḷichil, a *Dēvadāna Brahmadēya* in *Amūr koṭṭam of Jayakoṇḍachōḷamaṇḍalam* for maintaining the perpetual lamp. It can be dated on paleographic grounds to 11th century CE (Appendix p. 160).

INSCRIPTION ON THE SLAB

A lone inscription on a slab refers to donations made for the *kārtigai* festival. The inscribed slab might have once been a part of the flooring in the cloister *maṇḍapa*. The slab was subsequently used as part of the *balipīṭha*.

The flat slab was found on the north western side of the *mahāmaṇḍapa* at a depth of 2.00m, and it measures 1.65 x 0.40 x 0.10m. The inscription is engraved horizontally. This inscription records that, an individual Kīraippiriyan, donated 2 *kaḻañju* of gold for the celebration of *kārtigai* festival. It appears that the individual mentioned herein may be the same as the one mentioned in the inscription of *Nandivarmaṉ* II (Inscription No.1-above) as Kiraipiriyan of Māmallapuram. In lieu of the interest of the gold deposited the above festival was conducted by the *Gurukkal*. It is paleaographically assignable to c 8th century CE (Appendix p. 154).

INSCRIPTIONS IN THE PATAVETTAMMAN TEMPLE:

All the inscriptions have been noticed on the northern wall of the temple. The stone slabs probably formed part of the Subrahmaṇya temple but might have been incorporated in the construction of the *Amman* temple by the villagers. As all of them are fragmentary they are not informative.

Inscription No: 1

This fragmentary inscription mentions *Amūrkoṭṭam* and *nāḍu*. Written in characters of the 13th century CE, it records the gift of eight cows for the burning of the evening lamps.

Inscription No: 2

This merely mentions the word lamp in characters of 13th century CE.

Inscription No: 3

Fragmentary inscription and the details are not clear.

Inscription No: 4

A 13th century inscription, this refers to Subrahmaṇya and a person named Sentirattu Urudaiyar.

Inscription No: 5

This is a fragmentary inscription referring to the signatories Arputakkuttan and Narpattennaayiravan in characters of the 13th century.

Inscription No: 6

The inscription mentions the word karuppur in 13th century characters.

CHAPTER VIII

TECHNICAL REPORT

Chemical composition of lime plaster sample from Śāḷuvaṉkuppam, Dist. Kāñchīpuram, Tamil Nadu

Sample No	Location & Site	Physical Observation	Moisture %	Organic materials %	Loss on Ignition %	Acid Insoluble matter including silica %	R2O3 %	Calcium Oxide %	Magnesium Oxide Mgo %	Total %
I	From inside the sanctum sanctorum at the laterite level Eastern wall Śāḷuvaṉkuppam	Grayish-White Powdery mass mixed with granule like sand and gravel etc.	0.18	0.55	6.33	81.76	1.91	6.6	1.87	99.14
II	Outside the sanctum sanctorum on the northern side Śāḷuvaṉkuppam	About 2-16 mm thick conch shell and gravel pieces visible light brownish in color	0.44	1.24	22.1	43.27	2.57	25.76	1.2	96.58

[1] Sample analysed by Dr. Manager Singh, Suptdg. Archl. Chemist, ASI, Science Branch, Dehra Dun.

Chemical composition of lime plaster sample from Shore Temple, Māmallapuram, Dist. Kāñchīpuram, Tamil Nadu

Sample No	Location & Site	Physical Observation	Moisture %	Organic materials %	Loss on Ignition %	Acid Insoluble matter including silica %	R2O3 %	Calcium Oxide %	Magnesium Oxide Mgo %	Total %
I	Inner circumbulatory shore Temple, Māmallapuram	Looking like Lime mortar dirty yellow-brownish in color mixed with fine sand and gravel conch shining shell are also visible	1.30	2.75	26.08	36.38	3.49	26.6	3.2	99.80
II	From *prākāra* wall shore Temple Māmallapuram	Lime Mortar sample one side flat white in colour other side brownish black due to dried vegetational growth on it. Conch shell are visible lime along with sand and gravel pieces.	2.14	4.48	16.96	52.46	1.95	21.56	0.40	99.95

CHEMICAL ANALYSIS OF LIME PLASTER SAMPLES FROM ŚĀḶUVAṆKUPPAM AND MĀMALLAPURAM TAMIL NADU.

Śāḷuvaṇkuppam Sample No.1:-

From the chemical analysis of sample No.1 collected from inside the sanctum sanctorum at the level eastern wall. It is observed that the sample is very hard siliceous matter with inclusion of about 6.6% calcium oxide into it. This calcium oxide has carbonated to calcium carbonate. There is also no addition of any organic matter into the sample. The sample seems to be hard siliceous/carbonaceous in nature.

Śāḷuvaṇkuppam Sample No.2:-

Sample No. 2 collected from outside the sanctum sanctorum on the northern side. The analysis reports shows it is a good lime plaster with planned mixing of lime and sand to prepare the mortar. In appearance and chemical composition the lime plaster shows similarity to the plaster of Māmallapuram.

Māmallapuram Sample No.1:-

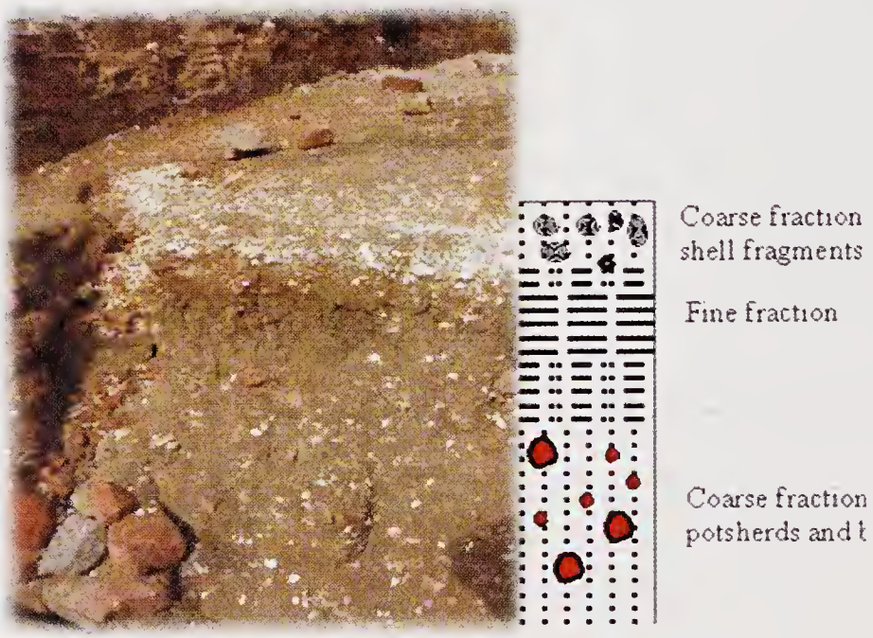
Collected from inner circumbulatory wall Māmallapuram, shore temple shows characteristic very similar to plaster works of that period. Analysis report indicates a slightly dolomitic lime source; quite different to *Śāḷuvaṇkuppam* was used in the preparation of plaster. The plaster has also been found added with little quantity of organic matter during its preparation.

Māmallapuram Sample No.2:-

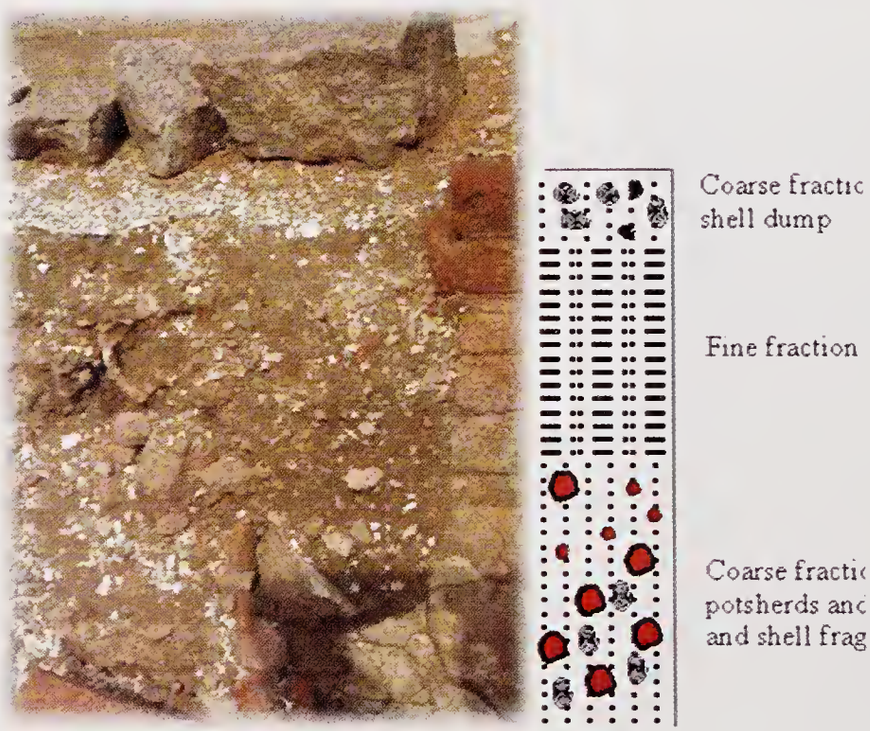
Collected from *prākāra* wall shore temple *Māmallapuram,* shows the source of raw material used for preparation of plaster may be different to the sample no.1 as the lime is not dolomatic in nature. Slightly higher quantity of organic matter was added in the preparation of plaster along with addition of larger quantity of sand than to the sample no.1.

Report on Sedimentology and Micromorphology of the historical site of Śāḷuvaṇkuppam[2]

A. Excavated litho section facing west.



B. Litho section facing north



2. Report prepared by Dr.Hema Achyuthan, Asst. Prof. Dept. of Geology, Anna University, Chennai.

Remarks: Shell are made of *meretrix, mytilus, ostrea, unio, vnericardia, tellina, laternula, arca, nucula, pteriacea, inoceramus, cardium, pectin, venus* and oyster shells of which arca is about 42% of the collection. These shells are characteristically bottom dwellers, adapted to life in brackish and normal marine waters. The shells are thick and aragonite in composition. It is a shell dump layer probably for storage. These shells are abundantly occurring in the shell beds at *Muttukadu* and probably this site was a major source for shells that were used as lime mortar for plastering the bricks that were used above the laterite bricks. In the excavated section no distinct layers of storm surges or tsunami were discerned.

C. Plastering of the brick wall is with lime mortar.



Remark: the cement is made of calcium carbonate and broken shell fragments.



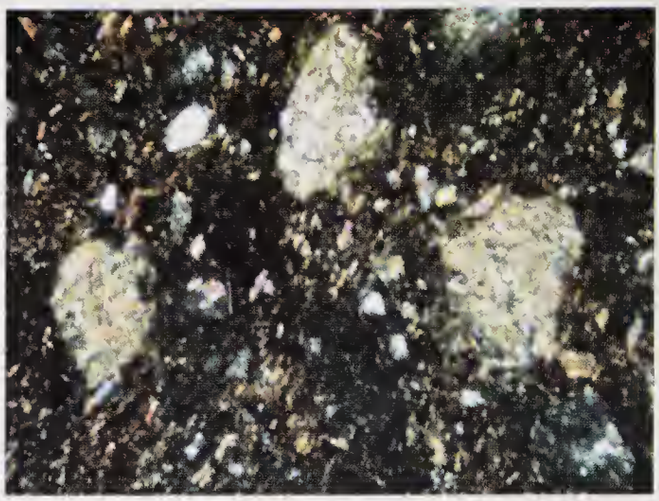
D. Well section with thick rim pottery. With outer markings of paddy. Highly Carbonized inner layer dated to 880 ± 55 14C yrs. BP.

E. Thin Section of the pot sherds in Plane Polarised light.

F. Micromorphology of the rim reveals highly carbonized matrix with coarse quartz grains and quartzite as rock fragment. The margins of the grains are well discerned in the matrix indicating very low temperature heating.

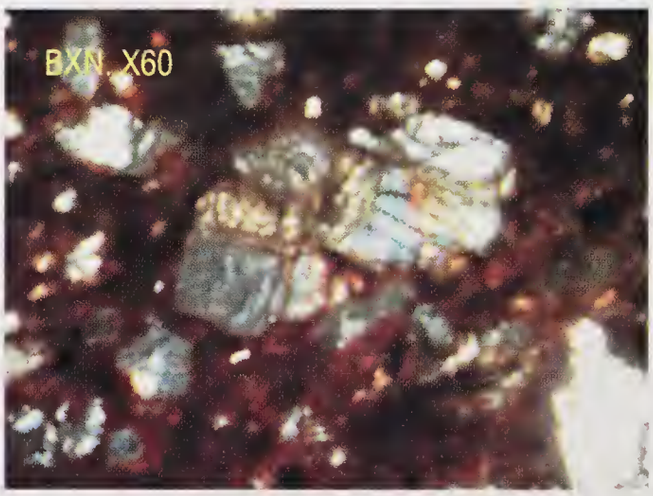
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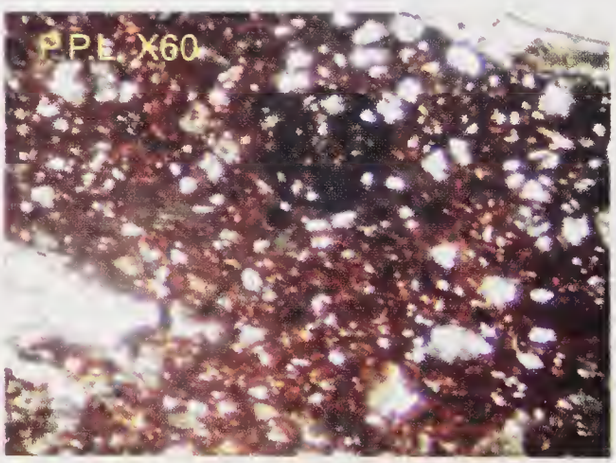


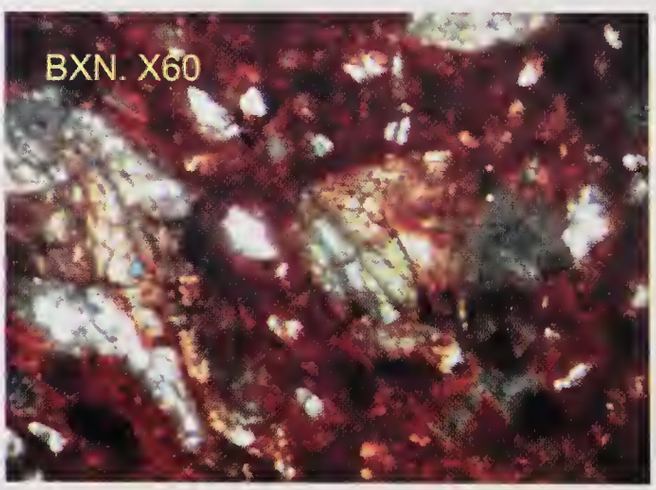


F. Close up of the paddy markings on the surface of the pottery of the well rim





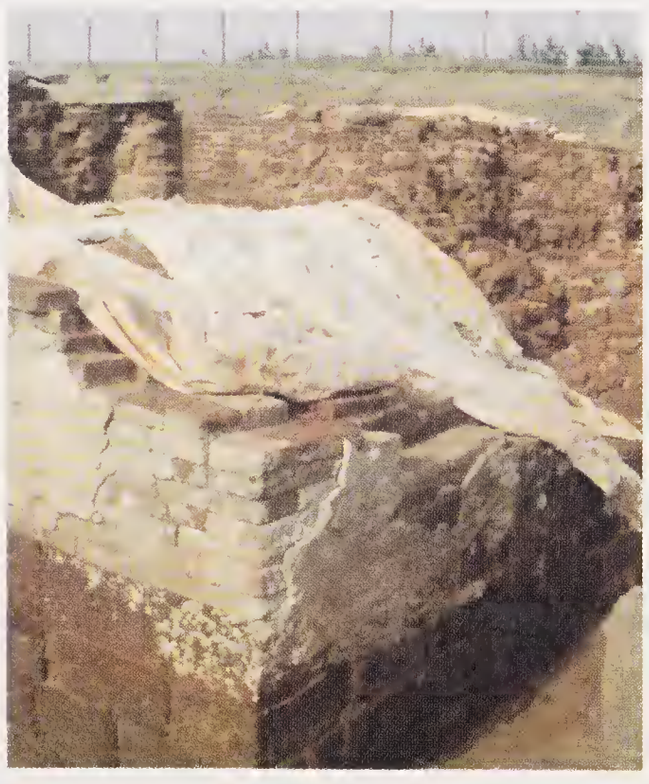




G. The foundation is made of laterite bricks. Thin section study of the laterite bricks indicates high content of hematite, magnetite, kaolinite patches and the porosity of the laterite varies from 5 to 10% only. The laterite could have been brought from a near by source.



H. Thin section of the bricks exhibit coarse grain quartz (60 to 75%) feldspar (8 to 10%) garnet (3 to 5%) and quartzite and rock fragments (8 to 10%). The grains are cemented by iron oxides and the heating temperature of the bricks is lower and not more than 750°C to 800°C. The laterite formed on the sandstone bedrock is cut into bricks, then probably heated at low temperature to give strength to the bricks. The foundation laterite does not have lime plaster while the one which is exposed is plastered. A brick layer below the laterite bricks was used to ensure evenness of the ground.





I. Tafoni on the huge natural rock boulder indicates major wind direction to be north south and/or North west and south east. The rock granite and granite gneiss with garnet and sillimanite. The rocks are coarse-grained, pink, exhibit graphic texture and is sheared along the major north-south direction.

Summary:

1. No evidence of Tsunami sediments in the excavated site.

2. Alluvium from the R.Palar has been placed in the garbhagṛiha. The sediments are coarse-grained, ill-sorted, sub-angular, negatively skewed with less percentage of mafics.

3. The occurrence of the shells has been used as lime cementing mortar.

4. Summary based on potsherds

a. Ceramic tradition is of particular interest in view of economy. Understanding ceramic technology helps in reconstructing cultural history. One of the goals in this study has been to elucidate pottery making technique and occurrence of shells in temple construction this site being on the coast.

b. Potshards were present below 45cm. Due to disturbance by human activity, trenching and filling with debris is clearly visible from the lito sections.

c. Potshard studied here was collected from a depth from 30-80 cm. down to the base level which represent the litho section 1 mt. thick facing west and north. Maximum dimension of potshard ranges from 0.5 to 6 cm. in length with mean thickness of 0.3 to 0.5 cm. However the well rim is around 2.2 cm. The colour of the potshards varies from reddish yellow on the surface to brown and they are largely coarse red ware.

d. The coarse red ware pottery shows surface features indicating it was handmade, while the finer red ware shards were probably manufactured using a combination of wheel and handmade techniques. Both types of pottery was fired at low temperature. The well rim were also heated at low temperature and hand made using less amount of fuel (paddy markings).

e. Representative 10 potsherds were studied for petrological analysis. Small pieces measuring 2 X 2cm, were cut from each shard and subject to a test in order to determine porosity, mineral composition and texture.

5. The thin sections were examined under a Euromex petrological microscope. The thin section analysis revealed the following.

a. Lower layer of the site, i.e., 60-80 cm. the pottery is predominantly fine grained. The grain sizes range from 40 to 110cm. Coarse grains dominated in the upper levels and the grain size frequency is bimodal. The grains tend to be sub-angular to sub-rounded. Though the coarser material is well rounded to sub-rounded some of the quartz grains are frosted and stained by iron.

b. Orthoclase feldspar, garnet and muscovite are present and vary from about 2 to 4 % in the upper layer to 5 percent in the lower. Some of the muscovite flakes are altered. Coarse quartzite grains occur as rock fragments (12-14%).

c. Coarse quartz grains comprises of 15 to 16%. They are fractured and stained with iron. Pores are in the form of voids- regular to irregular in shape and many times are lined with grayish fine silt components or black soot.

d. The occurrence of charcoal in pottery fabric as observed in the well rim, don't contain shells.

e. Two type of matrix can be identified in the pot sherds. One is composed of iron oxide which may be a mixture of hematite and goethite. Minerals of the iron oxide group in fine grained clay form pseudo grains, which are sub-angular to very well rounded. The ratio of clay matrix to the coarse and fine fraction varies from 7:3 to 4:6. In places the matrix exhibits micro-fracture and flow structure, and traces of cellular structures represent vegetal materials (well rim), which have been burnt during firing.

f. It is known that the crystalline structure of kaolin is destroyed at about 600°C. The absence of the kaolin group of minerals in the potsherds examined so far indicate that firing temperature would have reached this level.

g. Diffused illite (micaceous clay minerals) occurring suggests that the temperatures at which firing took place may not have been less than 600°C.

h. The crystalline form of quartz undergoes a change at 573°C. The fact that the quartz grain in the samples described above is unaltered makes it clear that, on the whole, maximum temperature could not have exceeded 600°C to 700°C.

i. As pointed out earlier, traces of orthoclase feldspar occurred through out sequence. Although low type of feldspar has inversion point of 900°C the effects of this alteration have not been detected in the pottery. The analysis carried out renders it possible to determine that the firing temperatures were fairly low.

6. Thin section study of the laterite bricks below indicate:

a. Ferrugenised coarse sandstone bedrock,

b. Fe - impregnated and indurated sediments including clay, sandy sediments,

c. Laterite bricks of complex sedimentary and pedogenic origin (like the pisoliths).

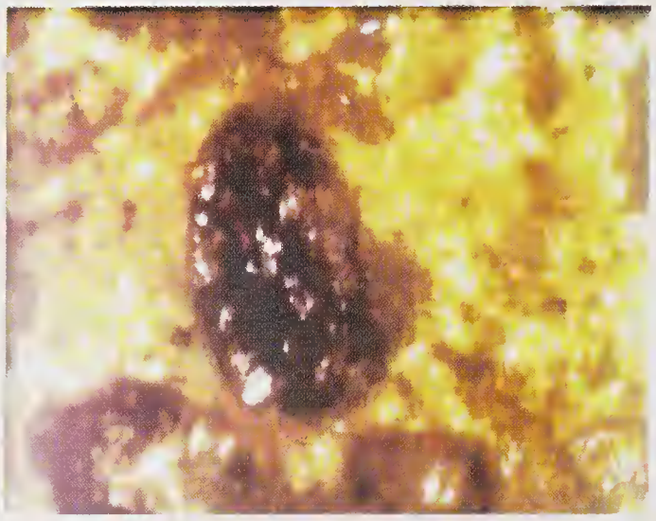
Type a and b laterite bricks have simple fabric and Feoxide mineralogy dominated by hematite, magnetite and gibbsite Al(OH)3. Fe oxides have variously replaced mica and feldspar. In bedrock ferricrete samples it is common to observe the physical disintegration of primary micas due to infiltration of iron bearing solutions along cleavages, fractures

and subsequent crystallization of Fe oxide minerals. Thin section data of laterite bricks indicates that multiple phases of iron oxide mobilized and precipitated from the original rock was particularly selected for the making of the foundation bricks.

Detrital mineral with the laterite bricks

Quartz grains of varying shapes are found embedded in the iron oxide-kaolinite matrix. The primary mineral composition contains only few easily weathered minerals: around 5 % feldspars and 5% phyllosilicates, which are mostly muscovite. In the heavy minerals separation the minerals identified were tourmaline, zircon, rutile and garnet.

Summary: They have been cut from the laterite rock exposures near source.



Pisoidal structure in iron oxide cement. P. P. L Magnification X 60

Results of sedimentological studies at the Śāḷuvaṇkuppam archaeological site, Māmallapuram[3]

INTRODUCTION

Māmallapuram was an important port built during the reign of the *Pallava* kings (Raman, 1998). Early navigators called it a town of seven pagodas. Many scholars believe that the sea had encroached on these temples in the past; the only surviving structure is the Shore temple (a world heritage site) that escaped the December 26, 2004 tsunami because it stands behind a shore protection wall. The other temples are completely submerged today, but their top parts were visible until a few centuries ago (Chambers, 1788). William Chambers, an early British explorer, attributed the submergence to "overflowing sea" caused by an earthquake (Chambers, 1788). The underwater exploration revealed remains of huge stone constructions, dating to 8th century A.D., at 6-15 m below present sea level (Sundaresh et al., 2004). Considering an average rate of erosion of 55 cm/year, the shoreline 1500 years ago must have been about 800 m seaward of the present coast (Ramaiyan et al., 1998).

The 2004 tsunami, inundated as much as 3-6 m above datum, eroded beaches creating scour ponds along the Māmallapuram Coast (Chadha et al, 2005; Fig. 1). Scouring also exposed buried rock sculptures and temple basements, and the Archeological Survey of India (ASI) therefore excavated two sites here. At one of their sites, about 300 m south of the Shore Temple, basement of a stone temple from the *Pallava* period (8th century A.D.) was uncovered. The other site is at *Śāḷuvaṇkuppam* about 1.5 km north of the Shore Temple and about 150 m from the sea. Here we examined multiple cultural levels with clear-cut working levels and associated sedimentary horizons. The results of study were published and what is given in this report owes much to our published paper (Rajendran et al., 2006).

3. C.P. Rajendran, Centre for Earth Sciences, Indian Institute of Science, Bangalore 560012, (With inputs from Kusala Rajendran, Terry Machado, T. Satyamurthy, and P. Aravazhi)

Stratigraphic Studies

The ASI site at Śāḷuvaṇkuppam (referred here as Trench 1 or T1) exposed a brick basement, overlain by remains of stonewalls and fallen pillars (Fig. 2). From the varying styles of construction, type of materials and the nature of artifacts, three phases of construction are evident. The youngest structure (Phase III), built in granite belongs to the *Pallava* period (8th-10th centuries A.D.) and its base is about 2 m below the present ground surface. Phase II (pre-*Pallava* period; 6-8 century A.D) is represented by an underlying brick basement. During the pre-*Pallava* period only bricks were used for construction of temples and it was after 8th century that the use of granite became popular. A much older level of construction using an earlier genre of bricks exists at a depth of about 3.2 m below the present ground level. Only a part of this structure (early 4th century A.D.; phase I) has been exposed at the time of our inspection (November. 2005), on the eastern side of the trench.

It is possible that each of these three settlements was destroyed, and that the succeeding generations used the previous ruins as basements. The question we posed is that whether possible that successive sequences of sea surges could have destroyed each of these constructions. The tsunami (December 26, 2004) inundation data shows that a previous tsunami of similar size could have reached 400-600 m inland and destroyed many of the man-made structures located on a previously extended coast. The destruction of the granite temple probably postdates A.D. 925 - the approximate age of the inscriptions (an earlier form of Tamil) on the fallen stone pillars. Since all the granite pillars occur to the west side of the temple complex, they may have been destroyed by sea surges from the eastern side.

At the time of our inspection the excavation of the site was only partial and the stratigraphic sections of the eastern and southern walls within the temple precincts were left undisturbed and were fully left exposed (these were later removed to expose the rock monolith). The sections consisted of several layers of sand, mixed with brick debris, shells and pottery shards. Human modification and leveling of the ground is evident on the eastern (seaward) side. We noted three layers of highly compacted sand, consisting of brick debris, plaster and shell remains, at various depths (L1, L2 and L3). Corresponding to various phases of construction, they probably represent working levels for each settlement. These layers may have been compacted to level and stabilize the ground as well as to provide protection from the sea. Since such layers are absent on other sides, protection from the sea may have been a major motivation behind making these compacted layers (Fig. 3A).

A tsunami may account for a layer of fine sand that extends along two of the trench's walls. This layer, marked as S1, shows varying thickness (14-10 cm) from the wall on the south to the east before it tapers out. A thinner version of this layer (1-2 cm) was seen on the northern wall, but its eastward extension was barely traceable. Fine convoluted laminations and grading

within this layer suggest its deposition under the influence of water. Small fragments of bricks and clasts, probably derived from the brick debris are embedded within this layer. Restricted spatial extent of the sand and its prominence on the seaward side suggest possible wave action. The maximum thickness, 14 cm, was observed close to a 4m. tall granitic outcrop that marks the southern limit of the T1. Perhaps, this rock monolith (see Fig. 2) must have caused deflection of the waves, leading to more scouring and also provided a sheltered depositional niche for the retreating water.

After the inferred tsunami and selective scouring, more brick debris were probably dumped as a means of protection. Some portions of the top most layer of debris consist of granite chips, which are totally absent from the lower levels. This change in the content of debris is noteworthy because, as mentioned earlier, granite became a choice of construction only after the 8th century. Thus, it appears that the original brick fill was reinforced after the sea surge and only these portions, notably those above the suspected tsunami deposit, contain the granite fragments. Charcoal pieces obtained from the sand layer (S1) provided an age of 955 ± 30 B.P. (A.D. 1019–1161), and probably post-dates the construction of the stone temple (Fig. 3A). The flooding event may have occurred between the inscriptional age of A.D. 925 and the calibrated charcoal date of A.D. 1161. At deeper levels, we logged another layer of fine-grained sand (10 cm), possibly of marine origin at a depth of 3.0 m from the surface (S2). This discontinuous layer of sand occurs above the ruins of the oldest brick structure (phase I) and as in the case of S1, this one is also prominent only on the southern wall, with a slimmer eastward extension. We presume that the presence of the granite outcrop may have provided favorable conditions for the selective deposition of this layer as well.

Ten meters south of the Trench 1, a separate trench (Fig. 3B) revealed another discordant layer of sand, presumably of marine origin. This 2.5 m deep trench (referred as Trench 2 or T2) bottomed on the extension of the oldest brick structure observed in the ASI site. The layer of sand, with maximum thickness of 10 cm was exposed in the southeastern corner of the trench. This layer marked by fining upward sequence contains clasts from the underlying layer; it also shows laminations suggesting deposition controlled by the flow of water (Fujiwara et al., 2000; Moore at al., 2006). We believe that this layer may be comparable to sand layer (S2) observed in T1. Its occurrence at a cultural level suggests that it was deposited by an unusual sea surge in an area that is otherwise protected from the wave action. However, in this trench (T2) we did not observe the extension of the younger sand layer (S1) present at T1. Perhaps, this deposit was preserved selectively and is specific to that site as it was trapped within the debris.

We obtained two sets of radiocarbon ages to constrain the time of deposition of the lower layer of sand from the ASI site as well as the trench and the charcoal samples yielded ages of 1674 ± B.P. (A.D. 321- 427) and 1581 ± 35 B.P. (A.D. 405-564). The

older age is obtained from the lower levels of Trench 1 (S2) and the younger age is from the corresponding deposit at the nearby trench.

EVIDENCE FOR A TSUNAMI?

Although the long-term sea level rise has been viewed as the main cause of the and submergence and destruction of temples along the coast of Māmallapuram about 1000 years ago (Suderesh et al., 2004), did predecessors to the tsunami of December, 2004 play a decisive contributory role? Did massive short-lived coastal inundations in the historical past struck a body-blow to the coastal community that existed those days? The anomalous sand layers described above suggest the possibility of at least two events of large coastal flooding at this site: one event around 10th century A.D. and the other around 5th century A.D. We tentatively ascribe them to giant tsunamis, not to storms, because the Māmallapuram Coast has generally been sheltered from the region's cyclones and the patterns have not shifted over time (Mescarenhas, 2005). Because of the apparent intensity of these historical tsunamis, we believe these were generated at the Sumatra-Andaman subduction zone rather than the other sources like Makran in the Persian Gulf.

It is important to expand the search for anomalous sand layers to other areas along the east coast, particularly closer to ancient cultural settlements, and to date the layers well. It is also important to characterize them in terms of flow hydraulics, sedimentology, mineralogy and lithology, so as to discriminate them from other types of coastal sedimentation. The December, 26 event provides us a unique template for such studies. That the Māmallapuram site is in a beach environment severely limits the preservation potential of the sedimentary deposits, and the possible reworking can destroy or obscure the characteristic features. Companion studies are therefore needed to distinguish storm deposits from tsunami deposits, and to identify paleoseismic evidence in Aceh (Sumatra) and in the Andaman-Nicobar Islands to constrain the causative earthquakes.

Acknowledgments:

We thank the Archeological Survey of India, Department of Science and Technology, Government of India and Centre for Earth Science Studies for the support. Rafter Radiocarbon Lab, New Zealand provided the AMS dates; D. Raju and Anil Earnest helped us in the field investigations. We thank Brian Atwater and A.S. Gaur for their comments.

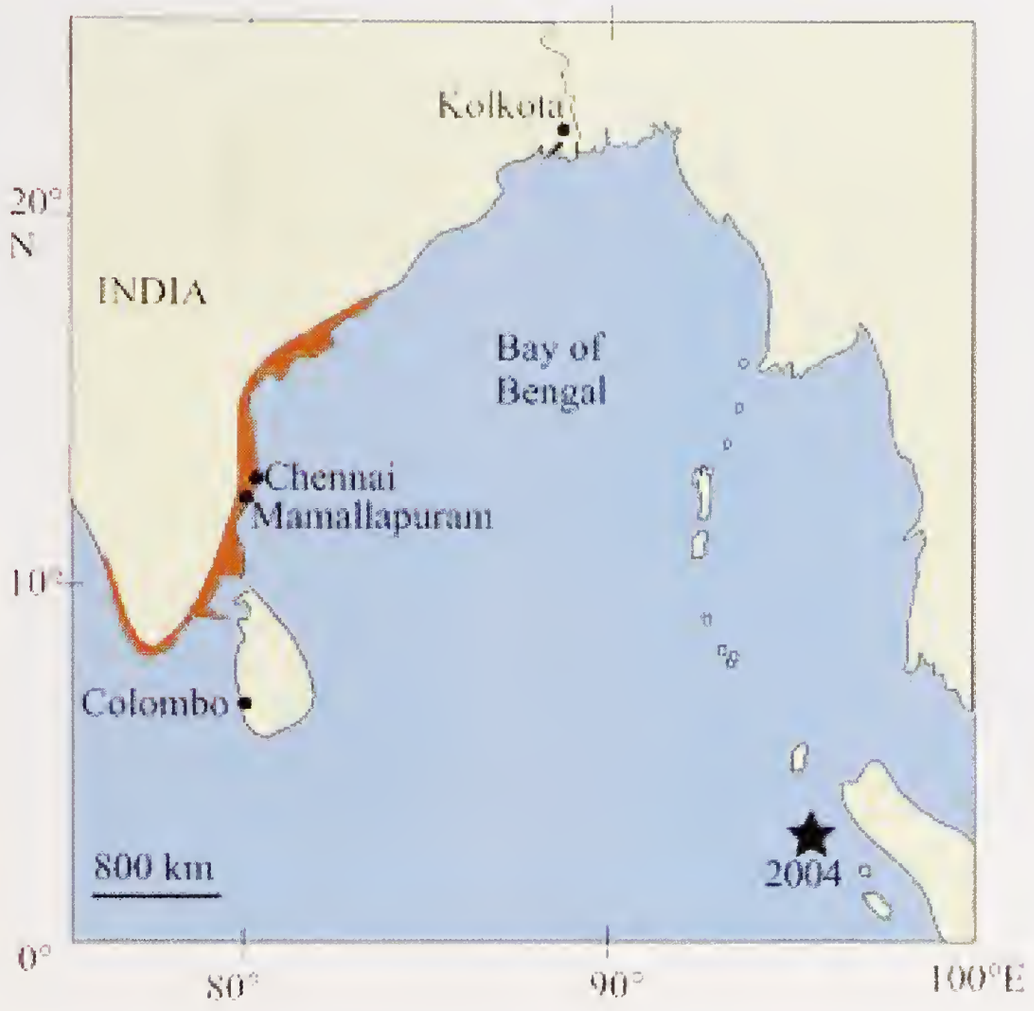


Fig. 1 Map showing areas along the east and southwest coast of India affected by the December 26 tsunami. The star shows the location of the 2004 mega thrust earthquake (Stein and Okal, 2005)).

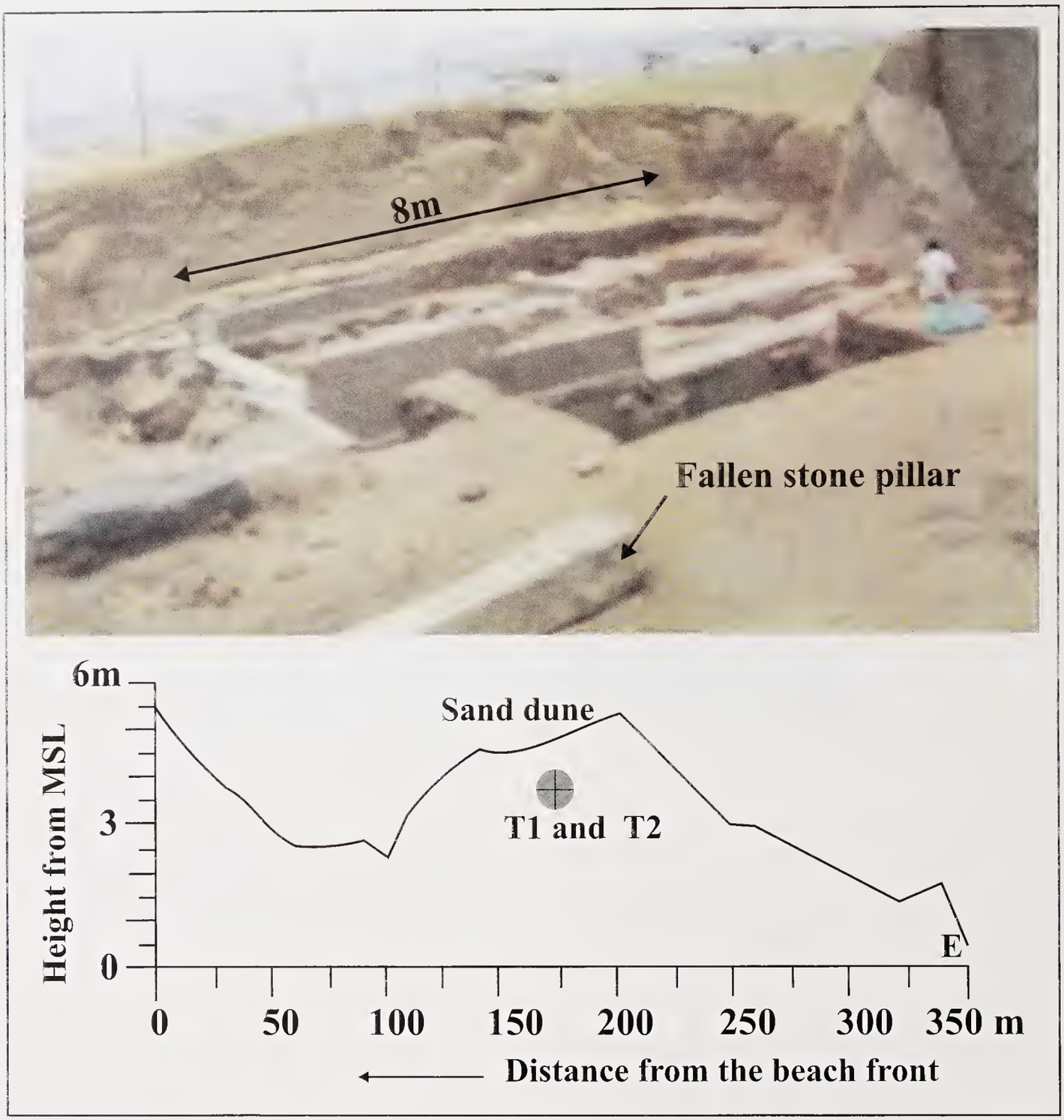


Fig. 2 Excavation site (Trench 1) at *Śāḷuvaṇkuppam*; bottom panel: Elevation profile of area showing the location of the site. T2 is located 10 m south of T1. Our inspection was conducted during November 2005 when the excavation by ASI was not complete.

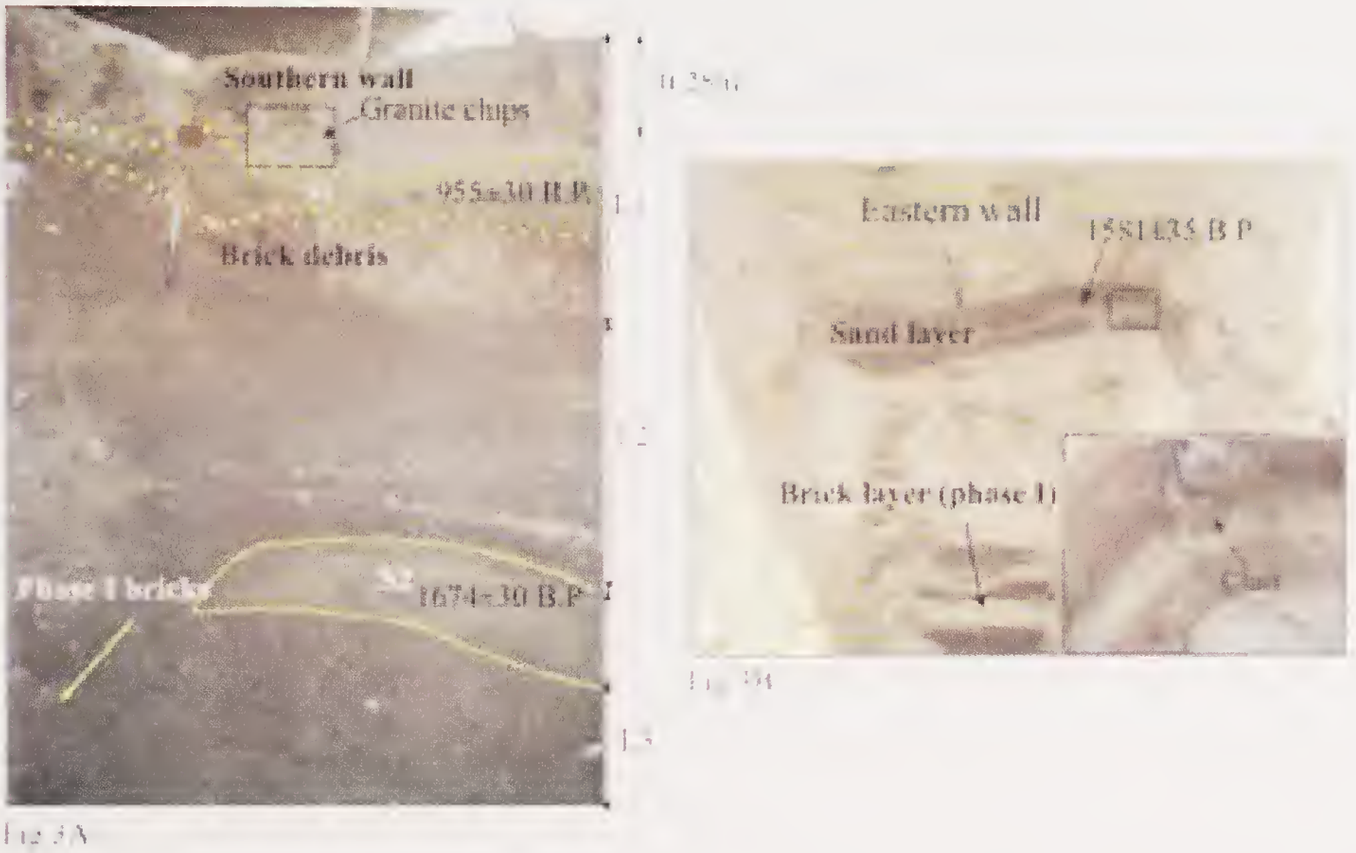


Fig. 3 (A). Stratigraphy of the eastern wall at Trench 1 (T1); (B) Stratigraphy of the eastern wall of the trench 2 (T2); inset: close-up of the area shown in the box.

CHAPTER IX

CONCLUSION

The Subrahmaṇya temple at Śāḷuvaṇkuppam is the earliest brick temple to be excavated in Tamilnadu and it reiterates the view corroborated through inscriptions and also by extant temples in Toṇḍai-maṇḍalam area like Kuram, Kalambakkam etc that brick as a medium of construction was in vogue in the Pallava period.

The excavated complex threw up a very complicated picture of structural activity that had taken place probably due to the use of brick as a medium of construction; close proximity to the sea; lack of direct royal patronage and primarily the evolving trends in the establishment of well laid out principles of construction and religious tenets. Finally the construction activity also tends to coincide with the growth of the temple alongside the growth of the settlement.

The major hurdle in trying to establish a proper structural development lay in the fact that the excavation was carried out in sandy soil where it was very difficult to maintain stratigraphic levels and also that the temple had undergone a number of structural modifications within a relatively short period of time.

The excavated remains at Śāḷuvaṇk-uppam which was known as Tiruviḷichil in inscriptions, speaks of the existence of a temple dedicated to Subrahmaṇya. Based on the development of the temple complex, seen through the excavations, three major phases of constructional activity can be discerned.

Broadly the structural activity can be classified as under:

Phase I Early Pallava period – Pre 6th century CE

Phase II

Sub Phase A Pallava period – 6th – 8th century CE

Sub Phase B Late Pallava period – 8th – 10th century CE

Phase III Chōḷa period – 10th – 13th century CE

The earliest period of activity is the Early Pallava phase. As evidenced by the excavated remains, this phase had only a moderate structure constructed of brick, built abutting a natural stone boulder. It may be of relevance to reiterate that the exposed stone boulder is naturally shaped like a seated elephant. This brick structure is rectangular on plan and facing north. A platform of dressed laterite stones over which might have been laid four courses of brick, forms the nucleus of the initial structure. The outer walls of this structure have been partially plastered with lime.

The technique of construction is very significant. Over a bed of river sand was raised four courses of dressed laterite and above it was placed four layers of brick. Laterite being porous by nature is ideally suited for coastal conditions. The core between the inner and outer veneer of laterite has been filled with river sand.

It is of relevance to note that a stepped tank like structure of granite and a screen wall of laterite were discovered to the west of the Shore temple. Attributed to a time frame earlier to the Shore temple, it reveals that for its proximity to the sea, laterite was used due to its intrinsic qualities. The use of laterite even in this structure for the same reason is worth mentioning. The absence of granite slab at the plinth level, a feature continuously adopted by Rājasiṁha gives this excavated structure at Śāḷuvaṇkuppam a date earlier to Narasiṁhavarmaṇ II Rājasiṁha.

The discovery of a conical jar in the present structure at the bottom most level of the laterite course and that of a Roman coin of Arcadius (395 – 400 CE) at the top levels of the same course enables one to indirectly date this structure to pre - 6th century CE. The large size of the bricks 40x21x7cm also indicate a similar period for the structure.

The discovery of a Pallava coin with the legend "lakshita" – the title of Mahēndravarmaṇ I (600 - 630 CE) further strengthens the view that this structure was already in existence during this period.

In the absence of any image and the lack of other identifying features and also due to the plain and simple nature of the structure, it is very difficult to establish the true status of the earliest phase of this structure – whether secular or religious.

The reporting of a terracotta plaque (Fig.21.2) identified as *kuravai koothu*, a dance associated with Murugan makes one ponder if at this stage it was dedicated to the local deity Murugan.

The second phase of temple activity ensures a far more accurate understanding of the structure. Through the earliest inscriptions of this phase – (i) of Nandivarmaṇ II (731 – 796 CE.) and (ii) palaeographically dated to the same period which were found at the top levels, indicate that a temple dedicated to Subrahmaṇya was consecrated here.

During this phase the basic structure of the temple had not been disturbed, but only the height of the temple had been raised. Together with this, a pillared cloister *maṇḍapa* and a *mukhamaṇḍapa* had been added. The crowning glory and most unique contribution of this temple to the history of architecture in Tamilnadu, is the erection of a stone spear or *vēl*, an attribute of Subrahmaṇya on a brick platform. Further north is the *balipīṭha* displaying all the features of an *adhiṣṭhāna* moulding and with a granite base slab as the bottom course, a feature associated with Rājasiṁha. Such a *pādabandha* adhiṣṭhāna is also seen at Mukunda Nayanar temple in Māmallapuram, Tirupulisvara temple at Vāyalūr and Virattanēśvara temple at Tiruttani. While the first two are associated with Rājasiṁha the last one belongs to the period of the later Pallava ruler Aparājita. The pillars of the cloister *maṇḍapa* and the

spear are similar in profile – hexagonal in the middle and rectangular at the top and bottom. Similar type of pillars have been noticed at Mukunda Nāyanār temple, Māmallapuram, Tirupulīśvara temple, Vāyalūr, Konar temple, Tirumalpur(seen in old photographs), Śiva temple, Kūram and Virattanēśvara temple, Tiruttani. Only in the *maṇḍapa* of the Kūram temple are such types of pillars still seen *in situ* and in the Tiruttani temple, similar type of pillars have been incorporated into a *maṇḍapa* of a later period. In all the other places, they have been noticed as stray finds stacked in the temple. While the Kūram temple is attributed to Paramēśvaravarmaṇ I, all the other temples are associated with Rājasiṁha or the later Pallavas.

The inner side of the cloister *maṇḍapa* has been paved with granite and the outer wall of the cloister *maṇḍapa* in the north has a *pādabandha adhiṣṭhāna* (a feature of Rājasiṁha period). Lateral entrance to the temple has also been provided at this stage probably in keeping with the by then well established *Āgamic* injunctions.

The basic structure of the temple without being destroyed was further raised to the extant levels in all probability due to damages caused by sea surges. The lateral entrances were now provided with steps, some of which are broken pillars. A new *mukhamaṇḍapa* further north and an additional rectangular structure on the north was also added. The intense constructional activity is seen through the presence of a storeroom (layers of clay and lime were encountered) on the northwest and the rudimentary lime kilns on the cloister *maṇḍapa* floor.

The continued absence of any ritualistic elements in the *garbhagṛiha* poses the question whether this structure was a temple? When the availability of a large number of terracotta lamps is viewed along with the inscriptions, which enumerate the donations of gold for the burning of perpetual lamps; and the celebrations of festivities like *kārtigai* it can be presumed that it was a very popularly propitiated temple. The Subrahmaṇya cult gained significance in the Pallava period by 7th century CE. The canonization of Subrahmaṇya can also be attributed to this time frame as seen in Tirumurugarrupadai.

The final phase of activity established this temple in its stone form with a *garbhagṛiha, ardhamaṇḍapa, mahamandapa* and *mukhamaṇḍapa* raised and constructed with material from the earlier phases, underscores the temper of the inscription of Kullōttuṅga III which states, that as the temple was badly in need of funds, and for which the temple lands were sold. The spear in the mukhamaṇḍapa was buried upto its neck and had obviously lost its significance, but it was not disturbed. Even at this stage no modifications to the structure were made to enable the temple to discharge its ritualistic functions. The entrance to the temple was restored to the original direction i.e. north. Probably the same image was being used which might have perished with age.

Some scientific analysis were carried out with samples obtained from this site and the date to a large extent corroborates archaeological evidence.

The radiocarbon dates from the chareoal pieees obtained from the sand layer S2 at a depth of 3m provided an age of 405-564 CE. This date corresponds with Phase I of the temple as already identified.

The radio carbon date of 1019 – 1161 CE , from eharcoal samples taken from the upper layers, probably indieates the beginning of the final phase of temple reconstruetion. The boulder inscription of Kulōttuṅga III dated 1215 CE reeords his efforts to rebuild the temple. These two dates eomplement the struetural aetivity of the temple in the last phase.

Lime samples from inside the *garbhagṛiha* just above the laterite level, and from the outer wall of the *garbhagṛiha* on the north both from the exeavated site at Śāḷuvaṇkuppam were eompared with the lime samples taken from the inner eireumambulatory and *prākāra* wall of the Shore temple, Māmallapuram. The ehemieal eomposition of the samples from the inner eireumambulatory of the Shore temple and the outer wall of the *garbhagṛiha* (northern side) of the exeavated site show similarity in appearanee and ehemieal eomposition. The teehnique of lime plaster preparation also appears to be similar. This similarity might indieate that the brick structural temple at Śāḷuvaṇkuppam must have also been of the same period as the Shore temple of Narasiṁhavarmaṇ II Rājasiṁha.

The propitiation of Subrahmaṇya as a God assoeiated with the Vedie Brahmanism as followed by Rājasiṁha and the popularity of the cult due to the Bhakti movement in Tamilnadu at this stage and the presenee of Rāshṭrakūṭa affiliations in all probability led to the intense activity witnessed by this temple betwcen 6th and 10th centuries CE. Thc prevalence of temples dedieated to Subrahmaṇya in nearby Tiruppōrūr and Anur is noteworthy sinee they are also of the same period.

The briek temple complex refleets the tenor of the establishment of both religious tenets and principles and the experimentation adopted in raw material and construetion during the Pallava period. It also reiterates both through direet and indireet evidenee that this temple complex was primarily Pallava in its origin and development.

POSTSCRIPT

Articles have been written on the excavation based on geological analysis and these have been placed in the Technical Report section. These articles give an idea about the sea surges/ tsunami which might have occurred in the east coast of India and might have been the cause for the destruction of the temple in different time frames. Also the radio carbon dates obtained from samples of this site almost coincides with the chronological dates of archaeological data.

It is pertinent to note that, the site was only partially exeavated when the authors of the above artiele were undertaking the geologieal study.

APPENDIX - I

List of Temples where brick has been used in the Pallava period in Toṇḍai-maṇḍalam*

S.No.	Site	Taluk	District	Temple Name	Remarks
1	Kūram	Kāñchīpuram	Kāñchīpuram	Śiva Temple	The foundation of the temple is of brick.
2	Tirukkalukkunram	Tirukkalukkunram	Kāñchīpuram	Vedagirisvara Temple	It is presumed that the adhiṣṭhāna mouldings, pilasters and the prastara mouldings were originally composed of brick and stucco as would have been the case in regard to the talas above.
3	Uttiramērūr	Uttiramērūr	Kāñchīpuram	Sundaravarada – Perumal Temple	The walls of the aditala and of the mukhamaṇ-dapa as well as the talas above are in brick and mortar built over a stone adhiṣṭhāna.
4	Uttiramerur	Uttiramerur	Kāñchīpuram	Kailasanatha Temple	The mañcabandha upapītha with plain galapādas and the bold padabandha adhiṣṭhāna are fashioned from granitic stone; the structure above is brick and plaster.
5	Kalambakkam	Tiruttani	Tiruvallur	Tirunagesvara Temple	The foundation of the temple is of brick.
6	Sumangali	Cheyyar	Tiruvannamalai	Ruined temples	The temple wall is coursed ashlar masonry forming the inner and outer veneer with a brickwork core.
7	Tiruvadigai	Cuddalore	Cuddalore	Tiruvirattanesvara temple	The stone kapōtabandha adhiṣṭhāna was added in the 13th century (proved by an inscription) as a revetment or protective facing for the much eroded brickwork of the original adhiṣṭhāna.

Michael W. Meister,** ***Encyclopedia of Indian Temple Architecture, South India Lower Dravidadesa 200 B.C.- A.D. 1324,
American Institute of Indian Studies, 1999

APPENDIX II

Selected Brick Sizes from Early Historical Sites – Tamil Nadu State and Puducherri (UT)

No	Site Name	Brick Size	Period
1	Kāveripaṭṭinam) (TN) Kilayur (wharf) Vanagiri (inlet sluice) Pallavanesvaram (Vihara)	61 x 40.5 x 7.7cm 40 x 26 x10cm 42x24x10 cm	1- 3 century CE 1-3 century CE 4 – 6/7 century CE
2	Kāñchīpuram (TN)	28x19x16 cm, 56x23x8 cm, 40x18x 6 cm	1 BCE-4 century CE
3	Korkai (TN)	45x23x7.5 cm	1 BCE-4 century CE
4	Mangulam (TN)	35 x 17 x 5.5 cm 35 x 18 x 6 cm 36 x 19 x 7 cm	3 BCE – 3 century CE
5	Arikamedu (Puducherri UT)	35.56 – 38.10 x 20.32 – 21.59 or 26.67 – 27.94 x 6.35 – 7.36cm; 33.02 – 34.29 x 22.86 x 6.86cm; 35.56 – 39. 37 x 22.86 – 24.13 or 26.67 x 27.94cm x 6.86 – 7.1cm 38.1 – 40.64 x 26.67 - 27.94 x 6.86 – 7.62cm	1 – 2 century CE
6	Porunthal (TN)	48 x 24 x 8cm 42 x 21 x7 cm	1 BCE- 1 century CE

APPENDIX - III

COPPER COINS FROM ŚĀḶUVAṆKUPPAM EXCAVATIONS

S.No.	Dynasty	Location	Depth	diameter	Weight	Remarks
1	Roman	Southern side of the western cloister *maṇḍapa*.	2.50m	1.6cm	1.30gm	Ob: Bust only – Arcadias (?) Re: Not clear
2	Roman	Western side of the main shrine near the outer wall of the g*arbhagṛiha*	2.70m	1.6cm	2.69gm	Ob: Bust of Arcadias (395-400CE) Bust turned to right with aquiline nose. Shown wearing a head band, with legend on either side of the bust. Re: Two standing figure each holding a standard on the left hand. One holding a ring on the right. Legend on the right side
3	Pallava	Surface find? Adjacent to the boulder	---	1.8cm	2.07gm	Ob: Standing bull Re: Not clear
4	Pallava	South-west of the main shrine	1.68m	1.4cm	1.02gm	Ob: Standing bull facing right with legend on the top left in Pallava Grantha characters which reads lakshita Re: Central depression – corroded (?)
5	Pallava	South-west of the main shrine	2.60m	1.9cm	2.68gm	Ob: Standing bull facing right within a dotted border Re: Not clear

6	Pallava (Broken)	South of the spear on the stone platform	2.70m	1.3cm	0.79gm	Ob: Bull Re: Not clear
7	Pallava (heavy coin)	Northern side of the *garbhagṛiha* of Phase I and beneath the a*rdhamaṇḍapa* of the Phase III	3.45m	2.3cm	9.64gm	Ob: Bull (not clear) Re: Chakra
8	Chōḷa (Ceylon man type	Eastern side of the main shrine	0.80m	1.6cm	4.18gm	Ob: Standing figure of king with head to right, and flower on left. The right arm is extended outwards and downwards; lamp on left and pellets on right field. Re: seated figure of king with left arm raised and right arm resting on the right knee; in the right field Nagari legend.
9	Chōḷa (Ceylon man type	Stone debris and brick wall (southern entrance)	2.90m	1.6cm	3.10gm	Ob: Standing figure of king with head to right, and flower on left. The right arm is extended outwards and downwards; lamp on left and pellets on right field. Re: seated figure of king with left arm raised and right arm resting on the right knee; in the right field Nagari legend.
10	Unidenti-fied	Surface find	---	1.9cm	3.27 gm	Ob: Not clear Re: Not clear

11	Unidentified	Surface find	---	0.9cm	0.24gm	----
12	Unidentified (Broken)	South-eastern side of the boulder	2.50m	1.2cm	0.41gm	Ob: Not clear Re: Not clear
13	Unidentified (Broken)	Between boulder and cloister *maṇḍapa*	2.73m	1.2cm	0.78gm	Ob: Not clear Re: Not clear

APPENDIX - IV

List of Roman Coins Found in Excavated Sites – Tamil Nadu

S.No.	Site	District	Period
1	Alagankulam	Ramanathapuram	Copper coin -Valentine II (383 – 395 A.D. Copper coin -Arcadius (395 – 408 A.D.) Copper coin CONCORDIA AVG Copper coin – SALVS REIPVBLICAE (402 -408 A.D.) Copper coin – GLORIA ROMANORVM (402 – 408A.D.) Copper coin – GLORI AROMA NORVM (408 – 423 A.D.)
2	Karur	Karur	Augustus I (27 B.C. – A.D. 14)

APPENDIX – V

List of Roman Coins Found in Tondaimandalam

S.No.	Site	Period
1	Cuddalore	Valentine I/II (?) (321 – 375 A.D./ 371 – 392 A.D) Eudoxia (?) (400 A.D.)
2	Mahabalipuram	Theodosius I (?) (346 – 395 A.D.) Valentine (?) (321 – 375A.D./371 -392 A.D.) Eudoxia (?) (400 A.D.)
3	Tirukkoyilur	Issuers (?) 4th AD.
4	Mambalam	Augustus I (27 BC-14 A.D)
5	Tondamanathan	Augustus I (27 BC-14 A.D) Tiberius (14 – 37 A.D.) Claudius (41 – 54A.D.) Nero I(54 68 A.D)
6	Saidapet	Unidentified

Appendix – VI

PUBLISHED INSCRIPTIONS AT MĀMALLAPURAM

Pallava Inscription

No	Place of inscription	King	Date	Language and script	Remarks
1	On the top of the niche in the north wing of verandah of the Ādivarāha temple	Narasiṁhavarmaṉ I (630-668A.D.)	----	Tamil, Pallava *Grantha*	It reads as *Sri Simhavinna pottrathirajan* **(ARE, 1922,no 661; SII, xii, no 17)**
2	Gaṇēśa ratha	Paramēśvaravarmaṉ I (669-700A.D.)	-----	Sanskrit, Pallava *Grantha*	Records that the temple was caused to be built by the king surnamed *Atyantakāma* to be the abode of Sambhu (śiva). The name of the temple is given as *Atyantākamapallavēśvaragṛiḥam* and his *birudas* are also engraved viz., *Paramēśvara,Śrīnidhi,Śrībhara, Raṇajaya, Taruṇāṅkura, Kāmarāga* etc.**(ARE, 1907,no. 531; SII, i, no. 18)**
3	In the Dharmarāja maṇḍapa	Narasiṁhavarmaṉ I (630-668A.D.)	-----	Sanskrit, Pallava *Grantha*	A duplicate of the above (**ARE, 1907, no. 532; SII , i , no. 19)**

4	On the top of the niche in the south wing of the verandah of the Ādivarāha temple	Mahēndravarmaṉ II (668-670 A.D.)	-----	Tamil, Pallava *Grantha*	It refers the king name *Sri Mahendra Pottrathirajan.*(**ARE, 1922, no. 662; SII, xii, no.18**)
5	In all three storeys of the Dharmarāja ratha	Paramēśvaravarmaṉ I (669-700A.D.)	-----	Sanskrit, Pallava *Grantha*	Enumerates the surnames of the king – *Srī Narasiṁha, Prithivīsvara, Bhuvanabhajana, Srīmēgha,Trailokyavardhana, Atyantakama,Anekapaya,Sthirabhakti, Madanabhirama,Apratihatasāsana, Kamalalitha, Ameyamaya,Sakalakālyaṇa, Nayanamanōhara, Vama,Atimana,Parapara, Anupama,Nayankura, Lalita, Sarvatōbhadra, Srīnidhi, Niruttara., Vibranta and Satyapārakrama.* Mentions *Atyantākamapallavēsvaragṛiham* and *Raṇajaya* (**ARE, 1907,nos. 512-29; SII, i, nos. 1-17**)
6	Ramanuja *maṇḍapa*	Paramēśvaravarmaṉ I (669-700A.D.)	-----	Sanskrit, Pallava *Grantha*	The imprecatory verse declares "six times cursed be those in whose hearts does not dwell Rudra (siva), the deliverer from the walking on the evil path."(**ARE, 1907, no.533; SII, xii, no.22**)

7	On the floor of the Ādivarāha temple	Paramēśvaravarman̤ I (669- 700A.D.)	-----	Sanskrit, Pallava *Grantha*	It records the imprecatory verses "six times cursed be those in whose hearts does not dwell Rudra (siva), the deliverer from the walking on the evil path." It is exactly identical with those found in the Ganesa Ratha, Dharmaraja mandapa and the Ramanuja maṇḍapa.**(ARE, 1922, no. 665; SII,xii, no 117)**
8	On the plinth of the two platforms excavated in the Western side of the 2nd courtyard of the Shore temple	Narasiṁhavarman̤ II (691-728A.D.)	------	Sanskrit, Pallava *Grantha*	It records the birudas of the king viz. *Apratima, Avanibushaṇa, Akalaṅka, Dharaṇīchandra, Arimardana,Atulabala, Kulatilaka, Atyantakama, Aparajita, Chandraradha-sekhaabhima, Guṇalaya, Srivallabha, Abhimana, Raara-sikhamani,Chaudasani,Udayachandra,Rajasiṁha,Ranajaya,Srībhara, Chitrakarnuukha, Ekavira, Sivachuḍamani, Kalakala, Abhirāma,Raṇavīra, Yuddhārjuna, Narēndrasiṁha* etc., **(ARE, 1912, no. 566; EI, xix, no. 18(A))**
9	On the beam below the cornice in the north wall of the Shore temple	Narasiṁhavarman̤ II (691-728A.D.)	-----	Sanskrit, Pallava *Grantha*	Reads as *sribha ras – chitrakarnuukah,* which seems to be a *biruda*.**(ARE, 1959-60, no. 263)**

10	On the pitha of the Bhuvarāha sculpture at the Shore temple	Narasiṁhavarmaṇ II (691-728A.D.)	-----	Sanskrit, Pallava *Grantha*	Records the *birudas* like S*ri Rajasimha, Sri Ranajaya, Sribhara,* and S*ri chitraka mukha*. **(IAR 1991)**
11	On the two side of a slab lying in the courtyard of the Ādivarāha temple	Nandivarmaṇ Pallavamalla II (731 – 796)	65th regnal year (796 A.D.)	Tamil, Tamil	It records that one Idaivalanchan Kandan belonging to the nagaram of Māmallapurm bought some land paying gold from kandan, the son of Ilam- Paduvumar, the headman of Kunrattur in *Amur nadu* and presented the same as grazing land for cattle. Among the boundaries of the land two tanks called *Koneri* and *Mandaitalaivaneri* are mentioned.**(ARE, 1922, no.666; SII, xii, no. 38)**
12	Dharmarāja Ratha	--------	------	Sanskrit, Pallava *Grantha*	Reads *Mahamalla*. Paleographically assignable to 7th to 8th cent A.D.**(ARE, 1958-59, no.384)**
13	On a pillar of the rock cut maṇḍapa (south west of the Gopis churn)	--------	------	Sanskrit, Pallava *Grantha*	Records the label *srivamankusa (*bearer of handsome elephant goad) Paleographically assignable to 7th to 8th cent A.D.**(ARE, 1907, no. 530; SII, xii, no.23)**
14	On the portal of the niche to the right of the Varāha cell in the Ādivāraha temple	--------	------	Sanskrit, Pallava *Grantha*	It gives the ten *Avataras* of Vishnu, viz., the fish, the tortoise, the boar, the Narasimha, the dwarf, Rama (Parasurama), Rama (Dasarathi Rama), Rama (Balrama), Buddha and Kalkin. Paleographically assignable to 7th to 8th cent A.D.**(ARE, 1922, no.663; SII, xii, no. 116)**

15	On the rock towards the north of Ganesa Ratha	--------	------	Sanskrit, *Grantha*	It records the name *vikri* (the extant portion) another label engraved nearby in slightly later Tamil characters reads as *Nārāyaṇa*. Paleographically assignable to 8th cent A.D.(**ARE, 1959-60, no. 264**)
16	Same place	--------	------	Sanskrit, *Grantha*	It records the name *Bhuvana.* Paleographically assignable to 8th cent A.D.(**ARE, 1959-60, no.265**)
17	On the eastern side of the Gopis churn,	--------	------	Sanskrit, *Grantha*	It refers to *Śrī–kaviti*. Paleographically assignable to 8th cent A.D.(**ARE, 1959-60, no. 266**)

Chōḷa inscriptions

No	Place of inscription	King	Date	Language and script	Remarks
18	On the south base of the Shore temple	Rājarāja I(985 – 1014)	25th regnal year (1010 A.D.)	Tamil,Tamil	It records a *vyavasthai* made by the *nagaram* and the *perilamai* of Māmallapuram, which met in the flower garden to the south of the temple of Jalasayanadevar in the presence of Pudukkudaiyan Ekadiran Aimbadenman, who was the settlement officer of the *Amur kottam*. **(ARE, 1887, no. 2; SII, I, no. 40)**
19	On the north base of the same temple	Rājarāja I(985 – 1014)	26th regnal year (1011 A.D.)	Tamil,Tamil	It seems to record a gift of gold by the *nagaram* and the *perilamai* of Māmallapuram, a *nagaram* of *Amur nadu* and *Amur kottam* to the temples of Kshatriyasimha pallava Isvara deva, Jalasayana, Rajasimha pallava –Isvara deva, and Pallikondaruliya –deva. **(ARE, 1887, no. 3; SII, i, no. 41)**
20	Inside the Shore temple	Rajendra II (1052 – 1064 A.D.)	9th regnal year (1061 A.D.)	Tamil,Tamil	It registers that the *Mahasabha* of Siridavur alias Narasimha mangalam donated 2000 *kuli* of land as Tiruvidaiyattam to the god of *Tirukkadalmallai.* **(ARE, 1887, no.1;SII, I, no42)**

21	On a niche in the Varahasvamin temple	Rajendra II (1052 – 1064 A.D.)	9th regnal year (1061A.D.)	Tamil,Tamil	It records tax free land by the *nagaram* and *perilamai* of Māmallapuram alias Jananathapuram to the god of Sri Paramesvara Mahavaraha Vishnugriha, at Māmallapuram a *nagaram* in *Amur nadu* in *Amur kottam*, a sub division of Jayakondacholamandalam. Among the boundaries of the gift land are the temple of Urgamudaiyamahavishnukkal and Māmallai peruman koyil. (**ARE, 1890, no. 54; SII, iv, no377**)
22	On the niche in the same temple	Rajendra II (1052 - 1064 A.D.)	------	Tamil,Tamil	It records a gift of land as tax free to the temple of Paramesvara mahavaraha Vishnu grihattalvar at Māmallapuram by the village of ***Tiruveḷichil,*** a *devadana brahmadeya* in *Amur nadu* in *Amur kottam*, a sub division of Jayangonda chola mandalam. Also refers to the gift of 90 sheep to the temple. (**ARE, 1890, no. 54A; SII, iv, 378**)
23	Stone slabs in the possession of Shri. Srinivasagopalachari	Kulōttuṅga I (1070-1122 A.D.)	-------	Tamil,Tamil	It contains portions of the historical introduction of the king, and seems to record a gift for the services to god Tirumudalisvaram udaiyar in Māmallapuram. (**ARE, 1961-62, no. 297**)

24	On the west wall of the Vāhana Mandapa near the Sthalasayana perumal temple	Vikrama Chōḷa (1118-1135 A.D.)	-------	Tamil,Tamil	It contains the historical introduction of the king commencing with the words Pumalai midaindu etc., of the inscription of the king. (**ARE, 1932-33, no. 116**)
25	On the stone slab in the possession of Shri. S.T.Srinivasagopala chri	Vikrama Chōḷa (1118-1135 A.D.)	-------	Tamil,Tamil	It also contains the historical introduction of the king commencing with the words Pumalai midaindu etc., of the inscription of the king. (**ARE, 1961-62, no. 298**)
26	On the east wall of the Gangaikondan maṇḍapa	Kulōttuṅga II-(?) (1130 – 1150 A.D.)	-------	Tamil,Tamil	It registers the gift of 12 kasu deposited in Sribandara Vaduga Naranan, a merchant of Anaikkarai puttur in *kanru nadu* in *Manaiyir kottam* for land sold by the *nagaram* and *perilamai*. (**ARE, 1901, no. 310; SII, vii, no. 536**)
27	On the left wall of the Gangaikonda maṇḍapa	Kulōttuṅga II-(?) (1130 – 1150 A.D.)	-------	Tamil,Tamil	It mentions *Amur nadu* in *Amur kottam* in Jayangondachola mandalam. (**ARE, 1887, no. 4; SII, i, no. 89**)

28	On the stone slabs in the possession of Shri. S.T.Srinivasagopala chari	Kulōttuṇga II-(?) (1130 – 1150 A.D.)	-------	Tamil,Tamil	It contains a portion of the historical introduction of the king. **(ARE, 1961-62, no. 299)**
29	On the stone slabs in the possession of Shri S.T.Srinivasagopala cahri	--------	-------	Tamil,Tamil	It seems to record a provision made for services to some deity. **(ARE, 1961-62, no. 300)**
30	On the stone slabs in the possession of Shri S.T.Srinivasagopala cahri	--------	-------	Tamil,Tamil	It records the sale of land for gold to provide for offerings to some deity. Mentions …variyaperumakkal. It seems to be the concluding portion of a record. In 12th cent.A.D. Characters. **(ARE, 1961-62, no. 301)**

Telugu Chola inscriptions

No	Place of inscriptions	King	Date	Language and script	Remarks
31	On the south wall of the central shrine of the Sthalasayana Perumal temple	Gandagopaladeva (1250)	2nd regnal year 1252 A.D.	Tamil,Tamil	It registers a gift of money by a *bhatta* of Kuttanpakkam for a twilight lamp in the temple of Ulaguyyannira perumal at Kadalmallai.(**ARE, 1932-33, no. 114**)
32	On the south wall of the central shrine of the Sthalasayana Perumal temple	Gandagopaladeva (1250)	-------	Tamil,Tamil	It records the gift of 12 *madai* by a certain Adinarayanan for burning a twilight lamp in front of the god (ula)guyya vanda perumal .**(SITI, I, no. 504)**

Pāṇḍya inscription

No	Place of inscriptions	King	Date	Language and script	Remarks
33	On the south wall of the mandapa, in front of the central shrine in the Sthalasayana Perumal Temple	Jatavarman Sundara Pandya I (?) or II (?)	16th regnal year (1288 A.D) or (1292 A.D.)	Tamil,Tamil	Registers the grant of the village Chedirayanallur founded in the name of Vadagopuramudaiyan Singapperumal alias chedirayan of Irayankudi in Pandimandalam for the maintenance of a feeding house conducted by the *bhattas* of Ayyanankakara Chaturvedimangalam who were the priests of the temple of the god of Tirukkadalmallai at Māmallapuram, made by the assembly of Paiyanur in *Amur nadu* in *Amur kottam*, a district of Jayakondacholamaṇḍalam. **(ARE, 1932-33, no. 115)**
34	On a stone slab in the possession of Shri. S.T.Srinivasagopalachari	--------	In 13th cent A.D. character.	Tamil,Tamil	It seems to record some grant to god Perumal Ulaguyya ..... for various services. **(ARE, 1951-52, no. 303)**

B) PUBLISHED INSCRIPTIONS AT ŚĀḶUVAṆKUPPAM

Pallava and Chola inscriptions

No	Place of inscription	King	Date	Language and script	Remarks
1	On the northern wall of the Śāḷuvaṇkuppam Atiraṇachaṇḍesvara Cave temple,	Narasiṁhavarmaṇ II (691-728A.D.)	------	Sanskrit, Pallava Grantha	It records that the temple was built by king Atiraṇachaṇḍa and was called after him Atiraṇachaṇḍēśvara..**(SII, i, no. 23)**
2	On the southern wall of the same Śāḷuvaṇkuppam cave temple	Narasimhavarman II (691-728A.D.)	------	Sanskrit, Pallava Grantha	Records that the temple was caused to be made for Sambhu by the king Atiranachaṇḍa and was named Atiranachaṇḍēśvara. It has the titles of the king like; *Atyantakama, Kamaraga, Srinidhi, Sribhara, Ranajaya, Anugrasila, Kalakala, Samaradhanamjaya, and Samgramadhira,* the three first are found in the Kāñchipūram inscription.**(SII, i, no. 21)**

3	On the northern wall of the same temple	Narasimhavarman II (691-728A.D.)	------	Sanskrit, Early Nagari	It gives the name of the temple as Atiraṇachaṇḍēsvara griham.(**ARE, 1907, no. 535; SII, i, no. 22)**
4	On a rock to the north of the cave temple	-------	------	Tamil, Tamil	It records the gift of one *Ma* land for food offering and a gold *kalanju* for a lamp to the god *Chattapperumanar* by a person named *Madisuda –Bhatta.*(**SII VOL.IV Of 380)**
5	On a rock to the north of the cave temple	Kulōttuṇga III (1178 – 1218 A.D.)	37th regnal year (1215 A.D.)	Tamil ,Tamil	It records a sale of some land belonging to the temple of Subrahmaṇya devar at Tiruviḷichilur, a *devadana brahmadeya* in *Amur kottam*, a sub division of Jayankonda chōḷa mandalam by the temple authorities to Karuppurudaiya Nambi Arpudakuttan. The land was sold it is said since the temple was badly in need of funds. (**ARE, 1890, no.57; SII, iv, no. 381)**

C) NEWLY DISCOVERED INSCRIPTIONS AT ŚĀḶUVAṆKUPPAM

Pallava inscriptions

No	Place of inscription	King	Date	Language and script	Remarks
1	Engraved on the pillar found on the western side of the cloister *prakara* during excavation	Nandivarman II(731 – 796 CE)	12th regnal year (743 CE)	Tamil ,Tamil	Records that 10 *kalanju* of gold was donated to the temple of Subrahmaṇya devar at Tiruviḷichil in *Amur kottam* by an individual Kiraippiriyan of Māmallapuram. Out of the interest, the festival during the asterism *Kartigai* in the month of *kartigai* to be celebrated and food offerings to be made on the occasion. The *Sabha* undertook to conduct the festival and provide food offerings. The members of the of the village gave an understanding to Kiraippiriyan to perform the above festival.

2	Engraved on the pillar excavated on the western side of the *Mahamaṇḍapa*	Kamphavarman (814 – 880 CE.)	17th regnal year (831 CE.)	Tamil ,Tamil	It refers to the gift of 16 *kalanju* of gold to the deity Subrahmaṇya situated in the village Tiruviḷichil by Vasanthanar, Kampa bhattan of *sandalya gotra*, and a resident of *Manaiyur nadu* in *Manaiyur kottam*. The total interest per year was 90 *nalis* of oil, accrued at the rate of 6 *nalis* per *kalanju* per year. The *Sabha* agreed to maintain the lamp without a break. She also donated a small house site and from the income sacred food offerings were to be provided. The *Sabha* agreed to maintain the food offerings annually. The name Nambi *bhattar* also finds mention.
3	On the pillar excavated near the western side of the cloister *prākāra*	Kamphavarman (814 – 880 CE.)	17th regnal year (831 CE.)	Tamil ,Tamil	It refers to the gift of 16 *kalanju* of gold to the deity Subrahmaṇya by Vasanthanar the wife of Siyasarman alias Kampa bhattan of *Sandalya*

					gotra of *Manaiyir kottam* in *Manaiyir nadu* for the lighting of perpetual lamps in the temple of Subrahmanya at Tiruviḷichil. The total interest per year was 90 *nalis* of oil, accrued at the rate of 6 *nalis* per *kalanju* per year. The *Sabha* agreed to maintain the lamp without a break. She also donated land to the temple from which food offering was made to the temple.
4	On the pillar excavated near the northern side of the stone spear	Kamphavarman (814 – 880 CE.)	18th regnal year (832 CE.)	Tamil ,Tamil	Records that Kopperuman, the wife of Siyasarman, of *sandalya gotra*, a resident of *Manaiyur nadu*, a sub division of *Manaiyur kottam* gifted 10 *kalanju* of gold to the deity, Subrahmanya in the temple at Tiruviḷichil. The *Sabha* agreed to provide 60 *nalis* of oil per year as interest on the gold and maintain the lamp throughout the year.
5	Engraved on small broken piece	Nrupatungavarman (854 – 880 CE.)	13th regnal year	Tamil ,Tamil	Incomplete. Name and regnal

	excavated on the north of stone spear		(882 CE.)		year of king mentioned.
6	On the slab excavated on the north western side of the *Maha maṇḍapa*		Palaeographically assignable to 8th century CE.	Tamil, Tamil	Records that, a certain individual Kiraippiriyan, donated 2 *kalanju* of gold for conducting the festival in the month of *Kartigai*, in addition to the money donated by his Guru and also for maintaining a perpetual lamp in the temple.
7	On the pillar found on the western side of the *Maha maṇḍapa*		Palaeographically assignable to 9th cent. CE	Tamil, Tamil	Records that Nambithaya bhattar donated 5 *kalanju* of gold. The *Sabha* agreed to supply 30 *nalis* of oil annually to the temple. The interest mentioned is 6 *nalis* of oil per *kalanju* per annum.
8	Found on the western side of the *Maha maṇḍapa*	---------	Palaeographically assignable to 9th cent CE	Tamil,Tamil	Records that Peruncatti Arrula Narayana Sarman regranted one piece of land to the temple for conducting a festival (*thalai pali*).

Rashṭraḳuta Iṇscriptions

No	Place of inscription	King	Date	Language and script	Remarks
9	Boulder (eastern side) adjacent to the excavated temple site.	Rashṭrakuta Ḳṛishna III(939 – 967 CE.)	4th regnal year 943 CE	Tamil ,Tamil	Refers to the gift of a land as *unnaligaipuram* (for sanctum services) to the temple of Subrahmaṇya at Tiruviḷichil in *Amur kottam* by the village.
10	On the pillar found at the south west corner side of the cloister *prakara*	Rashṭrakuta Ḳṛishna III (939 – 967 CE.)	21st regnal year 960 CE.	Tamil ,Tamil	A grant of 90 *kadi* of oil was made to the deity Subrahmaṇyadeva situated at Tiruviḷichil. The oil is to be handed over annually to the temple priests serving in the *garbhagṛiha (tiru unnaligai).* The *ur* further agreed to deposit 180 *nalis* of lamp oil on the gold received by it to the temple. The amount of gold received by the *ur* is not specified in the inscription. It was stipulated that the amount of lamp oil will be handed over to the temple authorities on three occasions in the months of *Masi, Vaigasi* and *Aiyppasi.* The amount of lamp oil payable during the months of *Masi* and *Vaigasi* are 10 nalis each and in the month of

					Aiyppasi, the amount is 4 *nalis*. It was further stipulated that the *ur* shall deposit the required amount of lamp oil even if the people (*kudi*) responsible to pay the oil have left the place. The inscription mentions that it is the order of the lord and whoever disrespects will be punished.

Chōḷa inscriptions

No	Place of inscription	King	Date	Language and script	Remarks
11	Engraved on the pillar found on the north of the stone spear	Rajaraja I (985 – 1014 CE.)	19th regnal year (1004 CE)	Tamil,Tamil	Records that the temple for Subrahmaṇya deva is situated at Tiruvelichiḷur in *Amūr kottam* in *Amur nadu*. The place is mentioned as a devadana brahmadeya and also a *bhattavritti*, granted to *bhattas*.
12	On the pillar found on the northern side of the main shrine	Kopparakesari – probably Rajendra I (1012 – 1044 CE.)	3rd. regnal year (1015 CE). Paleographically assigned to 11th /12th cent. CE.	Tamil,Tamil	Records the gift of land *Ainnurruva peraiyan kuli* to the deity Subrahmaṇya in Tiruveḷichil, a *devadana brahmadeya* in *Amur kottam*, a sub division of Jayakondacholamandalam for maintaining a perpetual lamp in the temple.
13	On the northern wall of the Patavettamman Temple in the village - Śāḷuvaṇkuppam		Paleographically assigned to 13th century CE.	Tamil,Tamil	Fragmentary. Mentions *Amūr kottam* and *nadu*, seems to records the gift of eight cows for maintaining the twilight lamp in the temple.
14	On the northern wall of the same temple.		Paleographically assigned to 13th century CE.	Tamil, Tamil	Fragmentary. Records about the lamp
15	On the northern wall of the same temple			Tamil,Tamil	Fragmentary. Not clear.

16	On the northern wall of the same temple		Paleographically assigned to 13th century CE.	Tamil,Tamil	Fragmentary. Records about Subrahmaṇya dēvar and a person Sentirattu Urudaiyar.
17	On the northern wall of the same temple		Paleographically assigned to 13th century CE.	Tamil,Tamil	Fragmentary. Records the signatory Arputakkuttan and Narpattennaayiravan.
18	On the northern wall of the same temple		Paleographically assigned to 13th Century CE.	Tamil,Tamil	Fragmentary. Records about Karuppur.
19	On the floor of the Atiranachanda Cave	Rajaraja I (985 – 1014 CE.)	.....	Tamil,Tamil	Fragmentary. Prasasthi is clear and only the Tiruviḷichil in amur koṭṭam is read.

PALLAVA INSCRIPTIONS

Inscription No: 1

Text:

Transliteration

1. svasti rī nandi
2. p pōttarai
3. yaṟku pan
4. niraṇḍāva(tu)
5. rī māmalla
6. purattu kīrai
7. ppiriya
8. tiruviḻichchil
9. sabhaiyō..
10. kkup pati
11. kaḻaju po
12. kuḍutta
13. āda po
14. liyūṭṭu ko
15. ṇ
16. ḍu
17. kāt
18. tigai
19. i kā
20. tti
21. gai nā
22. ṟu
23. via
24. ch se
25. itu
26. u
27. vo(m)

28.

29. m sabhai

30. yōm

31. ivvurm

32. kīraippri

33. yaukkukko(ṭ)

34. ṭikkuḍutt(m)

கல்வெட்டு - 1

1. ஸ்வஸ்தி ஸ்ரீ நந்தி
2. ப் போத்தரை
3. யற்கு பன்
4. னிரண்டாவ.
5. ஸ்ரீ மாமல்ல
6. புரத்து கீரை
7. ப் பிரியன்
8. திருவிழிச்சில்
9. சபையோ..
10. க்குப் பதின்
11. கழஞ்சு பொ
12. ன் குடுத்த
13. ன் அதன் பொ
14. லியூட்டு கொ
15. ண்ண
16. டு
17. காத்
18. திகை
19. ய் கா
20. த்தி
21. கை நா
22. ன்று
23. விழா
24. ச் செ
25. ய்து
26. ட்டு

27. வோ(மா)
28. னோ
29. ம் சபை
30. யோம்
31. இவ்வூரும்
32. கீரைப்பிரி
33. யனுக்குக் கொ(ட்)
34. டிக்குடுத்தோ(ம்).

Inscription No: 2

Text

Transliteration

1. svasti rī [kambha] va
2. mmarkku yāṇḍu
3. vatu tiruvi̲lichchil subra
4. hmaṇya bhaṭārarkku maṇai
5. yī sāṇḍayla ki̲lava
6. kayi.....sa makan
7. rī kambha bhaṭṭan brāhmaṇi va
8. santa piraṭṭiyār.....
9. kuḍutta patiā̲ru kalañju
10. poāl ka̲lañji vā
11. y aru nā̲lip pa̲liaiyāka ā
12. ṭṭaṇḍu tōrum toṇṇū
13. r̲u nā̲li eṇṇai ippo
14. koṇḍu ērikka.....
15. tthuk koṇḍu nan(tā) vi̲lakku
16. erippipōmāom.....
17. tiruvamirtukkuk kuḍutta
18. kī̲lan seruvil.....
19. na kallilum tiruva.....
20. vippomaom.....
21. .....

22. .....lai mel.....

23. tirukki kiḻavarāa.....

24. .....maṇṇa.....

கல்வெட்டு - 2

1. ஸவஸ்தி ஸ்ரீ (கம்ப வ)
2. ம்மற்கு யாண்டு பதினேழா
3. வது திருவிழிச்சில் சுப்ர
4. மண்ய படாரர்க்கு மணைய
5. யிற் சாண்ட யலன் கிழவன்
6. கயி..... சன் மகந்
7. ஸ்ரீ கம்பபட்டந் பிராமணி வ
8. ஸந்த பிராட்டியார்.....
9. குடுத்த பதினாறு கழஞ்சு
10. பொன்னால் கழஞ்சின் வா
11. ய் அரு நாழிப் பலிசையாக ஆ
12. ட்டாண்டு தோறும் தொண்ணூ
13. ரு நாழி எண்ணை இப்பொ
14. ன் கொண்டு எரிக்க.....
15. த்துக் கொண்டு நந்(தா) விளக்கு
16. எரிப்பிபோமானோம் ..
17. திருவமிர்துக்குக் குடுத்த
18. கிழான் செறுவில் ...
19. ந கல்லிலும் திருவ.....
20. விப்பொமனொம் .....
21. .....
22. .....லை மெல்.....
23. திருக்கின் கிழவரான.....
24. .....மன்ன

Inscription No: 3

Text

Transliteration

1. svasti rī kambha var

2. mmarku yāṇḍu patiēlā

3. vadu maṇaiyī kōṭṭattu ma
4. aiyi sāṇḍalya [ghon] kiḻa
5. vargalil [sī] yasarmarā
6. a rī kambha bhaṭṭa brāhma
7.ṇi vasantaā.....tiru
8. viḻichchil [rī su]
9. brahmaṇya bhaṭṭārarku
10. nontā viḻakkuk[ku] kuḍu
11. tta po pati[ā]
12. ru kaḻañju ippo
13. il kaḻañji vā
14. yau nāḻi eṇṇai
15. paliaiyāha[ā]ṭṭā
16. ḍu tōrum mu.....
17. aippasi se.....
18. dat toṇṇūr[ru]
19. nāḻi eṇṇai e
20. rivippōmāō
21. m sabaiyōm i
22. nta pirāṭṭiyār tiruva
23. mirtukkuk kuḍutta
24. kīḻan seṟuvi[l]
25. vanta bhoham[]
26. ttāṇḍu tō[rum]
27. iruttu kuḍu
28. ppōmāōm
29. sabaiyōm [//*]

கல்வெட்டு - 3

1. ஸவஸ்தி ஸ்ரீ கம்ப வர்
2. ம்மற்கு யாண்டு பதினேழா

3. வது மணையிற் கோட்டத்து ம
4. னையிற் சாண்டல்ய (கொங்)கிழ
5. வர்களில் (சி)யசர்மரான
6. ன ஸ்ரீ கம்பபட்டன் பிராம
7. ணி வசந்தனா...திரு
8. விழிச்சில் (ஸ்ரீ ஸு)
9. பிராமண பட்டாரர்க்கு
10. நொந்தா விளக்குக்(கு) குடு
11. த்த பொன் பதி(நா)
12. ரு கழஞ்சு இப்பொன்
13. இல் கழஞ்சின் வா
14. யறு நாழி எண்ணை
15. பலிசையாக (ஆ)ட்டாண்
16. டு தோறும் மு.....
17. அய்ப்பசி செ....
18. டத் தொண்ணூற்(று)
19. நாழி எண்ணை எ
20. ரிவிப்போமானோ
21. ம் சபையோம் இ
22. ந்த பிராட்டியார் திருவ
23. மிர்துக்குக் குடுத்த
24. கிழான் செறுவி(ல்)
25. வந்த பொஹம்(ஆ)
26. ட்டாண்டு தோ(ரும்)
27. இருட்டு குடு
28. ப் போமானோம்
29. சபையோம்(//*).

Inscription No: 4

Text

Transliteration

1. svasti rī
2. kambha varmmakku
3. yāṇḍu pati
4. eṭṭāvadu maṇai[yīr]

5. kōṭṭattu maṇai[yi]
6. .....ndan kiḻavarkalil ra
7. .....la iyasarma brāhmai
8. .....
9. nankaip perumāḻ
10. tiruviḻichchil subrahma
11. ṇya bhaṭārarkku nantaviḻakku
12. kku kudutta po pattu i-
13. p poṇṇālum vanta arupathiā
14. ḻi eṇṇaiyum āṭṭa
15. ṇḍu tōrum tiruviḻakku
16. muṭṭāmai eriya.....vai
17. ppōmāom sabhaiyōm.

கல்வெட்டு - 4

1. ஸவஸ்தி ஸ்ரீ
2. கம்ப வர்மற்கு
3. யாண்டு பதின்
4. எட்டவது மணை(யிற்)
5. கோட்டத்து மணை(யி)
6. .....ந்டங் கிழிவர்களில் ர
7. .....ல சியசர்மன் பிராமணி
8. .....
9. நங்கைப் பெருமாள்
10. திருவிழிச்சில் சுப்ரம
11. ண்ய படாரர்க்கு நந்தாவிள்க்கு
12. க்குக்கு குடுத்த பொன் பத்து இ
13. ப் பொன்னாலும் வந்த அறுபதினா
14. ழி எண்ணையும் ஆட்டா
15. ண்டு தோறும் திருவிளக்கு
16. முட்டாமை எரி.....வை
17. ப் போமானோம் சபையோம்.

Inscription No: 5

Text

Transliteration

1. Svasti rī yā
2. ṇḍu nirupatu
3. karkkup patimū
4. rāvadu kat

கல்வெட்டு - 5

1. ஸவஸ்தி ஸ்ரீ யா
2. ண்டு நிருபதுங்
3. கற்குப் பதின் மூ
4. ன்றாவது கத்

Inscription No: 6

Text

Transliteration

1. svasti rī kīraippiriya koṇḍan nannatya sivar kārtti
2. gaiyuukkun tiruvilavukkumahat tagal kurukkal vai
3. itta poōdu mudalāka vaitta poiru kaḷañju.

கல்வெட்டு - 6

1. ஸவஸ்தி ஸ்ரீ கீரைப்பிரியன் கொண்டந் நந்நத்ய சிவரே கார்த்தி
2. கையூட்டுக்குந் திருவிழாவுக்குமாகத் தங்கள் குருக்கள் வை
3. ய்த்த பொன்னாடு முடலாக வைத்த பொந்நிரு கழஞ்சு.

Inscription No: 7

Text

Transliteration

1. tirukki kiḻavar nam
2. pirāṭṭiyār nandaviḻakku

3. kkuk (kuk) kuḍutta po
4. aiñ kaḻañjun kaa
5. ñjiāl arunāḻippa
6. liaiyaka ā
7. ṭṭāṇḍu tōrum mu
8. ppatiāḻi.....
9. .....kuduppip
10. pomam
11. sabhaiym

கல்வெட்டு - 7

1. திருக்கின் கிழவர் நம்
2. பிராட்டியார் நந்தாவிளக்கு
3. க்குக் (குக்) குடுத்த பொன்
4. அஞ் கழஞ்ச்சுங் கழ
5. ஞ்சினால் அறுநாழிப்ப
6. லிசையாக ஆ
7. ட்டாண்டு தோறும் மு
8. ப்பதினாழி.....
9. .....குடுப்பிப்
10. பொமானோம்
11. சபையோம்

Inscription No: 8

Text

Transliteration

1. svasti rī yāṇ
2. .....vadu perucha
3. ṭṭi āaāna kaya
4. .....tasappaḍi
5. vivukku kuḍutta kō

6. ṭṭa[t]tu kiḻā seruvum....

7. maṇa kollaiyil k

8. tu kākkum paṭṭiyum

9. kuḍutta.....kāṇi....ko

10. r̲r̲a sarmmasauva viya

11. um vir̲r̲uk koṇḍa pa

12. ach cheruvum ittiruvi

13. ḻāvukkē kuḍuttarōm.

கல்வெட்டு -8

1. ஸவஸ்தி ஸ்ரீ யாண்ண
2. .....வது பெருஞ்ச
3. ட்டி ஆற்றழாந கயண்ண
4. .....தஸப்படி
5. விழாவுக்கு குடுத்த கோ
6. ட்ட(த்)து கிழான் செறுவும்.....
7. மண கொல்லையில் கானா
8. து காக்கும் பட்டியும்
9. குடுத்த....காணி.....கொ
10. ற்ற சர்ம்மசறுவ விய
11. றும் விற்றுக் கொண்ட பள்
12. ளச் செறுவும் இத்திருவி
13. ழாவுக்கே குடுத்தாரோம்.

RĀSHṬRAKUṬĀ INSCRIPTION

Inscription No: 9

Text

Transliteration

1. svasti rī kachchiyum tajaiyum koa kaara dvarkku yu nlvadu

2. mūrk kattu tiruviichchil ruaiya dvar subrahmaya bharar

3. uigaippuramāi.....ruaiym idvarku(//*)

கல்வெட்டு -9

1. ஸ்வஸ்தி ஸ்ரீ கச்சியும் தஞ்சையும் கொண்ட கன்னர தேவர்க்கு யாண்டு நாலாவது
2. ஆமூர் கோட்டத்து திருவிழிச்சில் ஊருடைய தேவர் சுப்ரமண்ய பட்டாரர்
3. உண்ணாழிகைப்புறமாய் ..... ஊருடையோம் இத்தேவர்க்கு(//*)

Inscription No: 10

Text

Transliteration

1. svasti rī kaara dēvarkku yā
2. ṇ̣u 21 āvadu āmūr kōṭṭat
3. tu tiruvi̲lichchil subrah
4. manyadēvar tiru uṇṇā̲li.....
5. pi̤diyarula...ka....tatu...il...
6. .....lu.....āhak ka̤dantu
7. eṇṇa̲likkallāl thoṇṇu
8. r̲ru kāḍiyum āṇ̣utōrum
9. nikar tigalum mahap pa̤da nisitam
10. a̲ru nā̲li a̲lantu kṳduppōmāha .
11. gal ka̤dappoṇṇāl.....
12. .....teai patinā̲likka.....
13. i nai [a]ippasiyarupati
14. nāiyum ha iya.....
15. iyum aippasi yarupa
16. tiā̲liyum āha....vu.....
17. nā̲li eṇṇaiyum rī k[o]
18. yilukkē koṇ̣u sendu.....
19. .....kuduppōmāom.....
20. ritta.....iyyutom tiruvāṇ̣u
21. .....varma vaiva tiruvi̲lichchi

22. ....nom iv.....pa.....

23. .....ittiruviḻichchil.

24. yom.....ivaha.....r

25. ppaya.....

26. yrum pe.....

27. .....vaum tiruveḻuttu

28. .....paum tdappaḍuvōm

29. .....ilt tirambinom kē[kaiyiḍai]

30. .....kumari iḍai se[i]tār sēta.....

31. vom [//*]

கல்வெட்டு - 10

1. ஸவஸ்தி ஸ்ரீ கன்னர தேவர்க்கு யா
2. ண்டு 21 ஆவது ஆமூர் கோட்டத்
3. து திருவிழிச்சில் சுப்ர
4. மண்ய தேவர் திரு உண்ணாழி.....
5. ற் பிடியருல... க...தது...ல்....
6. .....லு....ஆகக் கடந்து
7. எண்ணாழிக் கல்லால் தொண்ணூ
8. ற்று காடியும் ஆண்டுதோறும்
9. நிகர் திங்களும் மகப்பட நிசிதம்
10. அறுநாழி அளந்து குடுப்போமாக..
11. ங்கள் கடப்பொன்னால்.....
12. .....தென்னை பதினாழிக்க.....
13. ழி நாழி (அ)ய்ப்பசியறுபதி
14. நாழியும் ஆக சிய.....
15. ணியும் ஐப்பசி யருப
16. தினாழியும் ஆக.....உ.....
17. நாழி எண்ணையும் ஸ்ரீ கெ(ா)
18. யிலுக்கே கொண்டு சென்று.....
19. .....குடுப்போமானோம்.....
20. ரித்த.....ய்யுதொம் திருவாண்டு
21. .....வர்ம வழிவா திருவிழிச்சில்
22. ....நொம் இவ்.....ப.....

23. .....இத்திருவிழிச்சில்....
24. யொம்.....இவக.....ர்
25. ப் பய.....
26. யாரும் பெற்.....
27. .....வாழும் திருவெழுத்து
28. .....படும் தீண்டப்படுவோம்
29. .....ல் த் திரம்பினொம் கே(ங்கையிடை)
30. .....குமரி இடை செ(ய்)தார் செத..
31. வொம்(//*)

CHŌḶA INSCRIPTION

Inscription No: 11

Text

Transliteration

1. .....rī
2. rājarāja rājakēa
3. ri pama.....yādu 19 āvadu
4. āmūrk kōṭṭattu āmūr nāṭṭu
5. vrōm ikkōṭṭat
6. tut tiruvi̲lichchil subrah
7. maṇya.....
8. .....cha ṇḍya dēvar

கல்வெட்டு -11

1. .....ஸ்ரீ
2. ராஜராஜ ராஜகேஸ
3. ரி பன்ம.....யாண்டு 19 ஆவது
4. ஆமூர்க் கோட்டத்து ஆமூர் நாட்டு
5. வூரோம் இக்கோட்டத்
6. துத் திருவிழிச்சில் சுப்ர
7. மண்ய.....
8. .....ச ண்ட்ய தேவர்

Inscription No: 12

Text

Transliteration

1. svasti rī kōp
2. parakēsari (pamaku)
3. yāu mūrāvathu
4. (vadu) jeyako
5. a chōḻama
6. dalatthu āmūrkō
7. ṭṭattu āmūr[nāṭṭu]
8. dēvatānam brahmatē
9. yam tiruveḻichchil
10. (subramaya dēvarkku)
11. tiruveḻichchi....(kaḻi)
12. vali.....
13. ku.....n... m...
14. ṭttu. t tiruna-
15. ntā vilakkukki
16. kku .....
17. tira.....
18. sammatittē.....ka.....
19. ainnūṟṟu
20. va pēraraiyae
21. n.....kuḻikku iṟai.
22. .....

கல்வெட்டு- 12

1. ஸவஸ்தி ஸ்ரீ கோப்
2. பரகேசரி (பன்மற்)
3. யாண்டு மூண்றாவது

4. (வது) ஜயங்கோ
5. ண்ட சோழ மண்
6. டலத்து ஆமூர்கோ
7. ட்டத்து.....
8. தேவதாநம் பிரமதே
9. யம் திருவிழிச்சில் சு
10. (ப்ரமண்ய தேவர்க்கு)
11. திருவிழிச்சி...(கழி)
12. வளி.....
13. கு.....ந்.....ம்....
14. டத்து.. த் திருந
15. ந்தா விளக்குக்கி
16. க்கு.....
17. திர.....
18. சம் மதித்தேன் க...
19. ..ன் ஐந்நூற்று
20. வ பேரரையனே
21. ன் குழிக்கு இறை
22. .....

Inscription No: 13

Text

Transliteration

1. - mr kattu mr nu tvatnam piramatya i tiru...-
2. - ukku via pasu a ippasu eum kaikkou santi vikku...-
3. - aum ikkyil ki uaiya piramaha na...-
4. - .....villi..... pati.....an.....-

கல்வெட்டு - 13

1. - ஆமூர் கோட்டத்து ஆமூர் நாட்டு தேவதாநம் இ திரு.....-
2. - டுக்கு விட்ட பசு அ இப்பசு எட்டும் கைக்கொண்டு சந்தி விளக்கு..-
3. - டனும் இக்கோயில்ம் காணி உடைய சுப்ரஹம ந..-
4. - ..... வில்லி..... படி....னந்....-

Inscription No: 14

Text

Transliteration

1. (vi)akku ko-
2. ikk -
3. vilil a -

கல்வெட்டு - 14

1. (வி)ளக்கு கொ -
2. இக்கோ -
3. விலில் அ -

Inscription No: 15

Text

Transliteration

1. ippai aiven ivvr ka -
2. koa villi uai -

கல்வெட்டு - 15

1. இப்படி அறிவெந் இவ்வூர் க -
2. கொண்ட வில்லி ஆளுடை -

Inscription No: 16

Text

Transliteration

1. .....ruaiya supramaya tva
2. .....rkka kaavmm
3. .....sentiattu vruaiy.....-

கல்வெட்டு- 16

1. ....ருடைய சுப்ரமண்யதேவ
2. .....ர்க்க கடவோமானோம்
3. ..... செந்திறத்து வூர்ய்டையா.....-

Inscription No: 17

Text

Transliteration

1. aiyn aputakkttan euttu
2. ana npattoyira -

கல்வெட்டு - 17

1. டையாந் அற்புதக்கூத்தந் எழுத்து
2. றநான நாற்பத்தொண்ணாயிர -

Inscription No: 18

Text

Transliteration

1. .....aiya mapati karuppr -
2. .....epp.....-

கல்வெட்டு - 18

1. ....டைய மற்பதி கருப்பூர் -
2. ...எப்பேற்....-

Inscription No: 19

Text

Transliteration

1. savasti rī tirumaka pōla perunilach selviyu taakkēyurimai pūṇṭa
2. mai makkoḷa kāntaḷūr sālai kalamaṟuttaruḷi vēkai nāṭu kaka pā

3. ṭiyu uḷampapāṭyun taṭikaipāṭiyum kuṭamalai nāṭum kollamu
4. kalikamum eṇṭsai pukaḻ tara īḻamaṇṭalamum iraṭṭapāṭi
5. ēḻaraiyilakkamun tiṇṭiṟal venṟiththaṇṭāṟ koṇṭa taeḻl
6. vaḷruḻiyu..... ḻutakai viḷku yāṇṭē se
7. ḻiyarai tē.....sa kēsari pamarāa iī rāsarāsa tēva
8. rkku..... vatu āmūr kōṭṭa
9. ā..... tiruviḻichchil
10. rī (āti)..... (va)rattu mahā
11. tēvarkku.....ma.....
12. .....mu.....

கல்வெட்டு - 19

1. ஸவ்ஸ்தி ஸ்ரீ திருமகள் போல பெருநிலச் செல்வியுதனகெஉரிமை பூண்ட
2. மை மனக்கொளகாந்தளூர் சாலை கலமறுத்தருளி வெங்கை நாடுங் கங்க பா
3. டியு நுளம்பபாடியுந் தடிகைபாடியும் குடமலைங் நாடும் கொல்லமுங்
4. கலிங்கமும் எண்டிசை புகழ் தர ஈழாண்டலமும் இரட்டபாடி
5. ஏழரையிலக்கமுந்திண்டிறல் வெந்றித்தண்டாற் கொண்ட தன்னெழில்
6. வளருழியு .....ழுதகை விளங்கு யாண்டே செ
7. ழியரை தே.....ச கேசரி பன்மரான ஸ்ரீ ராசராச தேவ
8. ர்க்கு.....வது ஆமூர்கோட்ட
9. ஆ.....திருவிழிச்சில்
10. ஸ்ரீ (ஆதி).....(வ)ரத்து மஹா
11. தேவர்க்கு.....ம.....
12. .....மு.....

Appendix –VII

CONJECTURAL DRAWING

A conjectural isometric drawing of the temple based on the available plan on the ground and the conclusion drawn from the various structural activities has been prepared. The plan as envisaged here is of the period when the temple complex had all its components and when it had reached its zenith.(Phase II)

Basically the temple was facing north with a north-south orientation enclosing the natural rock boulder which too had a religious signifcance. Entering through the north one would fnd a *Balīpiṭha.* As one moves towards the temple there is the most important attribute of the deity Subrahmaṇya, the*"Vēl"* in stone placed on a brick paved platform. The above described features are all located in the open court yard which has a floor of rammed earth. Stone steps lead to the main temple complex with *garbhagṛiha, ardhamaṇḍapa, mahamaṇḍapa* and *mukhamaṇḍapa* enclosed by the cloistered *prākāra* wall. The temple complex also had an entrance from the south. Primarily the temple must have been simple and moderate in its construction as the evidence from the excavation shows. It might have been a rudimentary structure with a hut shaped (*kūtagāra*) *vimāna* similar to the Draupadi Ratha in Māmallapuram with a terracotta fnial on the top. The surface might have been decorated by stucco works, pieces of which were recovered during the excavations. Also the lime plastered surfaces carried some paintings. The other *maṇḍapas* were all flat roofed. The cloister around had a series of stone pillars supporting a tiled roof which again is conjectured due to the retrieval of roofng tiles from the site. The plinth of the temple was at a higher level and was reached through lateral stone steps.

Śāḷuvaṇkuppam: Conjectural Isometric view of the Excavated Temple site

Fig. 29

BIBLIOGRAPHY

Annual Reports on Indian Epigraphy.

Crole, C.S. 1888. *Chengalpattu District Manual*, Govt. Press, Madras.

Epigraphia Indica (for the concerned years).

Gopalakrishnan, M. 2000. *Gazetteers of India: Tamil Nadu State- Kanchipuram and Tiruvallur Districts*, Vol. I, pp.10-11.

Gopinatha Rao, T.A. 1993. *Elements of Hindu Iconography* , Vol. II, Part II, Delhi.

Indian Archaeology- A Review (for the concerned years).

Krishnamurthy, R. 2004. *The Pallava Coins*, Chennai.

Krishnamurthy, R. 2007. *Late Roman Copper Coins from South India*: *Karur, Madurai and Tirukkoilur*, Chennai.

Mahalingam, T.V. 1969. *Kanchipuram in Early South Indian History*, Madras.

Mahalingam, T.V. 1970. *Report on the Excavations in the Lower Kaveri Valley (Tirukampuliyur and Alagarai) – 1962-64,* Madras University Archaeological Series, No. 2, University of Madras, Madras.

Mahalingam, T.V. 1988. *Inscriptions of the Pallavas*, Indian Council of Historical Research, Delhi.

Mahalingam, T.V. 1989, *A Topographical List of Inscriptions in the Tamil Nadu and Kerala States, Chingelput District*, Vol. 3, Indian Council of Historical Research, Delhi.

Meister, Michael W. 1983. *Encyclopaedia of Indian Temple Architecture, South India, Lower Dravidadesa, 200 B.C- A.D. 1324*, American Institute of Indian Studies, Oxford University Press, Delhi.

Nagaswamy, R. 1968. *Mamallai*, State Department of Archaeology, Government of Tamil Nadu.

Nagaswamy, R. and A. Abdulmajeed. 1978. *Vasavasamudram,* A Report on the Excavation Conducted by the Tamil Nadu State Department of Archaeology, Government of Tamil Nadu, Chennai.

Nagaswamy, R. 1979, *Thiruttani and Velanjeri Copper Plates*, Tamil Nadu State Department of Archaeology, Madras.

Narasimhan, K.T. (Unpublished) *Excavations Report on Mahabalipuram (1990-2000),* Archaeological Survey of India, Temple Survey Project (SR), Chennai.

Narayana Ayyar, C.V. 1974. *Origin and Early History of Saivism in South India*, University of Madras, Madras.

Pillay, K.K. *A Social History of the Tamils*, Vol. I, University of Madras, Madras.

Rajamanikkanar, M. 2000. *Pallavar Varalaru*, Thirunelveli Saiva Siddhanta Works Ltd, Chennai.

Raman, K.V. Archaeological Excavations in Kanchipuram, *Tamil Civilisation,* Vol. 5, Nos. 1&2.

Ramaswami, N.S. 1989. *2000 Years of Mamallapuram,* Vol. I, Navrang, New Delhi.

Sarma, I.K. 1982. *The Development of Early Siva Art and Architecture, with special reference to Andhradesa*, New Delhi.

Sitharam, Gurumurthi (Ed.). 2008. *Mangulam Excavation (2006-2007)*, Department of Archaeology, Government of Tamil Nadu, Chennai.

Soundara Rajan, K.V. 1972. *Indian Temple Styles, The Personality of Hindu Architecture*, New Delhi.

Soundara Rajan, K.V. 1980. *Glimpses of Indian Culture Art and Architecture*, Delhi.

Soundara Rajan, K.V. 1994. *Kaveripattanam Excavations 1963-73, (A Port City on the Tamil Nadu Coast)*, Archaeological Survey of India, Government of India, New Delhi.

Soundara Rajan, K.V. 2006. *Nagajunakonda 1954-60*, Vol. II, Archaeological Survey of India, Government of India, New Delhi.

South Indian Inscriptions (for the concerned years).

South Indian Temple Inscriptions (for the concerned years).

Sridhar, T.S. 2005, *Alagankulam – An Ancient Roman Port City of Tamil Nadu,* Department of Archaeology, Government of Tamil Nadu, Chennai

Srinivasan, K.R. 1958. The Pallava Architecture of South India, *Ancient India*, No. 14, Archaeological Survey of India, Government of India, New Delhi.

Swaminatha Iyar, U.V. 1958, *Purananuru,* Kazhalagam, Part I, Madras, 1954.

Swaminatha Iyar, U.V. *Purananuru*, Kazhalagam, Part-I, Madras.

Swaminatha Iyar, U.V. 1957. *Paripadal*, Madras.

Swaminathan S. 2000. *Rashtrakuta Inscriptions in the Tamil Country,* Kannada University, Hampi.

Tamil Lexicon ,Vol. III, Part 1.

Tamil Lexicon, Vol. VI, Part 3.

Wheeler, R.E.M., A. Ghosh and Krishna Deva. 1946. Arikamedu: An Indo Roman Trading Station on the East Coast of India, *Ancient India*, Vol. 2, Archaeological Survey of India, New Delhi.

